Topic: LC-SPIRE-08-2020

Technical Annex 1-3

CEM-WAVE – Novel Ceramic Matrix Composites produced with Microwave assisted Chemical Vapour Infiltration process for energy-intensive industries

List of participants

Part.	Participant organisation name	Acronym	Type	Country
1	University of Pisa	UNIPI	UNI	Italy
2	National Research Council	IPCF-CNR	RTO	Italy
3	Fricke und Mallah Microwave Technology GmbH	FM	SME	Germany
4	Fraunhofer-Gesellschaft*	FHF	RTO	Germany
5	University of Birmingham	UoB	UNI	UK
6	Archer Technicoat Limited	ATL	SME	UK
7	National agency for new technologies, energy and sustainable economic development*	ENEA	RTO	Italy
8	Centre national de la recherche scientifique	CNRS	RTO	France
9	Politecnico di Torino	POLITO	UNI	Italy
10	IRIS Technology Solutions SL*	IRIS	SME	Spain
11	ArcelorMittal*	AMIII	LARGE	Spain
12	Steinbeis Advanced Risk Technologies*	R-TECH	SME	Germany
13	Kneia S.L.	KNEIA	SME	Spain

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LIST OF ABBREVIATIONS

CEM-WAVE most used acronyms: Artificial Intelligence (**AI**); Ceramic Matrix Composite (**CMC**); Environmental Barrier Coating (**EBC**); Green House Gasses (**GHG**); Key Performance Indicator (**KPI**); Life Cycle Analysis (**LCA**); Life Cycle Costing (**LCC**); Microwave assisted Chemical Vapour Infiltration (**MW-CVI**); Risk Analysis (**RA**); Radiant Tube Furnaces (**RTF**); Specific Objective (**SO**); Sustainable Process Industry through Resource and Energy Efficiency (**SPIRE**); Thermo-economic Analysis (**TA**); Weak Matrix Composite (**WMC**); Weak Interface Composite (**WIC**).

Management acronyms: Advisory Board (**AB**); General Assembly (**GA**); Consortium Agreement (**CA**); Communication and Dissemination Team (**CDT**); Project Coordinator (**CO**); Grant Agreement (**GA**); Innovation Management Team (**IMT**); Plan for the exploitation and dissemination of results (**PEDR**); Project Management Team (**PMT**); Research and Technology Organizations (**RTOs**); Small and medium-sized enterprise (**SME**); Steering Committee (**SC**); Valorization Committee (**VC**).

1. EXCELLENCE

1.1 Objectives

1.1.1 Rationale

In 2018 the European Commission set the strategy "A Clean Planet for all", a long-term vision for a European climate neutral economy. In December 2019 the European Green Deal (COM (2019) 640) was published, setting new ambitious priorities and aiming at Europe to become the first climate-neutral continent by 2050. Considerable effort is being made towards reaching this objective, such as research and innovation on technologies that will allow an intense exploitation of renewable energy, improvement of energy efficiency and development of clean, new low-carbon technologies.

Renewable energies are, by their very nature, inconsistent, fluctuating and potentially generating extreme conditions (such as the danger of material embrittlement caused by greater use of clean hydrogen). It is thus of outmost importance to adapt current processes to these future changes, to allow increased use of clean and renewable energy for industrial settings. This is particularly true in the so-called energy intensive industries, such as steelmaking. Novel materials able to sustain variable and harsh

conditions (high temperature, corrosive environments, etc.) are needed to optimise the shift in energy sources and, at the same time, guarantee high-energy efficiency and high-performance.

Materials with high-performance and energy efficient properties that are potentially able to withstand such demanding conditions do already exist but, due to their high production cost, they are currently not used in high energy process industry sectors. These materials are the **Ceramic Matrix Composites (CMCs)**.

In CEM-WAVE our aim is to introduce to European industries an innovative CMC production process, based on **Microwave assisted Chemical Vapour Infiltration (MW-CVI)** technologies. This novel process, while **maintaining the excellent properties of the currently used CMCs**, will, most importantly, **significantly reduce the costs for making components**, thus making CMCs **sustainable for process industries in energy-intensive sectors such as steelmaking**.

CEM-WAVE focuses on the application of this innovative production process to develop a CMC-based component that will, in turn, allow the creation of a small-scale innovative tube embedded with sensors. The small-scale CMC-based tube will be validated both in laboratory reproducing relevant environments of the envisaged future energy source scenarios and in an actual radiant tube furnace in the steelmaking industry with the aim of substituting the currently employed Inconel/Stainless steel alloys. The research and innovation work will be flanked by AI-aided modelling research to predict the material behaviour. Furthermore, Life Cycle Analysis (LCA), Life Cycle Costing (LCC), Thermoeconomic Analysis (TA), including the the waste management process, will be undertaken to guarantee that the project follows, at every step, the best directions in term of sustainability and future market uptake of the generated results.

1.1.2 General and Specific objectives

The **general objective** of CEM-WAVE is to **validate (TRL5) an innovative Microwave assisted Chemical Vapour Infiltration technology (MW-CVI)** to produce **Ceramic Matrix Composites (CMCs)** with a **significant reduction in production costs**.
This answers to the need of high-performance materials and components able to withstand fluctuating and extreme future processes conditions **that industries are going to experience due to increased use of renewable energy sources.**
The cost reduction derived from CEM-WAVE technology will make CMCs affordable to a large palette of industrial sectors currently not in condition to make use of this high quality, but currently expensive, material.

To reach the general objective the following Specific Objectives (SOs) have been identified. For each technical SOs (SO1-SO6), KPIs and related target values have been identified and detailed aiming at either a) assess the expected improvement in technical / process indicators, or b) assess the product / process / activity consistency with relevant standards and requirements.

SO1 – MW-CVI process optimisation for efficient non-oxide CMC preparation: the objective is to optimise the Microwave assisted Chemical Vapour Infiltration (MW-CVI) technology for non-oxide-based CMC production. In more detail, the expected KPIs related to SO1are:

Main indicators for SO1	
Key Performance Indicators	**Targeted Metrics**
• Processing time	• 40÷50 hours
• Density	• 2,4÷2,5 g/cm^3
• Porosity	• < 10%
• Fibre content	• ~30÷45 vol %
• Elongation at break	• > 0,4 %
• Young Modulus	• 250 GPa
• Tensile strength	• 300 MPa
• Flexural strength	• 400 MPa
• Interlaminar shear strength	• 30 MPa
Related WPs: 1, 2, 3, 4, 5	

SO2 – MW-CVI process optimisation for efficient oxide CMC preparation: the objective is to develop and then optimise the Microwave assisted Chemical Vapour Infiltration (MW-CVI) technology for oxide-based CMC production. In more detail, the expected KPIs related to SO2 are:

Main indicators for SO2	
Key Performance Indicators	**Targeted Metrics**
• Processing time	• 25÷40 hours
• Density	• 3,4÷3,5 g/cm^3
• Porosity	• 15÷20% for WMCs and <10% for WICs
• Fibre content	• ~40÷45 vol%
• Elongation at break	• ~0,5÷0,7% for both WMCs and WICs
• Young Modulus	• ~200 GPa (WMCs) and ~300 GPa (WICs)
• Tensile strength	• ~150 MPa (WMCs) and ~300 MPa (WICs)
• Flexural strength	• ~200 MPa (WMCs) and ~350 MPa (WICs)
• Interlaminar shear strength	• ~30 MPa (WICs)

Related WPs: 1, 2, 3, 4, 5

SO3 – Effective coating and joining techniques: the objective is to define the most suited protocols to perform first coating and then joining of both oxide- and non-oxide-based CMCs. In more detail, the expected KPIs related to SO3 are:

Main indicators for SO3	
Key Performance Indicators	**Targeted Metrics**
• maximum joining process T (non-oxide CMC) • maximum joining process T (oxide-oxide CMC) • maximum joint in service T (non-oxide CMC) • maximum joint in service T (oxide-oxide CMC) • minimum joint strength • maximum joined area • surface complexity • maximum EBC process T (non-oxide CMC) • maximum EBC process (oxide-oxide CMC) • maximum EBC in service (non-oxide CMC) • maximum EBC in service T (oxide-oxide CMC)	• 1500°C • 1100°C • 1200°C • 900°C • 15 MPa (or comparable to that of CMCs) • 100 mm^2 • flat/curved surfaces • 1600°C • 1100°C • 1700°C • 1400°C
Related WPs: 4, 6	

SO4 – AI-assisted multiscale predictive modelling: the objective is to develop a reliable multiscale predictive model of CMC tubes for radiant furnaces using, among others, AI tools. In more detail, the expected KPIs related to SO4 are:

Main indicators for SO4	
Key Performance Indicators	**Targeted Metrics**
• Scanning and segmentation of CMC specimens • Micro- and meso-scale geometrical and transfer properties evolution with infiltration • Infiltration simulations • Optimized infiltration conditions	• Quality and accuracy of the final numerical meshes • Consistency of results w.r.t. actual measurements (validation) • Validation w.r.t. experimental data on infiltration gradients • Verification of diminished infiltration gradients under suggested optimal conditions
Related WPs: 5	

SO5 – Process analytical technology (PAT) and health monitoring: the objective is to make a profitable use of CEM-WAVE produced CMCs as a sensor, on the basis of the CMC dielectric properties. The sensorial aspect will be two-fold: a) using MW during the CMC production to determine the optimal process conditions. B) during the use of the CMC in radiant tube furnaces (RTF) with combined use of IR imaging and electrical analysis. In more detail, the expected KPIs related to SO5 are:

Main indicators for SO5	
Key Performance Indicators	**Targeted Metrics**
• Sensor robustness • Tailored PAT sensors and software	• Resist process/use conditions, e.g. temperature >1000ºC • Imaging capabilities to detect defects and 2 versions for process control and health monitoring of CMC embedding data driven models for process optimisation
Related WPs: 3, 7	

SO6 – Small-scale version of the tube (Ox and non-ox CMC) validated in current furnace: the small-scale version of the tube will be produced with a length of around 200 mm, a wall thickness of 3÷9 mm and a diameter of 100 mm. It will display improved properties compared to tubes based on current materials and the technology will be validated to TRL5 in this relevant environment. In addition, the MW-CVI technology will have been upgraded by the use of the latest solid-state sources, which will improve its energy efficiency even further compared to current manufacturing practice. The small-scale tube to be tested will have to fulfil all the following Key Performance Indicators (KPIs):

Main indicators for SO6	
Key Performance Indicators	**Targeted Metrics**
• Heat transfer efficiency • Emissivity • Thermal conductivity • Max Temperature in application • Min Temperature in application • Lifetime • High temperature impact resistance • Corrosion resistance	• ≥30% • >0,8 • ~20÷30 W/m K (for non-oxide); ~ 2,3÷4,5 W/m K (for oxide); • ~1200°C • ~830°C • >20% (>10 years) • >160 J/m • 0,4 mpy

● Fatigue resistance	● 10.000 cycles
Related WPs: 7	

SO7 –Sustainable long-term solution: In order to bring the novel composites into the real full-scale market, sustainable environmental and economic solutions are intended to be met through three stages; (i) Initial guidance in material selection based on LCA/LCC analysis, (ii) Encouraging circular economy principles to steer and optimize the development of CEM-WAVE along the value chain including use-phase and end-of-life; and (iii) Risk assessment of the processes with respect to ISO31000 /31050 to identify the potential hazards to human health and environment, as well as identification and monitoring of the new and emerging risks posed by such technologies, in order to develop a reliable methodology for better risk management of high-performance materials and combined components. Proposed sustainable development methodology will enable quantification and measurement of eco-efficiency to identify solutions with low environmental impact and high economic viability. The SO7 will deliver a scenario model to identify suitable paths to market for the current project but also for future development projects, where CEM-WAVE CMCs will aim at higher TRLs.

Main indicators for SO7	
Key Performance Indicators	**Targeted Metrics**
● Reduction of CO_2 emissions and resource utilisation ● Energy reduction ● Quantitative risk assessment	● 20% decrease ● 30% decrease ● For minimum 5 alternative solutions
Related WPs: 8	

SO8 Maximise exploitation, dissemination and impact: It is essential for the results of CEM-WAVE to be used effectively for the benefit of European industries. In addition, it is important to position CMCs as competitive solutions for successful future exploitation, through which further business cases in other sectors and for other possible applications will be explored, appropriate IPR protection sought, and dissemination will be implemented as per the plan described in Section 2. Business models and all documentation (marketing, technical, product LCA, etc.) for the future transfer of the CEM-WAVE technology framework to other sectors will be generated. Moreover, a stakeholder group will be formed in the initial stages of CEM-WAVE whose members will act as a continuous sounding board for the development results and as exploitation and replication partners depending on the members' profile, need and interest.

Main indicators for SO8	
Key Performance Indicators	**Targeted Metrics**
● Number and expertise of members in the stakeholder group ● Dissemination activities organised and attended ● Meetings with the external AB ● Innovative business models developed	● At least 50 members in the stakeholder group ● At least 16 conferences and 10 articles in high-impact journals ● Organisation of 2 CEM-WAVE workshops ● 4 Business models validated
Related WPs: 9	

SO9 Address the barriers and facilitate the integration of the developed processes and materials, primarily in the steelmaking sector but also in other relevant sectors, and assess side-effects and risks: CEM-WAVE (WP8-9) will evaluate the current and optional future gaps that could be barriers for the inclusion of the results into the steelmaking application field along three main axes: post project financing, technical and political/regulatory barriers. Among such barriers are the lack of internalisation of externalities, resource pricing; lack of skills and investment in the steelmaking sector; insufficient enablers to improve cross-cycle and cross-sector performance due to non-alignment of incentives for transformation between actors within and across value chains and so on. A comprehensive risk register will be developed early in the project lifetime and appropriate mitigation strategies will be defined by the consortium members. Ultimately, analysis of post project financing, technical and political/regulatory barriers for steelmaking sectors will be carried out.

1.2 Relation to the Work Programme

CEM-WAVE addresses the topic LC-SPIRE-08-2020: "Novel high performance materials and components"; from a wider point of view, CEM-WAVE perfectly fits the context where the topic is placed, i.e. the Leadership in enabling and industrial technologies – Nanotechnologies, Advanced Materials, Biotechnology, and Advanced Manufacturing and Processing Work Programme under the Sustainable Process Industry (SPIRE) section.

CEM-WAVE focuses on the steel industry, one of the high-energy demanding sectors which are considered a key part of Europe's manufacturing base and which represents, together with other sectors such as cement and chemicals, 20% of European industry in terms of both employment and turnover. The steel sector is characterised by energy intensive technologies and thus, among other targets, it has to optimise its industrial processes to reduce energy consumption and increase the use of renewable sources to make it environmentally sustainable.

How CEM-WAVE will address the Specific Challenge is detailed in the table below:

Topic Specific challenge	How CEM-WAVE will address the challenge
Energy intensive industries will require a radical transformation of their production processes to reach carbon neutrality by 2050. Future low carbon technologies and processes should address fluctuating and extreme conditions, such as high temperature or corrosive environments, materials and components that will need to be able to be sustained.	Possible scenarios (WP7 – Task 7.1) to simulate the conditions where the innovative CEM-WAVE CMCs will be tested. The scenarios, all applicable to radiant tube furnaces, will reproduce the expected harsh conditions due to the introduction of hydrogen/biogas mixtures used as alternative clean energy sources for steelmaking processes.
In the same way, they also need to be designed for high-energy performance	Thermo-economic analysis (TA) and Energy optimization (Task 8.3) of processes using CEM-WAVE materials will be performed throughout the project.
Existing components materials and any combination thereof have however inherent limitations to meeting new extreme conditions. Stress resulting in degradation, corrosion, wear and/or deterioration can in particular lead to reduced plant efficiency even plant shutdowns or entire equipment failures. The challenge is therefore to develop new, or overhaul the performance of, materials and combined components.	Novel CMC components' properties will be studied and compared to the properties of currently employed materials in RTF (key part of the furnaces in the continuous annealing and hot dip galvanizing lines). CEM-WAVE components will target several Key Performance Indicators such as lifetime (Task 4.1 and Task 4.4), heat transfer efficiency (Task 4.2), high temperature impact (Task 4.1), corrosion resistance in relevant environment (Task 7.1) and health condition monitoring will be performed (Task 7.4). Different joining and integration technologies will be investigated (WP6) to produce complex shapes and for easier maintenance and repair in service.

How CEM-WAVE fits the topic scope is detailed in the table below:

Topic scope	How CEM-WAVE fits the topic scope
Proposals should develop and test high performance materials and combined components to withstand extreme and varying conditions that are expected in future processes, and improve their target performance for a long time	CEM-WAVE will develop CMC-based materials and components (WP1-3) produced by microwave assisted chemical vapour infiltration process (MW-CVI) upgraded with the latest fully tuneable solid state sources (WP2), improving heating uniformity of the sample with a faster and finer control of the operative parameters. The CEM-WAVE components will be first characterized (WP4), joined and coated (WP6) and finally tested in harsh environments (WP7) to have an expected lifetime at least 20% higher than currently used metallic-based materials in RTF even in future extreme and fluctuating conditions. Accelerated ageing (Task 4.4) and creep (Task 4.1) tests will be performed to precisely assess the increased lifetime of the samples produced in relevant environment.
Design, including through modelling and artificial intelligence, …	CEM-WAVE will develop new image-based (using X-ray CMT and SEM micrographs data) numerical modelling methodologies for the MW-CVI process, through AI-aided tools starting from material properties (internal geometry, heat and mass transfer properties and complex dielectric permittivity into the operative ranges of interest), to determine optimal conditions for infiltration time, material quality and energy consumption (WP5).
… development, processing and testing of highly innovative materials with improved properties in terms of, e.g., temperature and humidity resistance (corrosion, oxidation, thermal insulation), strength, functionality, weight, etc ..	CEM-WAVE will develop, process (WP3) and test (WP7) non-oxide SiC-based CMCs, oxide CMCs ($Al_2O_{3f}/YAG-ZrO_2$ and $Al_2O_{3f}/Al_2O_3-ZrO_2$), and joining (WP6) for complex shaped CMC components. All materials will be characterized (WP4) to have excellent thermo-mechanical properties up to at least 1200°C (creep resistance, thermal shock and impact resistance, pseudo-ductile behavior).
…. and components with graded and protective coatings, yield strength, hardness, and resistance to media relevant for specific industrial application.	CEM-WAVE products are themselves resistant to yield strength and with very high hardness. Proper environmental barrier coatings (EBCs) will be deposited on them to increase high temperature tolerance to combustion environments (WP6).
Components embedded with sensors to minimize industrial processing conditions constraints	CEM-WAVE will a) investigate (Task 3.3) a new line of the latest generation monitoring tools based on the variation of the dielectric properties along the densification of the composite, controlling the processing conditions constraints and avoiding risks mainly connected to overheating and plasma appearance; b) combined use of IR imaging and electrical analysis to perform structural monitoring on the walls of the ceramic composite tubes samples potentially leading to malfunctions during its use (Task 7.4).

Significant increase in lifetime of equipment by reducing damage and degradation such as wear and oxidation in high temperature demanding applications of productive processes	CEM-WAVE will employ proper EBCs to improve oxidation resistance and to avoid excessive wear mainly for high-temperature demanding applications thus increasing the lifetime of equipment of at least 20%.
Reduction of environmental impacts in terms of waste management and energy and resource consumption	LCA, LCC, TA and risk assessment will be performed in WP8 to assess, among other, environmental impacts of processes using CEM-WAVE material. Moreover, the increased thermal efficiency of CMCs and the introduction of ceramic inlays inside the end leg (made of oxide-oxide CMCs) will allow to reduce pollutants emission due to the lower outlet temperature.

As requested by the topic, CEM-WAVE includes a well-defined business case and an exploitation strategy, both thoroughly described in Section 2.

1.3 Concept and methodology

1.3.1 Concept

1.3.1.1 CEM-WAVE overall concept

The escalating problems resulting from climate change and especially to the environmental impacts of anthropogenic activities are getting advisedly exponential attention. This is reflected in the political direction of the recently established European Commission: the Communication called "The European Green Deal" clearly outlines the mid-term and long-term objectives, which can be synthesised in "to transform the EU into a fair and prosperous society, with a modern resource-efficient and competitive economy where there are **no net emissions of greenhouse gases in 2050** and where economic growth is decoupled from resource use". In order to reach those main objectives, several key elements have been identified. Two of them are shared by and at the basis of CEM-WAVE:

- Mobilising industries for a clean and circular economy: energy-intensive industries, such as steel, chemicals and cement, are indispensable to Europe's economy, as they supply several key value chains. The decarbonisation and modernisation of this sector are thus essential, as also testified by the recommendations published by the High Level Group of energy-intensive industries, which demonstrates the industry's commitment to these objectives [1]. Furthermore, EU industry needs 'climate and resource frontrunners' to develop the first commercial applications of breakthrough technologies in key industrial sectors by 2030. Priority areas include clean hydrogen, fuel cells and other alternative fuels, energy storage, and carbon capture, storage and utilisation. One of these priorities is towards clean steel breakthrough technologies leading to a zero-carbon steel making process by 2030. The proposal to support zero carbon steelmaking processes by 2030 is one of the actions expected to be taken in 2020.
- Mobilising research and fostering innovation: new technologies, sustainable solutions and disruptive innovation are critical to achieve the objectives of the European Green Deal; apart from the full range of instruments available under the Horizon Europe programme, partnerships with industry and Member States will support research and innovation on low-carbon steel making among others.

In such context, research and innovation are directed in several directions including change of processes which will have to use clean energy sources to get lower CO_2 emissions. For energy intensive sectors, where very high temperatures have to be reached (e.g. steelmaking), the industries will need to follow one or more low-emissions pathways. Concerning steel industry, the tender will be to use energy inputs provided from circular carbon technologies, fossil fuels with carbon capture and storage (CCS) and clean power for hydrogen-based ironmaking processes. **Clean hydrogen** is enjoying unprecedented political and business momentum, with the number of policies and projects around the world. Two key developments have contributed to the growth of hydrogen in recent years: the cost of hydrogen supply from renewables has come down and continues to fall, while the urgency of greenhouse gas emission mitigation has increased [2]. The hydrogen debate has evolved over the past two decades, with a shift in attention from applications for the auto industry to hard to-decarbonise sectors such as energy-intensive industries, trucks, aviation, shipping and heating applications. Ensuring a low-carbon, clean hydrogen supply is essential; current and future sourcing options include: fossil fuel-based hydrogen production (grey hydrogen); fossil fuel-based hydrogen production combined with carbon capture, utilisation and storage (CCUS; blue hydrogen); and hydrogen from renewables (green hydrogen). Regardless of the hydrogen source. Greener fuels containing hydrogen (mixtures of hydrogen and bio-methane), that will be used to replace current fuels, can cause **harmful phenomena such as hydrogen embrittlement for metals and hydrothermal attack for ceramics**. These conditions would not be sustained by the currently used materials. We thus need a novel material that will be able to endure in such future extreme and fluctuating conditions for a long time, and contemporarily maintain its high-performance properties. CEM-WAVE is based on the concept that **a material able to withstand such conditions while keeping its high performances for long times already exists, i.e. Ceramic Matrix Composites (CMCs)**: CMCs are a class of materials that show considerable promise to provide fracture-toughness values similar to those for metals such as cast iron, yet are much lighter and designed for applications in severe environments (often combining high temperatures, high stress levels and corrosive atmospheres). Thanks to their exceptional properties, including high toughness, low density, high temperature, wear and corrosion resistance, these materials are increasingly being considered for use in applications where reliable operations under extreme conditions are required. Currently they are being introduced into many other new fields and their range of applications will grow if

their manufacturing costs will be significantly reduced [3]. CEM-WAVE innovative idea is thus directed at **drastically reducing costs of CMC production to allow their sustainable use in sectors where their benefit would be enormous both from the environmental impact and from the productivity points of view**. To highlight the impact of CEM-WAVE, one of the most important European stakeholders on the ceramic market (Sacmi) wrote an endorsement letter.

Figure 1.1 *- CEM-WAVE concept: nowadays CMCs are used for niche sectors (aerospace, defense, luxury cars, nuclear). CEM-WAVE, thanks to the reduction of the process time and improvement of the efficiency, will guarantee a CMCs lower cost to be affordable for new markets and, in cascade, a lower cost also of raw materials.*

Starting from the results of HELM [4] (New technologies based on physical processing of materials for mechanical or electro-technical applications – FP7 – NMP – GA N. 280464), FP7 project in which 5 out of the 13 CEM-WAVE partners were involved, CEM-WAVE aims at advancing Microwave assisted Chemical Vapor Infiltration (MW-CVI) technologies. The use of these technologies will in fact (as described in *Fig.1.1*):

- Drastically reduce the time of CMCs processing, which is one of the main components of this product costs;
- The replacement of magnetrons with newer solid-state sources will grant a fast, precise and automatic tuning of operational frequency and phase with a high coupling level and a much more uniform temperature profile in the preform.

Thus, **CMCs produced with MW-CVI will be cheaper but, at the same time will have higher performances than the materials currently used.**

In the steelmaking sector, most blast furnaces (BF) and subsequently basic oxygen furnaces (BOF), currently reduce iron ore using a high-temperature synthetic gas derived from coal and coke as energy inputs. Specifically, in steel processing lines, indirect heating of products in protective atmosphere is carried out with radiant tubes equipped with gas-fired burners (around 50 burners per horizontal furnace and 150 burners per vertical furnace). These components are made different grades of Inconel / stainless steel, depending on the location (first leg or end leg). In CEM-WAVE we aim at validating the use of non-oxide and oxide CMCs in radiant tube furnaces instead of the currently used material. CEM-WAVE technology will reach TRL 5. Demanding for this concept to be feasible and to find applicable solutions, CEM-WAVE consortium planned the research and innovation activities as described below and in the *Figure 1.2*. Set-up of the materials and of the Microwave technology will be performed (WP1-2); the hearth of the project activity is the oxide and non-oxide CMC production processes (WP3), contemporary material characterisation (WP4), AI-assisted modelling (WP5) and CMCs joining techniques (WP6), all feeding each other in a circular process, will identify the best composition and process parameters for obtaining the best oxide and non-oxide CMC materials. Sustainability aspects, both in terms of market analysis and environmental impacts will be investigated, (LCA, LCC, TA and Risk Analyses - WP8). All these activities will be completed by Dissemination, Communication and Exploitation (WP9) and Coordination and management (WP10) activities. The involvement of several Research and Technology Organisations (RTOs) with diverse expertise at the international level in Materials Science, and past projects in collaboration with material or sectoral industries, will ensure the experience for effective technology transfer from research to industrial applications. On top of the well-defined business case, an exploitation strategy has been carefully planned in order to give to CEM-WAVE results larger possibilities of being used in other sectors where similar conditions are expected to be in place. The multi-disciplinary team (Material Science and & Engineering, Electronic Engineering, including experts in Electromagnetic wave propagation and their interaction with matter, Physicists, Chemists and Environmental Scientists) brings together European expertise in all the relevant fields required by the project. Between them, the consortium has the skills needed to achieve the

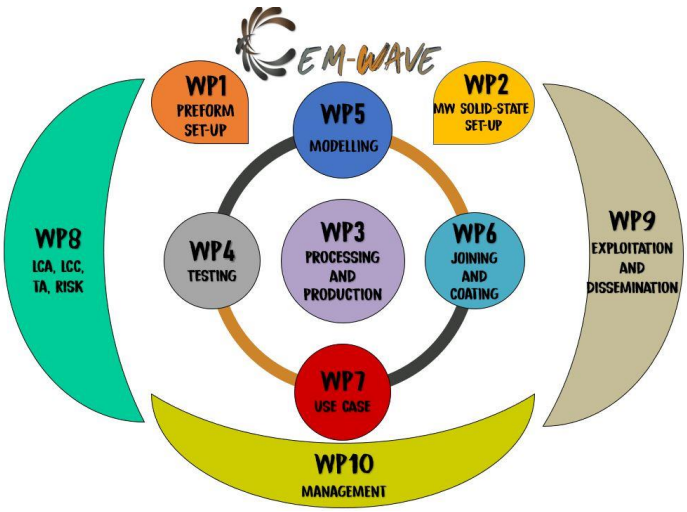


Figure 1.2 - *CEM-WAVE WPs strategy and configuration.*

project objectives and to develop and transfer the know-how to further enable industrial-scale exploitation, creating the required political and social environment underpinned by economics (business case) that encourages both transfer and adoption of these ideas. CEM-WAVE will enable this development and transfer of know-how through its value chain of actors. The participating academic and technology institutes (8 RTOs) includes both international experts in MW-CVI processing of non-oxide CMCs (UNIPI) and oxide CMCs (UoB), one of the leaders in all process chains of CMCs including fibre and EBC coating (FHF), renowned expert in experimental and numerical electromagnetic methods including high-temperature dielectric characterization of CMC materials (IPCF-CNR), the Italian largest institute on thermo-mechanical and non-destructive testing for CMCs (ENEA), internationally renowned group on micro/macro-scale modelling using AI-aided tools of CMCs (CNRS), and the top research teams on advanced joining and integration technologies for CMCs including testing and monitoring of the joined components (POLITO). The industry participants include the industrial final end-user (AMIII) which is the world's leading steel and mining company, one of the SME leading German suppliers of microwave ovens for industry and research (FM), a SME specialised in the development of CVD processes and the construction of industrial CVD equipment (ATL) and a SME expert in development of novel monitoring hardware and software solutions (IRIS). CEM-WAVE has identified the essential needs to consider regulative constraints (end-user), financial aspects (KNEIA), as well as environmental issues (R-TECH), together with all partners involved in the value chains, at an early development stage. The competence and expertise of the Consortium will be complemented by the Advisory Board (AB) which will consist of prominent experts, in MW-CVI techniques and in CMC materials both in academia and in industrial sectors. The preliminary composition of the AB is: Byungil Yoon (DACC CARBON Co., Ltd., South Korea), Alexey Kulik (Soft-Impact Russia), Alberto Ortona (SUPSI, Switzerland), Roland Weiss (Consultant for CMCs, Germany).

1.3.1.2 Positioning of the project

From an overall project point of view, CEM-WAVE will start at TRL 3 and will achieve TRL 5 by the end of the project. CEM-WAVE can be split in several sub sections, each of them characterised by a well-identified technology (Table 1.1).

***Table 1.1** – Positioning of the project*

Technology	Initial TRL	Final TRL	Description
Manufacturing of ceramic based fibre preforms based on prepregs and filament winding technology	3	5	Manufacturing of adapted fibre preforms, flat samples and tubes, by using coated oxide and non-oxide ceramic fibres (WP1). Adjustment of suitable porosity and density for the CVI-MW processing (WP3)
Coating of non-oxide and oxide ceramic fibres for a viable fibre-matrix interface in CMCs	3/4	5	Coating of ceramic fibres in a fast, robust and continuous roll-to-roll process. Design of multilayer coatings to provide a weak fibre-matrix bonding for damage-tolerant WIC. Coated fibres will be processed to prepregs in WP1.
Application of EBC coatings on CMC to improve life time of components	3	5	Deposition of multilayer EBC coatings on CMC by a slurry coating process. Coated CMC test samples and prototypes will be characterized and evaluated in WP4 and WP7 under relevant test conditions.
Production of SiC-based CMCs by MW-CVI upgraded with solid-state sources	3/4	5	Production of pilot-scale SiC-based CMCs samples with different geometries by upgraded MW-CVI process (WP3). Materialsand processing conditions will be optimized by multiscale modeling (WP5). The material produced will be tested (WP4) and then joined and coated (WP6) to be finally validated in relevant environment for the business case (WP7)
Production of oxide CMCs by MW-CVI upgraded with solid-state sources	3	4	Production of pilot-scale oxide-based CMCs samples with different geometries by upgraded MW-CVI process (WP3). The material and processing conditions will be optimized by multiscale modeling (WP5).The material produced will be tested (WP4) and then joined and coated (WP6) to be finally validated in relevant environment for the business case (WP7)
Characterisation of SiC-based CMCs durability and performances in combustion environments	3/4	5	Corrosion, thermophysical and thermomechanical tests in simulated working conditions (WP4)
Joining of CMCs produced and monolithic SiC	3/4	5	Preparation (WP6) and thermophysical and thermomechanical characterisations (WP4) for expected working environments
Monitoring of CMC process and use	3/4	5	Initial TRL based on the results of HELM project that will be further developed to reach in line monitoring capabilities (WP3 during process and WP7 during use). The single components such as sensors IR camera, software, etc. are market ready for other applications but need to be tailored to the use case especially in terms of withstanding the harsh conditions. Specific data driven models must to be developed for close loop control.

1.3.1.3 Other national and international R&I activities linked to CEM-WAVE

CEM-WAVE will build on prior state-of-the-art as well as Intellectual/Industrial Property belonging to several of the Consortium partners. A selection of the relevant projects that are linking research efforts to CEM-WAVE are listed in the following table (and further projects can be found in the respective partner profiles).

Table 1.2 *- National and international R&I activities linked to BIOHIGH*

Project	Partner(s) involved	Relation to CEM-WAVE
HELM, High-frequency Electro-Magnetic technologies for advanced processing of ceramic matrix composites and graphite expansion (FP7-NMP, 2011-2015-LARGE-5)	UNIPI, IPCF-CNR, FM; ATL, R-TECH	In HELM an innovative hybrid conventional/microwave assisted chemical vapor infiltration (MW-CVI) pilot plant was designed, developed, validating this technology for the production of SiC-based ceramic matrix composites (CMCs). The experience gained in the design of the MW-CVI reactor in HELM will be exploited for the further upgrade to solid-state sources.
High temperature composites using microwave enhanced chemical vapour infiltration (funded from the US Air Force Research (AFRL), 2016-2019)	UoB	A microwave assisted chemical vapour infiltration (MW-CVI) pilot plant was designed and constructed to allow the production of SiC-based ceramic matrix composites (CMCs)
EXCITE Processing and manufacturing of Ox Ox CMCs (funded from the Aerospace Technology Institute (ATI), 2018-2020)	UoB	A programme of research designed to learn how to produce oxide-oxide composites with both low cost processability and the required end-use properties.
MATISSE, Materials' Innovations for a Safe and Sustainable nuclear in Europe (FP7, 604862, 2013-2017)	ENEA, POLITO	In WP3 of Matisse project (FP7) high reliability SiC_f/SiC was studied for continuous operation at 1000°C for nuclear fuel cladding
Nextower, Advanced Materials Solutions For Next Generation High Efficiency Concentrated Solar Power (CSP) Tower Systems (H2020, 721045, 2017-2020)	ENEA, CertiMaC, POLITO	In Nextower project, SiC-based materials were characterized in simulated and real working conditions (thermal cycling up to 900°C)
EEE-CFCC (2016-2018), **FireMat** (2019-201) (POR-FESR, Emilia Romagna region).	ENEA, CertiMaC	Development of new low cost CMC as thermostructural fire resistant materials
CMC-Engine, Lightweight structures of SiC_f/SiC-Ceramic for gas turbines of the next generation (FHG funded program, grant number 833 886, 2017-2020)	FHF	In CMC-Engine project FHF developed a very good knowledge about the use of SiC_f/SiC-ceramic composite materials used in gas turbines. This knowledge can be useful as starting point for the production of the new CMCs technology.
CaGeFa, Development of fiber coating and fiber preforms by using textile technology for efficient CMC´s (funded by the German Ministry of Research and Development, grant number: 03X0143C, 2015-2018).	FHF	In CaGeFa project FHF developed a fiber coating technology that can be adopted for the preliminary fiber coating requested in CEM-WAVE project (WP1).
Ker_Twk, CMC for aeroengines (German funded LUFO project, 2018-2022)	FHF	In KerTwk CMC especially the complete SiC-process chain will be investigated with the focus on the fibre coating and infiltration processes.
TraTurb, CMC for turbines (German funded BMBF project, 2019-2022)	FHF	In TraTurb SiC-CMC are manufactured by using different fibre preforms and matrices, machined and tested by using international standards.
GaMA, new Gas routes for Aeronautic MAterials preparation (French Aquitaine Region & Safran group project, 2019-2022)	CNRS	In GaMa project new Gas routes for Aeronautic Materials preparations have been studied. The modelling approach adopted and the background acquired can be used for the muti-scale modeling of the MW-CVI process (WP5).
ADMACOM, Advanced manufacturing routes for metal/Composite components for Aerospace 16 (FP7 –FoF.NMP.2013 GA 609188, 2013-20)	POLITO, FHF	In ADMACOM innovative manufacturing technologies based on advanced design of interfaces and of joining materials between ceramic and metals materials for aerospace components, were developed. The knowledge can be transferred to the study of the new joining technique between ceramic components (WP6).

JOLIE, JOining of LIghtweight alloys to advanced FGM mEtal-ceramic materials (European Project MATERA –ERANET Ministry of Education, Universities and Research –MIUR 2227, 2013-2015)	POLITO	In JOLIE the research activity of POLITO was focused on the study of joining materials and processes for Al-alloy /ceramic brake systems. The knowledge can be transferred to the study of the new joining technique between ceramic components (WP6).
PROPAT, Robust and affordable process control technologies for improving standards and optimising industrial operations (H2020 SPIRE, GA 637232, 2014-2018)	IRIS	PROPAT developed novel sensors and analysers to measure properties of process streams and products, accurately and in real-time, integrating them into a global control platform for data acquisition and data processing. This background can be used as starting point for the development of new sensors to control and tune the operating parameters of process streams and products (WP2-3).
FIRE-RESIST - Developing novel fire-resistant high performance composites (FP7- 246037, 2011-2015)	R-TECH	In FIRE-RESIST project R-Tech made the LCA and Risk Management for the production of novel fire-resistant high-performance composites. The skills obtained can be useful for LCA and RA of the CEM-WAVE project (WP8).
caLIBRAte, Performance testing, calibration and implementation of a next generation system-of-systems (H2020 GA 686239, 2016-2019)	R-TECH	In caLIBRATE project R-Tech carried out Risk Assesment for the production of engineered nanomaterials. The skills obtained can be useful for LCA and Risk Analysis of the CEM-WAVE project (WP8).

1.3.2 Methodology

CEM-WAVE will be built carefully just like a spider does with the creation of its web *(Fig. 1.3)*. A work that is successful only with the integrated contribution of all the work packages, which represent the threads woven together.

More specifically, the ambitious objectives of the project will only be reached, because of the well-integrated planned activities of the inter-disciplinary CEM-WAVE consortium on the different aspects.

Four main groups of activities have been identified:

- laboratory Research and Technical Development - RTD (WP1 to WP6);
- industrial validation in relevant environment (WP7);
- LCA, LCC, TA & risk assessment (WP8);
- and dissemination and exploitation (WP9).

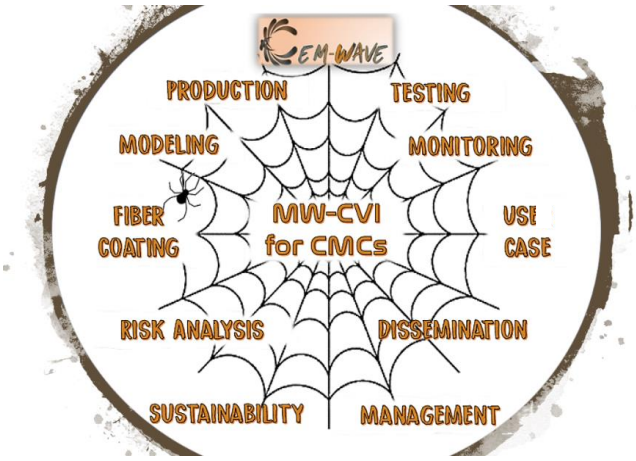


Figure 1.3 *– CEM-WAVE spider web construction. All activities are related like braided threads needing everyone's contribution for CEM-WAVE success.*

All activities will be supported by the management activities (WP10). The main approach behind CEM-WAVE is **the parallel validation of the two class of CMC material**s: a) **non-oxide CMCs** offer the better high-temperature performances in terms of thermo-mechanical properties and low density, while b) **oxide CMCs** are increasingly gaining attention due to their considerably lower cost and intrinsic oxidation resistance, albeit their maximum use temperature is generally around 1000 – 1200°C. The CEM-WAVE business use case well reflects these differences and has been conceived in order to make the best use for both classes of CMCs in the RTF in steel making.

AMIII USE CASE

CEM-WAVE methodology applied to the selected Use Case: *Figure 1.4* shows a radiant tube furnace with actual conditions and future conditions, (*i.e.*, in a hypothetical radiant tube furnace fuelled by a mixture of hydrogen/biomethane (H_2/CH_4) – these conditions will be studied in Task 1.1; in the figure a negative scenario is hypothesised). Since the conditions within the furnaces will be clearly variable, the radiant tube will be customised with different materials: several degrees in terms of properties and costs will be analysed to optimise the final costs of these components, by placing, in every section, the most convenient material possible. Indeed, a certain degree of tolerance will allow the materials in every section to resist to unexpected conditions or fluctuations. A hypothetical plan of the CEM-WAVE radiant tube configuration of the future is depicted in the right part of *Figure 1.4*. Different H_2/CH_4 will be investigated in order to assess CMCs behaviour in such working conditions during the project thanks to the innovative research on the set-up materials (WP1) on the process for producing both oxide and non-oxide CMCs (WP3) and the characterisation on the CMCs properties (WP4).

In order to make such a tube, CEM-WAVE includes innovative studies on joining of materials, especially between different types of CMCs (Ox-Ox – SiC-based CMC), tailoring both joining materials and processes for specific CMCs (oxide or non-oxide) and precise applications. CEM-WAVE joining activity will look at bonding techniques based on pressure-less, high-temperature, tailored glass-ceramic sealant, brazing by "in situ" high T composite joining materials suited for the bonding of non-oxide and oxide/oxide CMCs and ceramics (WP6). The use of CMC will only be sustainable if the process to produce such materials will be optimised to drastically reduce the costs of the CMC production following a twofold approach: integration of fully tunable solid-state sources and their automation, opening further new possibilities to the MW-CVI process (WPs 2 and 3). Thanks to these technologies it will be possible to exploit the material itself as a sensor during the MW-CVI process, thus allowing the obtainment of the desired properties and duration of the MW-CVI process (WP2). Building on the CMC manufacturing process monitoring strategies developed in WP3, the process analytical technology software platform will be tailored to additionally perform structural monitoring on the walls of the ceramic composite tubes samples to detect material structural and composition modifications and thus alerting the user for on-time maintenance of the tubes (WP7). To support the product design and optimisation, CEM-WAVE will take advantage of micro/macro-scale modelling assisted by one of the most powerful recent key technologies tools, *i.e.*, Artificial Intelligence (WP5). In more detail, it is envisaged to propose a consistent multiscale and multi-physics modelling frame making use of AI techniques for (i) the preparation of adequate computational domains from µCT scans of fibrous preforms and (ii) the acceleration of the balance equations solvers [5], and (iii) the optimization of processing parameters to obtain a pore-free infiltrated matrix. All these approaches will bring to a small-scale version of tube that will be tested in laboratory where (after ageing tests) it will experience conditions that will allow us to determine its feasibility in future plants powered by renewable energy sources; the same component will also be tested in current furnaces at AMIII where it will be placed on a specific support, to be also tested in current conditions. The latter is a fundamental part of the project exploitation strategy, which aims at a closer exploitation of the project results in current furnaces where our material can be introduced, bringing immediate advantages in terms of heat transfer efficiency and product lifetime (CMCs are expected to be better performant than currently used super-alloys on both aspects).The RTD work and the industrial validation will be supported by LCA, LCC, RA and TA (WP8), that will allow to keep the research and validation work on track with the fundamental requirements in terms of material requested properties, environmental impact, costs, energy efficiency and possible risks. Parallel monitoring of other possible commercial exploitation venues will be also part of our strategy methodology, that will be based on the constant analysis and monitoring of the results at every step and in every part of the project in order to identify any possible exploitable way for our materials, including other sectors.

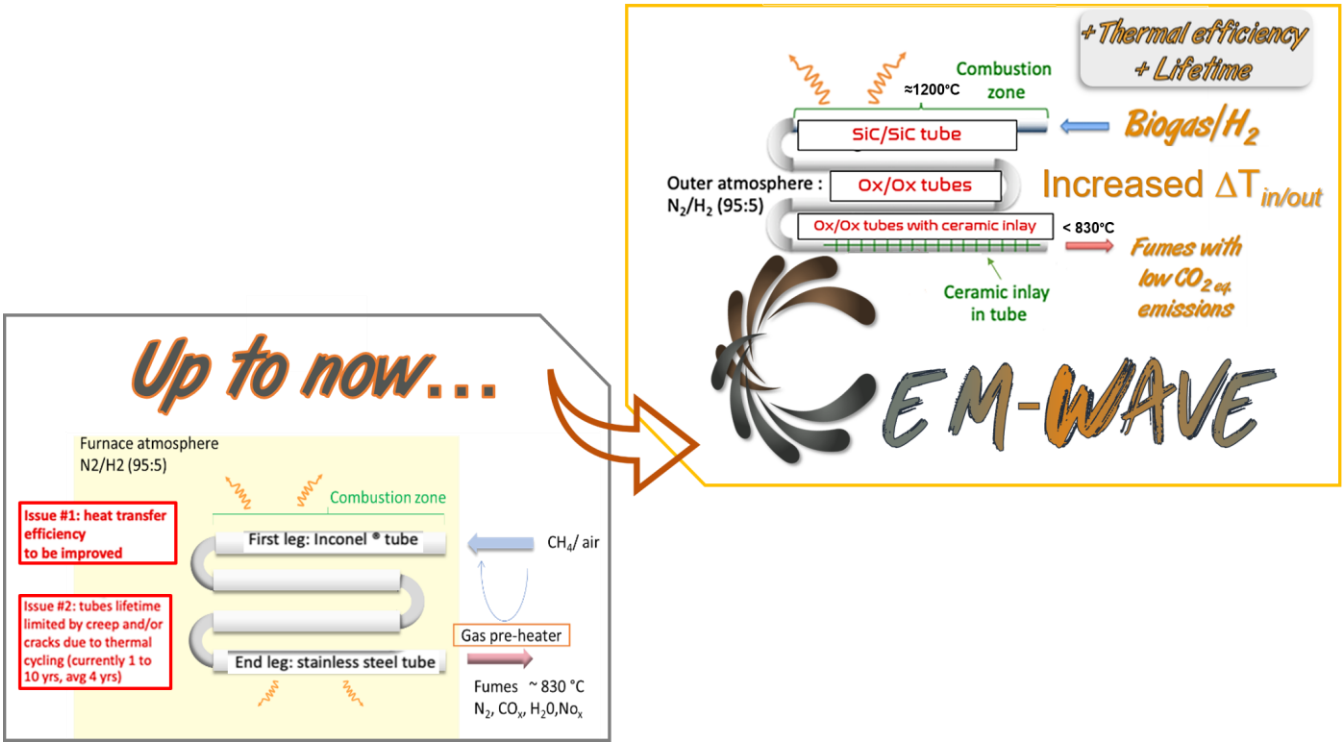


Figure 1.4 *- Comparison between the current radiant tube section adopted by CEM-WAVE use case (AMIII) and the CEM-WAVE solution for future RTFs.*

1.3.3 Gender dimension

The CEM-WAVE consortium takes a comprehensive approach towards gender aspects, which includes gender mainstreaming and positive actions. In accordance with the Articles 2 & 3 of the Treaty of Amsterdam (1997) and other EU policy directives (COM (96) 67 final) and reports (EUR 2002), the consortium is committed to follow the principles of gender mainstreaming. Gendered analysis stimulates gender-responsible science and technology, thereby enhancing the quality of life for both women and men

worldwide. The applicants are conscious of the strategic importance to identify and avoid gender bias in science and technology. To this purpose, gender aspects are tackled from 2 points of view:

1. In terms of project scientific and technical design and execution (methodologies, interpretation of data, dissemination of results). The applicants have considered that, due to the objectives of the project, the designing of methodologies and interpretation of data produced during the project should not be affected by gender considerations.
2. In terms of project management (researchers' recruitment, labour and family conciliation). The consortium will operate an equal opportunities policy for recruitment. The programme of work will not discriminate against persons of any gender and will aim to eliminate any relevant barriers and to encourage the participation of women in the programme of work and its outcomes. All efforts will be made to encourage participation by women and it will be instigated for all a work scheduling that allows for reconciliation between work and family life. Opportunities have arisen in relation with the project women scientists. The CO will be in charge of ensuring that gender issues are appropriately considered, for this reason a dedicated point will be included in the agenda of the consortium meetings.

This approach is coherent with EC policy "Strategic engagement for gender equality 2016-2019", published in December 2015. It is noteworthy that there is a strong presence of female researchers, technologists, and industrials in the consortium, a number of which are occupying key positions in the project. Several partners are represented by women participating in Project SC, and in WP and task leadership roles. A recruitment policy for future researchers and developers for the project is in place and is designed to ensure a reasonable level of balance between the sexes.

1.4 Ambition

There is an increasing need in the twenty-first century to develop stronger, tougher and more chemically resistant structural materials in order to meet the challenges arising from the rapid increase in demanding requirements across a wide and diverse range of applications, from construction to transportation, health to energy [6]. Ceramic Matrix Composites (CMCs) are of great interest as thermo-structural materials [7] and were developed initially for military and aerospace applications. Some of the most important market niches include: thermally loaded components in space applications such as propulsion systems or thermal protection systems [8–10], refractory materials for furnaces [11,12] and energy reactors [13,14], and ceramic brake systems [15]. Emerging sectors comprise: antiballistic plates [16], gas burners, high pressure heat exchangers [17], fuel and solar cells [18]. Currently they are being introduced into many other new fields and their range of applications will grow if their manufacturing costs can be significantly reduced [19]. The global CMCs market was valued at US $ 4,857.6 million in 2018 and is projected to reach US $ 11,516.1 million by 2026, growing at a CAGR of 11.4% from 2019 to 2026, helped in particular by the demand from the aerospace sector. They can be segmented into non-oxide and oxide CMCs [20] with the former offering higher temperature resistance (typically up to 1400 – 1600°C) and the latter intrinsically oxidation resistance, thus finding applications where this is a critical factor, albeit their maximum use temperature is generally around 1000 – 1200°C. The aforementioned applications will significantly fuel growth of the CMCs market. However, their cost has been a concern for many industries over the last decade. The prices of ceramic matrix composites are controlled, to a large extent, by the costs of ceramic fibers and energy intensive processing routes. Nonetheless, substantial opportunities will be presented by the energy and power sector, as it is a rapidly growing application segment [21].

1.4.1 Advances beyond the state of the art

1.4.1.1 Processing routes of non-oxide CMCs

To date, the greatest amount of developmental work has been focused on the non-oxide CMCs and, in particular, carbon (C) and silicon carbide (SiC) fiber reinforced SiC-based materials (C_f/SiC and SiC_f/SiC composites respectively), since non-oxide CMCs have attractive high temperature properties, including creep resistance, microstructural stability, excellent mechanical and thermal properties. CMCs reinforced with long fibers are commonly fabricated by infiltration methods [1], in which the ceramic matrix is formed from a fluid infiltrating into the fiber structure. These techniques differ from each other in the types of fluids and of the processes for converting the fluid into a ceramic:

- *Chemical vapor infiltration* (CVI): infiltration using a preceramic gaseous precursor (a vapor), which yields a ceramic as a result of chemical decomposition at elevated temperature;
- *Polymer infiltration and pyrolysis* (PIP): infiltration with a low viscosity preceramic organo-metallic polymer followed by pyrolysis when the polymer converts into a ceramic (multiple infiltration-pyrolysis cycles are required to reach the desired density);
- *Reactive melt infiltration* (RMI): infiltration with a liquid metal, which converts into a ceramic when reacting with a surrounding substance (e.g. *liquid silicon infiltration*, LSI, a type of RMI utilizing molten silicon).

Generally, all infiltration techniques employ the following fabrication stages:

- Preform preparation: At this stage the fiber reinforcing phase is laid up and molded to form a preform of the required shape, it can be made of stacked 2D layers of fabrics, needled '2.5D' layers, or woven 3D fabric layers;
- Interphase deposition: Where required, a coating (called interphase) may be deposited over the fiber surfaces before or after preform fabrication;
- Infiltration: The reinforcing fiber preform is infiltrated with a preceramic fluid as outlined above;

– Thermal processing: The preceramic fluid incorporated into the fiber reinforcing structure converts into a ceramic, filling the space between the fibers.

In CMCs, the fiber-matrix interfacial domain is a decisive component controlling the (thermo-)mechanical performance of these materials [22,23]. The favored damage-tolerant fracture behavior and high fracture toughness are achieved by a weak bonding between the fiber and the matrix [24], while this weak interface needs to be strong enough to have sufficient load sharing between fiber and matrix. This load sharing is of importance to achieve a high strength of the CMC materials [25]. In non-oxide CMCs the interfacial domain is adjusted by applying coatings, so-called "interphases", on the fiber [26]. Although many materials have been tested as fiber coatings [27], long-term stable damage-tolerant fracture behavior in non-oxide CMCs has only been achieved by using BN (boron nitride) or PyC (pyrolytic carbon) as interphases. A drawback of both materials is their limited oxidation and corrosion resistance; if the interphase corrodes via oxidation, the performance of the CMC degrades. To overcome this issue, additional layer(s) are applied on top of the BN or PyC layers [28]. These additional layer(s) (e.g. SiC, Si_3N_4, etc.), protect the underlying BN or PyC layer during further CMC processing and, to a limited degree, also during use of the CMC [29]. Moreover, these layers provide an additional possibility to adjust the bonding of the interface coating system to the matrix [26].

Applying the coatings on every single filament of a tow consisting of several hundreds of filaments with high quality is a great challenge. The coatings need to be homogeneous with a high degree of coverage. At the same time, bridging between the filaments needs to be avoided in order to be able to further process the coated tows by techniques such as filament winding.

The fiber coatings can be applied by various techniques, among which Chemical Vapor Deposition (CVD) and Wet-chemical deposition are the most common. Advantages and disadvantages of both CVD and wet-chemical coating technologies are compared in Table 1.3.

Table 1.3 *- Comparison of advantages and disadvantages of coating ceramic fibers by CVD and wet-chemical coatings technology.*

	CVD route	Wet-chemical route
Advantages	• Doping of coatings relatively simple • No additional pyrolysis steps necessary • Relatively low thermal load on fiber • No synthesis of precursors needed	• High variety of material systems can be deposited • Low investment costs • Coating set-up simple to modify • High degree of flexibility for process design • High process speed and through-put possible • Low maintenance costs • Technology easy to upscale and to implement in industrial production
Disadvantages	• Limitations in material systems that can be deposited • Raw materials often expensive • High investment costs for the technology • Low process speeds, low through-put • Complex infrastructure needed • Use of corrosive gases • Shrinkage, crystallization or phase transformations of the coatings during CMC processing or use possible	• Modification of coating materials needs modification of synthesis of precursors • Higher thermal load on fibers during pyrolysis of the coatings • Limited coating thickness

The fiber-matrix bonding must be sufficiently weak to allow debonding and consequently the relaxation of the stress field at the crack tip and the bridging of the matrix cracks by the fibers [30]. There are two methods of achieving this; for dense matrix CMCs (e.g. SiC-based composites) a low cohesion interphase coating (e.g. pyrolytic carbon or boron nitride) is applied to the fibres before the matrix is infiltrated. CMCs of this type are called Weak Interface Composites (WICs) and toughening mechanisms such as crack deflection at the fiber/matrix interface, crack-bridging by fibers and fiber pullout at the crack plane are operative. An alternative approach (e.g. for oxide-based composites) is to ensure that the matrix is weak. Such Weak Matrix Composites (WMCs) are characterized by the matrix being porous; the reduced matrix stiffness and strength allows damage-tolerant behaviour, even in the case of a strong matrix-fiber interface. Whilst both approaches work, for WICs the interface coatings may oxidize at high temperatures in air leading to brittle failure, whilst WMCs typically display a low overall strength or, if used at excessively high temperatures, the matrix can further densify leading to brittle behaviour. Concerning common non-oxide CMCs processing routes, Chemical Vapor Infiltration (CVI) technology is one of the most attractive industrially available methods [31,32] to produce fiber-reinforced CMCs. The main advantages are a) the possibility to manufacture complex net or near-net shaped components at relatively low temperatures for ceramic processing (typically $\leq$ 1000˚C), avoiding potential damage of the ceramic fiber textile structures typically used as the preform and b) the ability to control and modify the microstructure of the matrix. CVI methods can be divided into several categories, depending on: 1) the heating method, 2) the pressure regime, 3) whether the process is isothermal-isobaric (I-CVI) or there is thermal/pressure gradient across the perform (TG-CVI, F-CVI/P-CVI). Among them, I-CVI is the most commonly studied and used version of CVI [33], although it is recognized as being very slow (products can take up to 3 months to make) and hence expensive. Polymer Infiltration and Pyrolysis (PIP) is a method where a low-viscosity,

preceramic polymer is infiltrated into a C or SiC fibre preform, followed by pyrolysis in the absence of oxygen during which the polymer decomposes and converts into a polymer-derived ceramic matrix. Widely used high-performance silicon-derived polymer precursors available on the market are polysilazane (Ceraset VL-20, Kion Corp), which can be converted into SiCO, SiCN or Si_3N_4 with ceramic yields of up to 90%. The use of preceramic prepregs (C or SiC fiber textiles impregnated with a preceramic polymer) is a common method to achieve a stable geometry, however densification requires a multistep (sometimes up to 12 steps) iteration of infiltration and pyrolysis.
Liquid Silicon Infiltration (LSI) is used to fabricate SiC-based CMCs. The process involves infiltration of a carbon microporous preform with molten silicon at a temperature exceeding its melting point of 1414°C. The liquid silicon wets the open pores, driven by capillary forces, on the surface of the carbon preforms. Ensuring the correct pore dimensions is important to achieve complete infiltration. If the pores are too small the infiltration passages become choked off, resulting in premature cessation of the infiltration process [34]. Although having excessively large pores aids infiltration, it can result in incomplete chemical interaction and the formation of a structure with high residual free silicon and unreacted carbon. In contrast to the composites fabricated by PIP or CVI, ceramic matrices formed by LSI are fully dense (typically >99% of the theoretical density).
Some of the main advantages/disadvantages related with the various infiltration methods are summarized below (Table 1.4):

Table 1.4 *- Advantages/disadvantages of non-oxide CMC infiltration technologies.*

	CVI	PIP	LSI
Advantages	• High purity matrices • Low temperature (prevents fiber damage) • Excellent mechanical properties • Often smooth process surfaces	• Low temperature (prevents fiber damage) • Good control of microstructure and composition • No free silicon in the matrix	• Low cost • Short production time • Low residual porosity
Disadvantages	• Very slow process (~several weeks) • High residual porosity (10-15%) due to bridging • High capital and production costs	• Multiple infiltration-pyrolysis cycles • High production cost • High residual porosity • Low maximum linear elastic levels	• High temperature and highly corrosive metal melt (may damage fibers) • Residual free silicon • Low mechanical properties • Rough surfaces need machining with diamond tools

1.4.1.2 Processing routes of oxide CMCs

The challenge for oxide/oxide composites is to provide an alternative to SiC_f/SiC composites with the aim of achieving a lower cost material, both by decreasing the cost of the constituent materials and manufacturing, and also by creating composites with improved thermal stability in air at elevated temperatures. Oxide/oxide composites typically consist of a porous alumina (Al_2O_3) or alumino-silicate (Al_2O_3-SiO_2) matrix reinforced by alumina fibers, although a small amount of work has also been undertaken into mullite fiber reinforced composites. The latter reveal that mullite-based fiber has potential to offer slightly higher temperature capabilities. Whilst both fibers tend to start losing strength by 1200°C, alumina fibers can lose 90% of their strength by 1400°C whilst for mullite fibers it was only 50% [35]. There is also evidence that some mullite fibers can retain useful strength to 1500°C [36]; further work on developing enhanced mullite fibers is still needed though. Meanwhile, the majority of research has focused on alumina fibers and the Nextel fibers produced by 3M in particular, though other alumina fibers are starting to be produced and investigated for the production of ox-ox composites [37]. There are four Nextel fibers, 720, 610, 440 and 312 in order of decreasing temperature capability. The structural grade fibers, 610 and 720, have crystal structures based on α-Al_2O_3 and do not contain any glassy phases, allowing the fibers to retain their strength at higher temperatures. Nextel 610 is a fine-grained single-phase composition of α-Al_2O_3 with the highest strength and modulus; however, since it is essentially a single phase, the strength decreases rapidly at higher temperatures due to grain growth. Its maximum working temperature is limited to 1100°C for long-term applications. Nextel 720 is α-Al_2O_3 with SiO_2 added to form a mixture of alumina and mullite. It has better strength retention at temperature and increased creep resistance due to reduced grain boundary sliding as a result of the mullite pinning the grain boundaries and hence reducing grain growth during thermal aging. Nextel 720 has proven thermal stability up to 1200°C.
Nextel 440 and 312 are cheaper and offer even lower temperature capability. Made from varying percentages of Al_2O_3, SiO_2 and B_2O_3 (boria), they contain both crystalline and glassy phases due to the presence of the boria. The glassy phase helps the fiber retain strength after exposure to high temperature because it slows the growth of the crystalline phases that weaken the fiber. However, when the fiber is stressed at high temperature, the glassy phase weakens the fiber due to viscous flow much like a glass fiber. A major driver as ox-ox composites move towards production is achieving the cost reductions required to ensure that the solutions are market-competitive with titanium and other high temperature alloys. This has led to investigations into the use of both lower grade fibers and lower cost fabric design. Concerning the latter, this means using those based on higher-denier fabrics. Results suggest that a transition to heavier denier yarns and architectures has a relatively minor effect on mechanical properties or thermal stability whilst offering lower cost composites [38]. Development of dense oxide CMCs is a challenge since the oxide ceramic fibers can only be thermally loaded to about 1200°C [36,39,40] and the process for the fabrication of the dense

matrix has to fulfill the requirements for industrial implementation. Because of these challenges, the development of fiber coatings for oxide CMCs lags behind the developments for non-oxide CMCs. Rather, porous fiber coatings based on ZrO_2, Y_2O_3, ZnO, Al_2O_3 or $Y_3Al_5O_{12}$ have been investigated to provide a weak interfacial domain in oxide CMCs [41-43]. It has been shown that the favored debonding between fiber and matrix occurs during mechanical load, but the friction between fiber and matrix was too high and therefore the fiber pull-out effect was strongly reduced. Moreover, at high temperatures coarsening of both the coating microstructure and porosity occurred and thus the interface system cannot be considered to be thermally stable for long-term, high temperature applications [41,44]. Another concept is the use of the so-called "fugitive coating". In this case, the fibers are coated with a thin layer of carbon, which is oxidized, and therefore evaporated, during further processing of the CMC [45,46]. In this case, a thin gap is formed between fiber and matrix. Load sharing between fiber and matrix depends on the thickness of the carbon layer and on the surface roughness of the ceramic fiber as well as on the variation of the fiber diameter along the fiber axis. Load sharing between fiber and matrix occurs due to contact points between fiber and matrix. It has been assumed that the ceramic fibers are damaged at the contact points with a negative impact on CMC properties and life time [40]. Oxides with a layered crystal structure like phyllosilicates, beta-aluminates and magneto-plumbites have also been investigated as interphase materials for oxide CMCs [47-50]. Unfortunately, it was not possible to adjust the favored texture of the coating materials. Due to high processing temperatures, degradation of the ceramic fibers was observed. Interesting results have been achieved with fiber coatings based on rare earth phosphates, e.g. Me_3PO_4 [51-54]. Studies with different types of these coatings indicate that energy-dissipating mechanisms like debonding, fiber pull-out and crack deviation can occur under mechanical load [55-59]. It should be noted, however, that the latter research was based on CMC fabrication routes that are not viable for industrial implementation.
The most common processing route used to create oxide matrix CMCs is via the impregnation of oxide ceramic powder slurries into oxide fiber preforms. This requires significant control of the rheology of the suspension in order to achieve the correct impregnation characteristics [60]. After drying the slurry, the resulting powder can be sintered using low cost, conventional, air atmosphere furnaces. However, since even the best alumina fibers can only withstand temperatures to around 1200°C and alumina generally requires about 1400˚C to sinter, either very fine alumina powders, sintering aids (e.g. mixtures of CuO & TiO_2 [61], though these can affect other properties such as strength) or the use of sol-gel processes are typically required. One example of the latter involves the infiltration of liquid alkoxide precursors into the fiber preforms to form a polymeric oxide gel network *in situ* via hydrolysis and polycondensation reactions [30]. The primary issue with sol gel routes is that the first step of gel formation and its subsequent pyrolysis are characterized by a low mass yield of the ceramic matrix, leading to a high level of shrinkage (80% is not uncommon). A major issue with the standard route to the fabrication of ox-ox composites is therefore the difficulty of achieving sufficient control of the density of the matrix due to the need to use higher temperatures than the fibers can withstand. Alternatives such as using sol-gel based processes, ultra-fine particles or sintering aids all have their limitations, not least of which is the fact that they are slower and/or more expensive and/or affect the final properties of the composites produced. An alternative that has not been significantly investigated before is to use chemical vapor infiltration (CVI) to deposit an alumina or alumino-silicate matrix within oxide fiber preform of similar composition. The major advantage of using CVI is that, as described earlier for silicon carbide, the processing temperature is low, typically < 1000˚C and therefore the fibers are not damaged. In addition, it is possible to obtain good control of the microstructure developed. Whilst very little work has been focused on using CVI to prepare oxide-oxide CMCs, significant work has involved the deposition of alumina-based coatings using the closely related technique of chemical vapor deposition (CVD). Both CVI and CVD can usually use the same chemistry, allowing approaches originally developed for one technique to be adopted by the other. In the case of the deposition of alumina or alumino-silicates, it is quite common to use the chlorides [62] although research has also been undertaken using chemicals such as aluminum tri-sec-butoxide (ATSB) or aluminum tri-isopropoxide [63-64].

1.4.1.3 MW-CVI of non-oxide & oxide CMCs

In microwave-assisted CVI techniques, sample heating is obtained by means of high-power microwave irradiation, which releases energy in the sample through dielectric absorption. This approach enables volumetric heating of the sample, with heating rates essentially limited only by the available microwave power. Moreover, the combination of volumetric heating with thermal losses from the sample surface towards the colder walls of the reactor results in the development of an inverse temperature profile in the sample, i.e. the center becomes hotter than the surface. The temperature difference between the center and the surface of the component can be quite high, considering the high efficiency of the thermal irradiation governed by a T^4 dependence. Controlling these temperature gradients is key to controlling the MW-CVI process. Based on the 'inside-out' nature of the deposition, premature pore closure due to crusting can be completely avoided, allowing densification to occur an order of magnitude or faster than with isothermal CVI (I-CVI). Experimental work has yielded this inside-out densification pattern in a number of different ceramic systems, including SiC, Si_3N_4, Al_2O_3 and ZrO_2. By means of the MW-CVI process, high purity and high density SiC matrices have been obtained and preforms of very complex geometry can be successfully infiltrated at operating temperatures between 900°C and 1000°C. Since the pressure required by the process is low (10-100 kPa), the risk of damaging the fibers or of seeing reactions between them and the matrix is limited. Previous work carried out by Binner et al. [65] based on SiC_f/SiC components (50 mm diameter, 10 mm thick) has shown that, by using microwaves to enhance the CVI process, fabrication times can be reduced from several hundred hours to around 100 hours [70] and down to only ~25 hours when RF-CVI is used to produce C_f-ultra high temperature ceramic matrix composites (UHTCMCs) [68]. Similar results were obtained with the MW-CVI pilot plant developed in the European project HELM for the production of SiC-based CMCs (HELM: High-frequency Electro-Magnetic technologies for advanced processing of ceramic matrix composites and graphite expansion) [69]. Unlike

existing lab-scale MW-CVI equipment, the design of the HELM MW-CVI pilot plant in Pisa has been developed with the specific idea of allowing further industrial scale-up. This approach has led to some complex engineering problems, which have been tackled in an original way. In particular, the inner chamber and all the components exposed to high temperature and extreme chemical environment have been made out of conductive graphite. Moreover, this pilot plant can use a combination of conventional and microwave heating in order to tailor a suitable temperature profile in the preform to be infiltrated.
The objective was the production of SiC-based CMCs achieving a reduction in processing time, as well as the identification of strengths and weaknesses of the MW-CVI process. Among the latter, the formation of hot spots in the sample and the appearance of plasma regions were regarded as the most detrimental for the efficiency of the chemical process and the achievement of a continuous process. Traditional approaches to prevent the occurrence of hot spots consist of the rotation of the object to be heated or the inclusion of mode stirrers, i.e. mobile metallic surfaces that dynamically stir the electromagnetic field in the reactor. Both of these solutions, however, have severe limitations: they involve moving metal parts that, may get stuck as a consequence of dust accumulation in the environment of the reactor and they provide a limited capability to control the actual field intensity distribution. Many of these challenges have been successfully overcome; SiC_f/SiC preforms with dimensions of 100 mm diameter by 10 mm thickness, twice those of typical lab-scale samples, have been infiltrated successfully, confirming the advantages expected from a MW heating. Specifically, it was possible to work continuously with reproducible and optimal operating conditions in terms of transmitted/reflected power via the careful design of the inner MW cavity so that it acts as an overmoded resonator at the operating frequencies. Infiltration tests showed an improved reaction efficiency, even allowing for the increase of sample dimensions, without plasma formation and a good compromise in terms of operating pressure. Deposition rate data resulted in a potential reduction of total infiltration time with a predominant infiltration of SiC on central laminae of the sample, confirming the inside-out densification pattern and scaling-up of this technology. Research activities on MW-CVI for non-oxide CMCs will focus both on the electromagnetic aspects of the microwave heating and on the chemical aspects of the CVI process. Starting from the state-of-the-art results obtained in the framework of the HELM project by the UNIPI and IPCF-CNR teams in Pisa [70] the pilot-scale MW-CVI reactor, operated so far with magnetron sources, will be innovated with the introduction of fully controllable solid-state sources (see section 1.4.1.4 below). The aim of this upgrade will be twofold: first, it will enable a complete reproducibility and control of the heating conditions, minimising the risk of unpredictable hot spots formation and the activation of plasma phenomena. Then, it will allow the implementation of sophisticated heating procedures based on the 'probe and pump' technique, in which a low power sweep over the frequency range of interest (the 'probe') will determine the most convenient frequencies at which the high-power 'pump' must be done. Moreover, the possibility to obtain a more homogeneous heating of the sample by means of a sequential 'pump' irradiation at different frequencies will be investigated for the first time. The chemical aspects of the MW-CVI process will be investigated in detail with the aim to optimize the reaction parameters as operating pressure, infiltration temperature, reagents ratio and flows. A specific study will be devoted to the effect of a BN/PyC coating of the non-oxide CMC fibers, which guarantees the desired pseudo-ductile behavior, increasing at the same time the electrical conductivity of the sample. In principle, a thicker BN/PyC coating is desirable in order to improve the mechanical properties of the infiltrated CMC. However, an excessive amount of carbon in the CMC can increase the conductivity at a level in which the electrical conductivity is so high to shield the penetration of the electromagnetic fields in the sample, in analogy to the behavior of the conducting materials. The optimal trade-off between mechanical and electrical properties will be determined measuring the dielectric properties of the BN/PyC coated CMC fibers at the infiltration temperatures. These measurements will be based on a rigorous technique, recently demonstrated by the IPCF-CNR team in Pisa [71] which is capable to accurately determine the high temperature dielectric properties of samples, also in their conducting state. On the basis of the measured dielectric properties, an extensive analysis on the maximum allowed dimensions of the preforms to be infiltrated will be conducted, with the help of a rigorous numerical modeling based on the Comsol Multiphysics software. Concerning microwave-enhanced CVI of oxide-oxide composites, just two previous research publications have been found [72-74], undertaken by the team led by Prof Lynn Johnson at Northwestern University in the USA. In both cases a single mode microwave cavity was used, which very much limited the size of the samples that could be processed. Nevertheless, matrix growth was observed to occur from the inside towards the outside, i.e. an inverse temperature profile was created as observed in other research on both microwave- and RF-enhanced CVI undertaken by Binner's team, which is now located at the University of Birmingham in the UK [66,68,74-76]. One issue that was faced by the Northwestern team, which will also be faced by the Birmingham team in the present work, is that alumina fiber preforms do not readily absorb microwave energy. In the first of their two publications, the Northwestern team doped the preforms with a fugitive carbonaceous additive carbon to facilitate initial heating by microwaves. This consisted of impregnating the preform with a sucrose solution, freeze-drying it and then pyrolysing at 1000˚C under nitrogen to form carbon. In the second paper a susceptor was used in the form of a sleeve of strontium-doped lanthanum chromite. Whilst both approaches worked, the concept of using a fugitive additive will be more applicable in the proposed research program since susceptors are known to degrade rapidly with use meaning that they need replacing regularly, which decreases the economics of the process [77].
In the proposed research, the investigation will utilize a state-of-the-art multimode microwave cavity. This will allow large samples to be made and a greater range of processing variables to be investigated. The work will focus on the use of a range of different alumina and alumino-silicate ceramic fibres. Whilst the bulk of the work will be based on the use of the Nextel fiber range, work will also be undertaken using newer fibers that are available from Germany and elsewhere. Both 2D and 3D woven structures will be utilized and facilities exist at the University of Birmingham to measure the gas permeability of the preforms, very useful when developing a process route based on the principle of gas flow through the preform. Knowing permeability as a function of gas flow

rates can greatly assist with modelling work (see section 1.4.1.6 below). The work will focus on using aluminum and silicon chlorides, which have been successfully demonstrated to yield alumina and alumino-silicates as the matrix phase [72-73], whilst a range of different fugitive additives will be explored with a view to enhancing the coupling of the preforms to the microwaves. One major advantage for this approach for the production of oxide-oxide CMCs by microwave CVI is that the final product does not want to be fully dense, rather densities of 85 – 90% of theoretical are desired (see the discussion of Weak Interface Composites and WMC in section 1.4.1.1). Since the rate of deposition during any CVI process decreases exponentially with time, the lack of a need to obtain full density will significantly reduce the time required to produce each sample, greatly assisting in the ability to make progress. Having said this, as indicated previously, work will also be undertaken on the use of fiber coatings that can both absorb microwaves and also function as a weak interface with a dense matrix, allowing Weak Interface oxide-oxide Composites to be manufactured. A comparison of the different properties achieved in the composites will be most useful in identifying not only the best manufacturing routes but also the best properties that can be achieved together with an understanding of how to tune the properties for different applications.

1.4.1.4 Improvement of the MW-CVI process through solid-state sources

Recent advances in solid-state microwave sources [78] led us to analyze the use of this technology for the MW-CVI process. The reasons behind this choice lie in the rapid and precise control of the signal frequency, as well as a more stable and reproducible distribution of the EM field [79], with the important possibility, from an industrial point of view, to automate the whole infiltration process, which in turn opens further new possibilities to this technology [80]. The ability to generate and amplify MW signals using solid-state semiconductor devices is not new since this technology has been the cornerstone of modern wireless communications for many years. Solid-state MW sources have enormous potential beyond data transmission applications: they are increasingly being used also for the purpose of heat and energy generation, enabling greater efficiency and control than currently possible with conventional magnetrons [81]. One of the major deficiencies of magnetron tube-based MW sources is their lack of fine control over frequency, power and the phase of the emitted wave, which prevent the implementation of closed-loop control aimed at the minimization of the power reflected by the MW reactor. On the contrary, with solid-state MW energy sources, forward and reflected power levels can be easily adapted with closed-loop controls thanks to the inherent, precise control over the solid-state source emissions in terms of frequency, power and phase [82,83]. Solid-state MW transistors can also provide 10 times longer lifespans than magnetrons, ensuring significantly higher reliability and a reduced cost per working hour. In order to overcome the drawbacks of magnetron-based MW reactors, one of the innovations proposed in CEM-WAVE is the replacement of the high-power magnetrons with sources based on solid-state amplifiers employing LDMOS transistors. Currently, the solid-state power generators covering the two major ISM bands, 902-928 MHz (L-Band) and 2400-2500 MHz (S-band), are becoming more and more appealing from an economics perspective, thanks to the recent development of high-power transistors. These sources, now available also at the kW power levels necessary for MW-CVI, are frequency agile and capable of operating either at a single fixed frequency or in a frequency sweep mode that can cover the entire ISM band. The very short response time of the solid-state microwave generators and the possibility to pulse their output power can open unexplored possibilities in which the temperatures of the product can be rapidly controlled, avoiding hot spots and limiting the occurrence of plasma phenomena, thus reducing the risk of damage occurring due to overheating. Moreover, the solid-state generators can be connected to the reaction chamber (MW cavity) via standard coaxial cables, minimizing the space requirements.

The heating of the sample with solid-state sources is expected to improve the uniformity of the sample temperature, with the possibility to dynamically adjust the EM field in the reactor without involving any moving metal parts. In fact, the implementation of an automatic and extremely fast (millisecond range) frequency control, necessary to optimize the sample irradiation according to its dielectric properties – variable with the temperature and density - will allow a precise and stable heating power control during the process [84]. In the framework of the project CEM-WAVE, the benefits expected from the use of coherent sources will be maximized by means of a rigorous and detailed numerical modelling of MW-CVI reactors loaded with the samples of interest and based on the use of the software Comsol Multiphysics. In fact, thanks to the possibility to precisely select the operating frequency and so the mode granting the highest uniformity of the EM field, we will also be able to choose the best location of the sample inside the MW cavity and to predict the effects due to different sample dimension and geometry, as demonstrated in the framework of the project HELM In addition, the fast tunability typical of the solid-state sources will enable a more uniform heating of the sample by employing different resonance modes, excited in a sequential way during the infiltration process. Accordingly, solid-state sources are the ideal choice for a well-controlled and reproducible reaction, which translates to higher energy efficiency and better product quality. All these reasons, together with the steadily declining specific cost, explain the rapidly growing popularity of these sources in the community involved in industrial applications of microwave field-assisted heating.

1.4.1.5 Development of advanced process monitoring tools

In MW-assisted processes, one of the most critical reaction parameters to control is the temperature of the sample, particularly for samples larger than the employed wavelength. In these cases, the sample heating is prone to the formation of hot spots, mainly due to the standing wave pattern of the electric field in the reaction chamber. As a consequence, the monitoring of the spatial distribution of the sample temperature is a crucial step in the definition of an efficient MW-CVI process. In the most advanced approaches, the temperature of the sample is monitored by means of a thermal camera, which can give information on a relatively large part of the sample surface [70]. On the basis of the camera reading, the microwave power can be adjusted in order to reach the desired infiltration temperature. The use of a thermal camera in the monitoring of MW-CVI processes presents specific advantages with respect to the simple pyrometer. Indeed, the large view provided by the former is essential in limiting the

occurrence of hot spots. In the CEM-WAVE, thermal cameras will be employed to monitor the evolution of the surface temperature of the sample during the MW-CVI process. The images captured by the cameras will be acquired and processed in real-time by a proprietary software platform in order to obtain a 3D view of the sample surface temperature. The maximum temperature detected on the sample surface will be employed to correct automatically the MW power delivered by the solid-state sources, until the infiltration temperature will be reached. After the beginning of the infiltration process, the surface temperature monitoring will provide the necessary feedback to stabilize the sample temperature. A second innovation of the CEM-WAVE project, deriving from the use of thermal cameras and the related reconstruction of the surface temperature distribution, will concern the development of an algorithm of 'frequency jump', in which the high-power irradiation of the sample will be alternated on different frequencies, corresponding to modes with different field distribution. The use of such modes will help in obtaining a more homogeneous sample heating. Role of this algorithm will be to drive the power and frequency emission of the MW sources, in order to minimize the thermal gradients. The process monitoring strategies developed in the CEM-WAVE project will include the progress of the infiltration reaction. In conventional CVI processes, indications on the stage of the sample infiltration and thus on its density usually come from the previous experience. The use of reactor chambers acting as electromagnetic resonators opens the possibility of a direct investigation of the infiltration progress, based on the strict relationship between the resonance frequency of the electromagnetic modes and the dielectric properties of the sample that are, in turn, closely related to the density of the sample. In the CEM-WAVE project, the resonance frequency of selected modes will be continuously monitored in order to obtain, *in-operando*, a direct information on the dielectric properties of the sample and thus on its density. The information on the dielectric properties of the sample will serve to determine the stage of the process, as well as to identify possible anomalies (internal sample damages or hot spots). In the framework of CEM-WAVE, monitoring strategies will also be developed for the structural analysis of the walls of the CMC tube samples during the tests in highly demanding applications. The structural monitoring of the CMC tubes will focus on the detection of possible cracks and on the degradation of the tube material in tests representative of long and realistic applications. In particular, two combined strategies will be implemented. The first one, based on an IR camera, will monitor the temperature distribution on the surface of the tube, whose alteration during a prolonged use can indicate the presence of a local damage (crack) or a diffused component degradation (reduction in the tube density). A second approach will be based on the electrical resistance of the tube, determined along its length by means of graphite electrode placed at the beginning and the end of the tube, which is sensitive to both both the presence of extended cracks, which increase the electrical resistance, and to the density and thickness of the tube walls. The inputs of the two sensors will be correlated in order to minimize the risk of 'false positives'.

1.4.1.6 Micro/macro scale modeling approach of CVI processes

The CVI process and its variations [85] are based on the interactions between the transport of chemical species and their reaction, both inside and outside the porous medium constituted by the fibrous preform. The presence of thermal gradients strongly alters the reaction-to-diffusion ratio and can help a favorable situation development in which matrix infiltration starts from the inside and progresses towards the exterior of the porous medium. For instance, a criterion for a minimal thermal gradient allowing such a situation [86] has been derived with a simplified model considering homogenized materials properties evolving with the infiltration degree. This suggests that a consistent modeling approach to MW-CVI should consider several scales:

- The *reactor scale*, considering the whole chamber, the heating devices and the preform;
- A *macroscopic scale*, concerning the preform as a whole, considered as locally homogeneous;
- A *meso/microscopic scale*, related to the structural and textural details of the preform. The fiber diameter is of the order of about 10 micrometers, but larger scales related to the weave, stackings arrangement, etc., may also be present.

In the case of woven reinforcements, it is common practice to distinguish between the intra-bundle (microscopic) scale and the inter-bundle (mesoscopic) scale at which a Representative Volume Element (RVE) of the preform can be defined [87].

The principal phenomena that have to be accounted for at macro-scale are:

1. <u>Microwave heating</u>: this is achieved by solving electromagnetic equations [88,89];
2. <u>Heat balance</u> in the preform, including heating power in the volume, radiative and convective losses at the preform borders, and heat transfer connecting both [90];
3. <u>Gas transfer</u> in the infiltration chamber [91], then inside the porous medium, featuring multicomponent diffusion [92], viscous transport and Knudsen (rarefied) diffusion.
4. <u>Homogeneous and heterogeneous (gas-solid) reactions</u>, mostly at the internal surface of the hottest parts of the preform.
5. <u>Porous medium structural evolution</u>, which has an influence on heat transfer properties, internal gas flow, and the effective reaction rates; it results in less volume for the homogeneous reactions and more or less surface for the heterogeneous reactions. This structural evolution is in most cases much slower than the preceding ones. Accordingly, a usual assumption is to treat all heat and gas-related phenomena (transport and reactions) as being in a quasi-steady state; the porous medium properties may then be updated from knowledge of the latter.

Preform-scale (or macro-scale) models [93] have to be fed with several material properties related to internal geometry, heat and mass transfer, and, in the case of MW-CVI, complex electrical permittivity. This has to be achieved for several states of infiltration, which means that the porous medium evolution also has to be simulated [87]. Simple, analytical models were initially produced but have been progressively replaced by image-based models, first using a-priori image synthesis [94,95], then using direct X-ray CMT 3D images [96,97]. This requires some image analysis, in particular fiber and tow detection, the latter obtained, for example using local texture orientation analysis [98]. Efficient AI-based tools have been developed successfully for woven SiC-

fiber composite preforms [99], as seen in *Fig. 1.5*. The approach developed consists in a variational segmentation that combines µCT with a prior geometric model that is iteratively improved thanks to a specific optimization process. The fidelity of the models with respect to the input µCT is evaluated using a measure of similarity including both gray levels and local directions. This method allowed building realistic numerical models of woven fabrics that preserve the prescribed weaving pattern, and that are free of interpenetration, which makes them compatible with further numerical simulations. Using our approach, models of complex woven fabrics, but also of woven composites, could be consistently generated from µCT and can serve as reference models, e.g. to analyze in situ tests by providing digital twins. For the simulation of matrix infiltration by isothermal Chemical Vapour Infiltration (I-CVI), a two-scale approach has been developed, based on X-ray CMT images or on synthesized 3D images discretized on a regular voxel grid [96], as illustrated in *Fig. 1.6*. An intra-tow code allows for the precise simulation of the matrix deposition [100]; the deposit thickness variations are shown to depend on: (i) the diffusion-reaction competition, which implies a smaller deposit thickness in small pores, and (ii) the pore network connectivity. The results are interpreted as local laws for effective parameters (diffusivity, bulk reactivity) [101] in a large-scale solver based on Brownian motion [102], capable of simultaneously computing the matrix phase amount in the tows and the seal-coat matrix thickness outside the tows. This code accepts the local values of the volume fraction and of the orientation of the fibers. Both codes have been validated with respect to experimental data [103] in the case of carbon-carbon composite fabrication. So far, detailed image-based models have focused on I-CVI; more attention needs to be paid in the future on the heat transfer laws and the evolution of the effective complex permittivity as the preform is progressively filled by the matrix.

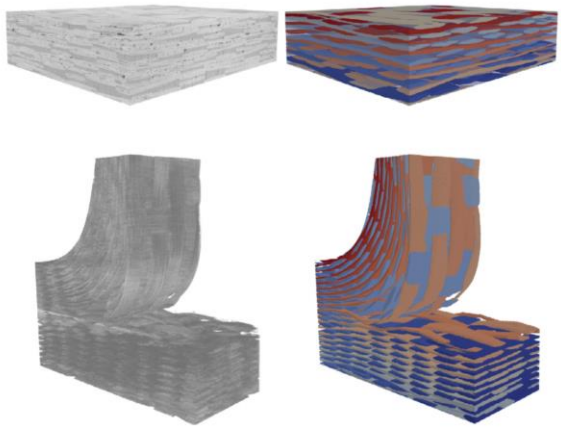


Figure 1.5 *- Illustration of the capabilities of our approach. µCT (left) and corresponding segmented model (right) for an infiltrated oxide/oxide composite (top) and for a fully woven CMC junction (bottom) [99].*

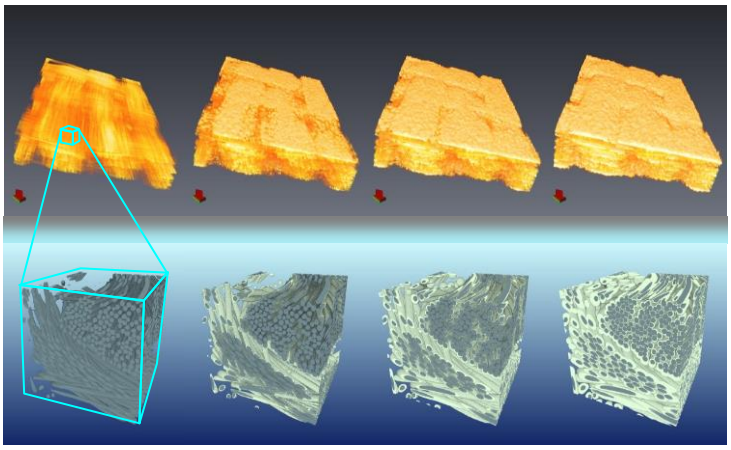


Figure 1.6 *- Macro-scale (top) and micro-scale (bottom) infiltration modelling by random walkers. Depending on the physicochemical conditions, there appear matrix thickness gradients at small scale (bottom right). The macro-scale code simultaneously handles intra-tow infiltration and seal-coat deposition (top right).*

1.4.1.7 Joining techniques of CMCs

Joining techniques remain a critical tool for reducing assembly time, cost and excessive use of materials. Japan and the USA have intensive activity on joining and testing of dissimilar materials; the Joining and Welding Research Institute in Japan, Pacific Northwest National Laboratory and NASA Glenn Research Center in USA are specialized in joining technologies and the testing of advanced materials. On the contrary, expertise on joining in EU is unnecessarily fragmented and confined in academia or private companies, seldom in contact to each other, limited by non-disclosure constraints, with consequent loss of efficiency, duplication of research and waste of money. A strong EU expertise in joining is particularly needed for the new type of lightweight composites (such as CMCs) for optimization of existing components, but also for the development of new products and new design solutions. In particular, the development of suitable joining processes fits into the context of developing high-performance and high efficiency components for the European steel industry. If the issues related to joining CMCs to themselves and to dissimilar materials are solved, they will be much more extensively used to produce both new components and 'hybrid' structures, the latter constructed of two or more different types of materials where each contribute its own unique properties that complement the properties of the other(s). Such components can be used for a range of applications in industries that include, aerospace, gas turbines for energy production, wind energy, furnace technologies, automotive, etc. Non-oxide CMCs are now more often used in components for energy production, defense, space and many other high-performance applications and have great potential to replace conventional materials such as metals and superalloys in many applications. However, the joining of CMC to themselves or to other materials remains a critical issue. The manufacturing of net-shape and ready to use ceramic components for advanced design solutions require large and complex parts, which can be built by joining and integrating discrete units of similar or dissimilar ceramics. Thus, joining of ceramics, and their integration remain fundamental issues to be addressed for the efficient use of these materials in many applications, whereas they could help in reducing fuel consumption, air pollution and in the optimization of existing components.Many techniques have been developed so far to join non-oxide CMCs, among them: mechanical fasteners [104], metallic brazing [105-109], diffusion bonding [107], glass-ceramic joining [110], the use of MAX-phases [111], transient

eutectic [112] and spark plasma sintering [113-114], Liquid Silicon Infiltration (LSI) [115], Affordable Robust Ceramic Joining (ARCJoint TM) [116], Solid State Displacement Reactions (SSDR) [117], and Transient-Liquid-Phase Bonding (TLPB) [118].
The "RM-Wrap" (RM= Refractory Metals, i.e. Mo, Nb, Ta, Zr) is a novel brazing technique where the refractory metal, shaped as a wrap, is used to contain one or more silicon foils. It is a pressure-less joining process performed at 1450°C in an inert, flowing argon, environment [109,119]. The wrap is necessary to prevent the molten silicon spreading too fast and leaking outside the joined region during the joining process. The joining materials are *in-situ* formed composites made of refractory metal disilicides including $MoSi_2$, $NbSi_2$, $TaSi_2$ or $ZrSi_2$ embedded in a silicon-rich matrix. Disilicides have high melting temperatures and good oxidation resistance at elevated temperature, making them a suitable choice to join ceramics. $MoSi_2$ has already tested as component of a brazing mixture for joining dense sintered SiC samples [120] following a pressure-less joining method based on the use of a Mo-silicides mixture. The RM-Wrap process is quick, free from undesired phases and has excellent reproducibility and versatility; the quantity of each metal can be modified and more than one refractory metal can be used in the same wrap. RM-Wrap has been proven to be very effective in joining SiC-based ceramic matrix composites and highly porous SiC substrates (SiC foams). In the case of ox/ox CMCs, the joining material must be oxidation resistant and able to perform well at high temperatures (e.g. > 800°C) without any degradation during service life. Furthermore, the selected joining material must have suitable wetting properties and a matching coefficient of thermal expansion (CTE), retain its strength at higher temperatures and should not react deleteriously with the composite during its service life. Several commercial metallic alloys, including silver and zirconium-based brazing materials, exhibit good wetting and joining behavior with ox/ox CMCs. However, they have a lower mechanical strength and poor oxidation resistance at 550°C in air [121], which is much lower than the expected working temperatures of ox/ox CMCs (1000°C – 1200°C). Diffusion bonding, on the other hand, requires the use of high pressures, which limits the shapes of the components that can be joined [122]. Furthermore, the joining temperature used with MAX phases present as the bonding phase is higher than the sintering temperature of ox/ox CMCs [111]. Therefore, there is a great demand for the development of reliable oxidation resistant, high-temperature, stable joining materials that will allow designers and engineers to successfully employ these CMCs in high-performance structures. Glass-ceramic materials have better oxidation resistance than brazing alloys [121] and can withstand higher temperatures for longer times in air [123-125]. To avoid thermal stresses, their CTE can be tailored according to the CTE of the substrate. Moreover, by selecting an appropriate glass composition, their volatilization in steam can be reduced and they are therefore a good choice for air, wet H_2 and steam operating conditions at high temperatures [126]. While existing joining technologies will not be ignored, the emphasis will be on new compositions, new processing routes as well as new joining strategies; new frontiers of research in joining activity are focused on tailoring of joining materials and joining processes for specific CMC (oxide or non-oxide) and precise applications. Research activity will look at bonding techniques based on pressure-less, high-temperature, tailored glass-ceramic sealant, brazing by "in situ" high T composite joining materials suited for the bonding of non-oxide and oxide/oxide CMCs and ceramics. To reduce the detrimental influence of thermal stresses, functionally-graded joints approaches could also be investigated. In addition, advanced surface engineering methods such as nanostructuring of surfaces by plasma etching or laser treatment, mechanical machining, thermal selective modification, etc., will all be studied. The combination of these techniques with innovative joining materials/processes will be exploited to maximize the interfacial strength of the joints.

1.4.1.8. Coating techniques of CMCs

Despite their remarkable suitability for high-temperature lightweight applications, SiC-based components suffer from rapid recession when used in combustion environments above 1100°C. This is due to the fact that volatile Si(OH)x species are formed when the superficial SiO_2-layer is in contact with hot water vapor, which is formed during the combustion of hydrocarbons and/or hydrogen. Therefore, Environmental Barrier Coatings (EBCs) especially on non-oxide CMCs are generally required to increase the resistance to water vapor corrosion [127,128]. EBCs generally comprise multiple layers, usually one bond-coat layer, up to several intermediate layers, and one top layer. The respective materials have to be carefully selected regarding their coefficient of thermal expansion (in order to have a high thermo-mechanical stability), chemical compatibility (in order to have a good bonding to the substrate and each other and to avoid contact corrosion), and phase stability. The latter is especially crucial for the top layer, which has to have a high hydrothermal stability, low diffusivity and a high resistance against CMAS (calcium-magnesium-alumosilicate species) corrosion. In addition to their chemical protection purpose, EBCs can also serve as Thermal Barrier Coatings (TBCs), providing a temperature gradient between the coating surface and the substrate surface and therefore raising the application temperature of SiC_f/SiC components even further. Present EBCs systems for SiC_f/SiC composites comprise a silicon bond coat, mullite-based intermediate layers and a rare earth silicate top coat [129]. Currently, the most common application technique for EBCs is atmospheric plasma spraying (APS) [130]. Nevertheless, APS is not applicable to coat complex shapes, as it is a line-of-sight process. Furthermore, the resulting microstructure usually exhibits cracks and a lamellar porous structure. In addition, volume changes due to shrinkage, crystallization or phase transitions during thermal cycling are likely to cause cracks [131,132]. Therefore, chemical [133,134] or physical vapor deposition [135,136] and slurry-based coating techniques are of increasing interest. The former allows for the production of well-defined layers, but is usually time-consuming and expansive. Slurry-based coating techniques like dip-coating, spraying or brushing, in contrast, are a cost-efficient, simple and economical approach to apply defined EBCs coatings [137,138]. Their main drawback, however, is the necessity to densify the respective layers in several thermal treatment cycles, which increases the thermal load on the substrate material and can lead to residual stresses.

In the framework of CEM-WAVE, burner operating parameters will be optimized in a way SiC is in the range of passive oxidation, guaranteeing long-term operation. In fact, making the SiC_f/SiC radiating pipe operating at surface temperatures higher than 1300°C is not expected to be convenient, due to increased production of NO_x.
At the FHF, EBCs will be deposited on selected oxide and non-oxide CMC samples via a slurry coating technique. The formulation of the slurries will be developed and the rheological properties adjusted for the coating process. The EBCs will be multilayer systems free of metallic silicon, comprising at least one bond coat and one top coat. The EBC will be designed to minimize thermal stresses during thermal cycling by adjusting the coefficient of thermal expansion to the CMCs substrate. The bond coat needs to be chemically compatible with the underlying CMC substrate. For non-oxide CMCs, the FHF has great experience in mullite- and zirconium titanate-based bond coat systems. As top coats, rare earth silicates will be considered. The top coats will be designed to raise the gas-tightness, which is an important requirement for good corrosion protection. This can be achieved by minimizing cracks in the top coat and by designing the microstructure of the coatings with closed porosity. The thermal treatment will be designed to minimize thermal stress to the substrate material. The overall objective is to provide an EBC system that protects the underlying CMC from corrosion during use.

1.4.1.9 Steelmaking industry & radiant tube furnaces

The industrialization has been powered by fossil fuels. In the steel industry this has involved using coal-based products, such as coke, to reduce iron ore in the blast furnace (BF). Although steel is less carbon-emitting per application than many other materials from primary sources, the sheer scale of global steel production means the industry contributes over three gigatons of CO_2 to global emissions annually. Global steel demand is forecast to increase from 1.7 billion tonnes in 2018 to over 2.6 billion tonnes by 2050 under current consumption patterns. Primary steel is produced by using two types of raw materials: hot metal or pig iron, and steel scrap, and through two types of processes: basic oxygen furnaces (BOF) or basic oxygen steelmaking and electric arc furnaces (EAF). In BOF steelmaking, about 75% of iron comes from the hot metal produced by BF process, the remaining 25% of the raw materials is steel scrap. EAF process uses 100% steel scrap in general, however, in some plants significant share of direct reduced iron (DRI) is used [139]. In steel processing lines, indirect heating of products in protective atmosphere is carried out with radiant tubes equipped with gas-fired burners [140]. One of the limiting factors for processing lines performances is the radiant tubes damage. This potential damage results from a combination of many parameters such as operating conditions (combustion performances, depending on burner design and control, furnace control, fuel quality and stability…) and design (material selection, geometry, manufacturing…). In current operating conditions, radiant tubes are exposed to severe thermal loading:

- High temperature up to material temperature limit;
- Heterogeneous thermal field resulting from combustion;
- Thermal cycles resulting from furnace control for various production and burner control;

These stresses account for the observed damage on radiant tubes resulting from coupled phenomena such as creep, thermal fatigue, and corrosion. The temperature increase needed to reach expected metallurgical performances for new advanced high-strength steels grades is limited by the mechanical strength of metallic material used for radiant tube.
Another issue is the efficiency of the radiant tubes. In all types of radiant tubes, the combustion takes place within the tube and the flue gas leaves the tube at the flue gas outlet. The efficiency of radiant tubes depends on the flue gas outlet temperature. The lower the flue gas temperature the higher is the efficiency of a combustion system [141]. The lifetime of radiant tubes and maintenance costs are also important for the economical application of the technology in heat treatment applications. Experimental investigations of radiant tubes in laboratory furnaces and under practical conditions in field tests are important in order to develop and improve radiant tubes. Now that the unintended consequences of using fossil fuels have become clear, the world needs to find a new way that enable further economic and social development, while minimizing environmental damage. This means we need to significantly reduce the carbon footprint of steel, which requires finding new ways to make steel in less emissions-intensive processes [142]. The BF process has been for the past two centuries the dominating ironmaking route to provide the raw materials for steelmaking industry. Nowadays, over 93% of the total iron production from ores is taking place via BF route. Actually, the reduction of iron ore to iron is predominantly achieved using high temperature carbon monoxide (CO), sourced from fossil fuels – coke and pulverized coal – which is also used as an affordable source of energy. The most important alternatives that are being investigated are: deriving CO from circular forms of carbon (bio-based and plastic wastes from municipal and industrial sources and agricultural and forestry residues), applying the process of electrolysis, or increase the use of high-temperature hydrogen gas (as reducing agent with respect to carbon monoxide in DRI process). The latter two pathways require vast amounts of electrical energy, which would all need to come from clean sources. In this hypothetical and yet very likely future, the most convenient fuel available in a steelmaking plant would be hydrogen or mixtures of hydrogen and biomethane, which will require materials not only able to sustain extreme conditions, such as high temperature or corrosive environments, but also phenomena like hydrogen embrittlement for metals and hydrothermal attack for ceramics. CEM-WAVE aims at substituting currently metallic alloys employed in radiant tube furnaces with CMC-based components which displays better thermo-mechanical properties, both in high temperature and corrosive atmospheres, as well as improved thermal efficiency and lifetime. A small-scale version of a tube exploiting the benefits of both non-oxide CMCs (higher thermo-mechanical properties) and oxide CMCs (inherent oxidation resistance and lower cost) will be produced by MW-CVI process to find the better performance/cost compromise and validating it in relevant environment expected by the increasing future use of renewable energy sources.

1.4.2 Innovation potential and coverage of the value chains

Furthermore, additional innovation potential results from the improved processes developed along the project, - using cutting edge technologies in the field of sensors, data analytics & modelling and AI that will be used as KER further detailed following each partner's exploitation intention.The knowledge and IPs generated by the project will be exploited by partner companies, in order to improve their competitiveness, differentiating their products in the market, thereby tapping into increasing market demand for high-performance CMCs.

Table 1.5 *- Progress of the CEM-WAVE project compared to other previously funded projects.*

Project Name, approach and technology	*Differences versus CEM-WAVE*
C³HARME (Next Generation Ceramic Composites for Combustion Harsh Environments and Space) funded under the H2020 NMP-19-2015 call. The main purpose of C³HARME is the design, development, manufacturing and testing of a new class of Ceramic Matrix Composites based on ultra-high temperature ceramic matrices reinforced with SiC or C fibers suitable for application in severe aerospace environments.	The innovation proposed in CEM-WAVE is related to the obtainment of high-performance CMCs capable of operating in various extreme conditions (high temperatures, corrosive environments, etc.). In fact, in a not so distant future, energy-intensive industries (e.g. steelmaking) will have to reduce CO_2 emissions, fossil fuels consumptions and waste production. As a consequence, these industries will have to operate with new mixtures of biogas/clean H_2 that will cause more operative extreme and fluctuating conditions. The materials currently used are not able to withstand future conditions. In CEM-WAVE, thanks to the use of solid state microwave technology and new joining techniques, by reducing the process time, it will be possible to obtain high and improved performance CMCs with lower costs. The lower costs will allow new market possibilities, widening the use of CMCs in a large number of sectors.
SIMUTOOL funded under FP7-SME (grant agreement number 286717) developed a simulation platform for the manufacturing of composites through microwave (MW) heating. The simulation will include an electromagnetic field coupled with heat transfer mechanisms that take place during the production process. One of the major outputs of the simulation platform has been the successful design of a ceramic matrix composite tool with a MW absorbing layer, in order to maximize the energy saving potential of the MW heating process.	SIMUTOOL focuses on macro-scale modeling of laminates and does not address the chemical vapor infiltration technique. In CEM-WAVE, in addition to modeling the electromagnetic reactor cavity, one of the main innovations in terms of modelling activities will be the AI-assisted modeling at both the micro and macro-scale level of the CMC obtained with the MW-CVI process. This will allow obtaining and predicting the final properties of the processed material.
ROLL-IT H2020-SME1 (Radically innovative technology for efficient manufacturing of preforms for ceramic matric composite (CMC) brake disks) aims to industrialize and commercialize a fully automated additive manufacturing techniques for shaping of near-to-shape preforms for CMC brake disks.	Thanks to the technology that will be developed in CEM-WAVE, it will be possible to obtain high-performance CMCs, even in complex shapes, assembled thanks to the new joining techniques that will also bring to an overall cost reduction of the CMC pieces.

Freedom to operate was already given a major insight as it is vital for the commercial exploitation of the results, and a patent search has been performed around each of our main technologies reflected in the SoA. An online worldwide patent search has been performed using the European patent office Esp@cenet engine covering the following combined search terms: CMC, MW-CVI, etc. The most relevant patents identified are reported in Table 1.6. It reflects most relevant patents in our main innovation areas highlighting differences with CEM-WAVE. The prior art is rich but there is space for further innovating beyond the state of the art that could be patented. The below mentioned patents will be taken into account during the project in two different ways: Using some of them and evaluate their applicability in the optimization of some approaches of our proposal; Improving and overcoming some of them as a result of the success of the CEM-WAVE project.

Table 1.6 *- Results of the patent search focused on the main innovation area of the novel IP considered to be generated in CEM-WAVE*

Patent	Description	CEM-WAVE differences
US2020024200 (A1) - Ceramic Composite Materials and Methods	Methods of making ceramic matrix composite materials are provided. The techniques include infiltrating a carbon nanoscale fiber network with a ceramic precursor, curing the ceramic precursor, and/or pyrolyzing the ceramic precursor. The infiltrating, curing, and pyrolyzing steps may be repeated one or more times. Composite materials also are provided that include a ceramic material and carbon nanoscale fibers.	Although CEM-WAVE also uses precursors for the CMC production, there is a fundamental novelty in the heating process that leads to the formation of SiC. Thanks to the use of the new solid-state microwave technology, it will be possible to obtain an enhanced heating of the material and therefore a greater reaction efficiency. Furthermore, there is an energy cost

		saving as the pyrolysis of the precursor is not necessary.
KR20190131834 (A) - Composite Ceramic Coating Materials For Boiler Tubes With Anti-fouling Characteristic And Coating Methods Using Thereof	Patent on a composite ceramic coating which can reduce fouling phenomenon while having good adhesion with a metal base material of a boiler tube. The present invention corresponds to 55-75% by weight aqueous alkali silicate solution; 20-40% by weight of heat resistant ceramic powder. This patent will provide a ceramic composite coating material for a boiler tube comprising 1 to 5% by weight of the active metal.	In CEM-WAVE thanks to the lower production costs of the CMCs, it will be possible to replace e.g. the steel parts in the radiant tube furnaces of the steel making industry (CEM-WAVE use case). In this way, using CMC tubes, the thermal efficiency improves and the fouling risk is avoided.
KR20140066812 (A) - Preparation method of ceramic-plate carbon composites using the microwave	The present patent can uniformly coat the surface of a plate carbon body such as graphite and graphene with ceramic through a sol-gel method by using uniform temperature gradient and rapid heating speed in a reactor using microwaves. Accordingly, the preparation method of a ceramic-plate carbon body composite using microwaves provided in the present invention can remarkably reduce time for synthesis in comparison with the conventional room temperature synthesis method or general heating method.	CEM-WAVE CMC production will be different. using the microwave, an infiltration of the sample (MW-CVI) and not only a superficial deposition (MW-CVD) will be obtained. Thanks to the new solid-state technology, the heating will be homogeneous so it will be possible to achieve an inverse temperature profile from the center of the sample outwards, which will allow to SiC to infiltrate starting from the heart of the sample.
CN108191460 (A) - Method for preparing C/SiC composite material	Patent on a method for preparing a C/SiC composite material. According to the preparation method, the highly-densified C/SiC composite material with better mechanical property is prepared with a precursor impregnation and cracking method repeatedly, compared with an existing reaction melting penetration method, the method cannot damage the surface of the C/C porous composite material, and performance of the composite material can be improved.	With the MWCVI processes adopted in CEM-WAVE, it will be possible to produce not only SiC_f/SiC CMCs but also C/SiC CMCs which was not attainable before using the old magnetron microwave technology. In this way the time reduction of the process is guaranteed and a cost reduction and a high lifetime of the final material is obtained.
CN108191460 (A) - Method for preparing SiC_f/SiC composite material	The invention provides a method for integrally preparing large complex SiC_f/SiC composite components, completes the connection of SiC_f/SiC composites in the manufacturing process, shortens the preparation period of the SiC_f/SiC composite components and lowers the cost. The connection method for the SiC_f/SiC composite in the invention can overcome the disadvantages of great brittleness and poor reliability of bonding, the surface shape changes of bolted components, and easy falling and poor reliability of riveting rivets in the prior art.	The connection (joining) developed in CEM-WAVE project is different. A specific ceramic inlay will be developed by sintering to be applied to the CMC preform with a view to obtaining a further improvement in thermal efficiency. In this way the joining technique adopted will be able not only to connect different pieces of CMCs but also to improve the thermal efficiency.

2 Impacts

2.1 Expected impacts

The CEM-WAVE project will result in the development and validation of an innovative CMC production process, based on Microwave-assisted Chemical Vapour Infiltration (MW-CVI) technologies. The proposed novel process, while maintaining the optimal features of CMCs (high toughness, low density, high temperature, wear and corrosion resistance) will significantly reduce the costs for making components, thus making CMCs sustainable for process industries in energy-intensive sectors such as steelmaking. Through a cross-sectoral, collaborative systemic approach, CEM-WAVE is expected to make measurable contributions in medium-term along with **all the expected impacts listed in the SPIRE-08-2020 topic.** Furthermore, CEM-WAVE covers half of the A.SPIRE sectors (Ceramics, Steel, Engineering, Chemicals - see *Fig. 2.1*).

Figure 2.1 *– CEM-WAVE coverage of A.SPIRE Sectors.*

Energy efficiency improvement of the target production and/or operation processes of at least 30%

World crude steel production reached 1.809 million tonnes (Mt) in 2018 [143]. Steel use is projected to increase steadily in the coming years to meet the needs of our growing population. Steel production is energy intensive. The iron and steel industry is the largest energy-consuming manufacturing industry in the world. Improvements in energy efficiency have led to reductions of about 60% in energy required to produce a tonne of crude steel since 1960. Energy constitutes a significant portion of the steel cost production, from 20% to 40%. Improvements in energy efficiency result in reduced production costs and thereby improved competitiveness [143]. World-wide average specific energy consumption for steel making is estimated to be 24 GJ/tonne. The most energy-efficient process requires 19 GJ/tonne for primary steel and 7 GJ/tonne for secondary steel. In the longer term, the specific energy consumption for making steel from iron ore can be reduced to 12,5 GJ of primary steel per tonne. A further reduction of up to 2,5 GJ of crude steel per tonne may be achieved when techniques are developed that can recover and apply heat from the hot steel at a high temperature. The specific energy consumption for secondary steel making can be reduced to 3,5 GJ/tonne by energy-efficient melting and shaping techniques [144].

Substituting CEM-WAVE CMCs to the currently used materials (super-alloys - Inconel) in radiant tube furnaces will allow a maximum operating temperature increase of ~100°C, due to an increased heat transfer efficiency, maximum operating temperature, better thermo-mechanical properties and corrosion resistance. As the radiation goes with the fourth power of the temperature (Stefan-Boltzmann Law) and since CMCs have a higher emissivity of Inconel, **an increased energy efficiency of ~31% is expected**. This energy efficiency will be enhanced by the **increased temperature difference** between the inlet and the outlet: the higher the temperature difference, the higher is the thermal energy exchange. This will, in turn represent, a significant reduction of the cost of steelmaking production, estimated to be between 7% and 13%, for steelmaking industrial end-users implementing CEM-WAVE results. Finally, due to the selection of energy efficient production technologies (MW-CVI), CEM-WAVE will target to achieve an improvement of energy efficiency of producing CMCs of at least 30% compared to the conventional process.

Reduction of CO_2 emissions and resource utilization by 20%;

Roughly 28 percent of global greenhouse-gas (GHG) emissions came from industry in 2014 [145]. Unless industry can lower its emissions, the world will struggle to reach the GHG reduction targets of 80 to 95 percent that governments set under the Paris Agreement of 2015. Half of industry's CO_2 emissions result from the manufacture of the four industrial commodities—ammonia, cement, ethylene, and steel. **Steel is a CO_2 and energy intensive, but highly competitive industry that also enables the uptake of major CO_2 mitigation measures in other sectors. On average for 2017, 1,83 tonnes of CO_2 were emitted for every tonne of steel produced. The CO_2 emission factor for secondary steel is about 35% that of primary steel production [146]. The steel industry generates between 7% and 9% of direct emissions from the global use of fossil fuel [147]**

Table 2.1 *- The CO_2 emission associated with the production of one tonne of steel varies depending on the production route [146]*

Production Process	CO_2 Emission (tonnes per tonne product)
Integrated steel mill	1,6-2,2
Scrap based production in a mini-mill	0,6-0,9
Scrap substitutes (DRI) in a mini-mill	1,4-2,0

The steel industry believes it is very important to be actively engaged as we continue to strive to move towards a low carbon society. The European steel industry is committed to reducing greenhouse gas emissions and making its contribution to meeting the targets of the Paris Agreement. The largest steelmaker in the world, ArcelorMittal, which is responsible for approximately 0,7 percent of all global carbon emissions, has set a goal of achieving carbon-neutral operations in Europe by 2050. 35 percent of emissions from these sectors come from burning fossil fuels to generate high-temperature heat [148].

CEM-WAVE addresses a future scenario in which renewable energy sources will be adopted in energy-intensive industries. Mixtures of biogases with clean hydrogen are the target of CEM-WAVE. **Thus, a reduction of at least 20% CO_2 emissions will be obtained using mixtures composed of 20:80 H_2/biogas.** To reach the topic expected impact, without excluding further improvements in CO_2 abatement by further increasing of H_2/biogas ratio, CEM-WAVE have set 20% CO_2 emissions reduction as KPI in Specific Objective 7. Moreover, the quality of the combustion efficiency (fuel efficiency) of radiant tubes will improve since the efficiency strongly depends on the flue gas outlet temperature of radiant tube systems. In order to achieve maximum efficiency, air preheating systems use the enthalpy of the flue gas for preheating the combustion air simultaneously reducing the flue gas outlet temperature of such systems. High efficiencies with less flue gas losses result in lower fuel consumptions and lead to a

reduction of the CO_2 emission per ton of product [141]. **Due to the improved thermal efficiency and thermo-mechanical properties the use of CMC materials in a radiant tube furnace while contributing to significantly increase the energy efficiency of both the production and operation process by at least 30%, will consequently reduce CO_2 emissions by at least 20%.**

The assessment of the environmental (LCA), energy and exergy impacts, and associated costs to the new developed production processes compared to the conventional ones will be thoroughly analysed during the project itself. Therefore, the success and impact of the project will be measured and valorised.

Increased lifetime of the equipment by at least 20%.

Since one of the limiting factors for processing steelmaking lines performances is the radiant tubes damage [140], CEM-WAVE aims at substituting currently employed metallic alloys with CMC-based components which display better thermo-mechanical properties (i.e. creep resistance, impact resistance), both in high temperature and corrosive atmospheres, as well as improved thermal efficiency and lifetime.

The research and innovation work will be supported by integrated sensors and AI-aided modelling to monitor and predict the material behaviour. **All these factors contribute to increasing by at least 20% of future radiant tube furnace when using CMCs components produced in CEM-WAVE**. Considering that the current lifetime of radiant tubes could be ranged between 4 and 8 years, the consortium expect to extend the lifetime of the selected equipment by at least 12 years.

2.1.1 Innovation and Economic Impact

The project, through a **systemic and multidisciplinary approach, will test and demonstrate possibilities to reduce the production costs of ceramic matrix composites** and to **establish new supply and value chains in the materials and composites manufacturing economy**. The key benefits of ceramic matrix composites over other materials includes high thermal resistance, hardness, corrosion resistance, light weight and non-magnetic in nature. Over years, ceramic matrix composites have gained significant importance in industrial applications due to their high thermal and physical properties over single phase ceramics and other materials. **Growing industrial demand for high temperature resistant, low-weight and density equipment is the primary factor driving the global ceramic matrix composites market**. CMCs are majorly used in the production of aero-entry systems, propulsion subsystems, high temperature radiators, insulations and coatings. Extensive usage of CMCs as an individual material component in the production of brake discs, clutches, braking systems is another factor driving the CMCs market. Increasing adoption of lightweight composites by the electrical vehicle manufacturers is expected to further boost market for ceramic matrix composites during forecast period. Additionally, the increased efficiency and lifetime of CMCs compared to currently employed metallic alloys in radiant tube furnaces, will open new possibilities in steelmaking (i.e. use of higher annealing temperatures, new chemistries to process) making further more competitive CMC-based components cost enabling their wider deployment to almost all industrial lines. **CEM-WAVE project has a huge innovation potential**, since CMCs represent very promising solutions for high temperature applications in strategic industrial sectors, such as transport and energy. Ceramic Matrix Composites (CMCs) occupy a fundamental position in the transport sector (e.g. for elements of combustion engines of aircrafts, valve-trains, turbine blades, exhaust systems, cars braking systems) and energy sector (e.g. refractory materials for silicon foundry furnaces, energy reactors, gas burners and high-pressure heat exchangers). Huge market opportunities are expected for CMC provided to overcome the **three major identified gaps**: **high cost**, **difficulty of processing** and **materials reliability**. New and more efficient manufacturing technologies can pave the way to improve material quality, reduce processing time, converge towards near-net shape fabrication, trim energy spent and abate production costs. CEM-WAVE addresses these challenges by design, modelling and artificial intelligence optimization of innovative high-frequency electromagnetic technologies for the production of CMCs: this will allow to drastically reduce the difficulties and costs of production while dramatically improving materials reliability. **CEM-WAVE will develop CMCs materials that can survive the harsh conditions of the future types of fuel that will be adopted in steelmaking plants (e.g. Hydrogen)**: by pursuing this, **the new materials will at the same time be able to resist in the current steelmaking plants and fit also current steelmaking industry needs** to find materials that increase the efficiency of the radiant tubes, by improving the heat transfer and increasing their lifetime. The durability and heat transfer characteristics of these materials will be tested in the project for future and current conditions of operations and this will be part of the exploitation plan of the results of the project. CEM-WAVE results will meet huge market opportunities for European enterprises contributing to enhance manufacturing capacities in Europe, and thus growth and jobs in Europe, in both the short and medium term. Next section addresses the direct targeted markets (CMCs and steelmaking) both at European and global level. Nevertheless, the consortium expects CEM-WAVE results will match the market and user needs in a cost-effective manner of other energy-intensive industrial sectors, as for instance the cement, metallurgical and chemical industries among the others.

2.1.2 Socio-economic impacts

Steel Industry - Global Market Overview: Steelmaking process involves the production of steel from iron ore and scrap. In this process, various impurities such as silicon, sulfur, nitrogen, phosphorus, and excess carbon are removed from the sourced iron. The global steel market is expected to reach USD 1,01 trillion by 2025. The steel industry is expected to witness a rapid growth owing to strong demand from construction, automotive and transportation sector over the forecast period. The overall iron and steel market is expected to grow due to the rapid urbanization and industrialization in the emerging regions such as Asia Pacific and Central & south America. Overall, steel production is the largest iron ore application [149].

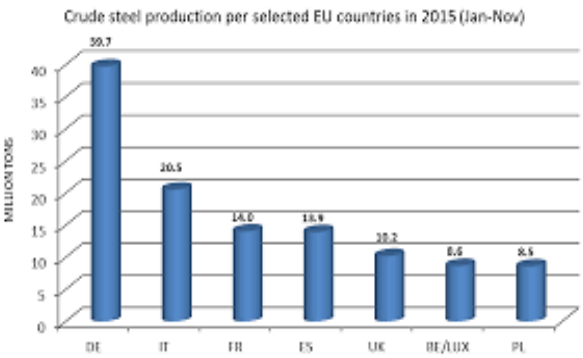


Figure 2.2 *- Steel Production per EU countries in 2015.*

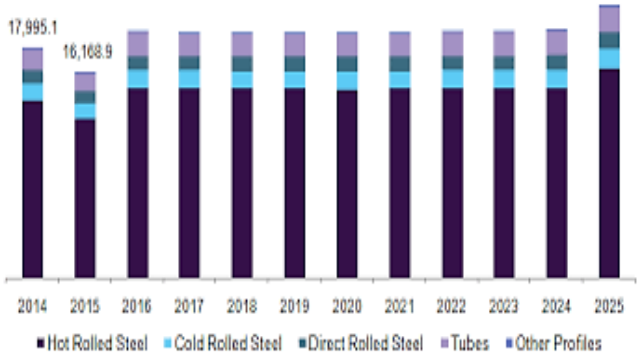


Figure 2.3 *- Steel Market Size.*

World crude steel production has been increasing steadily since the 70s, from 595 Mt in 1970 to 1414 Mt in 2010 (Worldsteel 2010). Geographically, crude steel production in 2009 was concentrated in Asia (comprising China, India, Japan, South Korea, Taiwan and South East Asian countries: 65,37% of the world total), and China alone accounted for 46,30% of the world total production. Other major producing areas were the European Union (11,33%), the former Soviet Union (8,03%), and North America (US, Canada, and Mexico: 6,65%). Production in developing regions was much smaller. At the global level, crudest steel is produced by primary routes (~72%), while the rest is produced by secondary steelmaking.

The steelmaking sector is the main market that CEM-WAVE addresses, offering unmatched opportunities for the project results exploitation and industrial uptake.

AMIII USE CASE

Radiant tubes are used in industry for heat treatment applications in which products are treated under a protective gas atmosphere within heat treatment furnaces. Therefore, the heating of such furnaces is performed with indirect fuel-fired systems or electrical heating elements. For indirect fired applications, the flue gas of the combustion process cannot enter into the furnace. The combustion takes place within radiant tubes and the heat is transferred—via radiation—from the outer surface of the tube to the process. The furnace capacity depends on the number of tubes installed and it is affected by leaks or cracks on the surface of the tubes. Depending on the size of the leak, it could cause various effects ranging from strip pollution due to combustion byproducts to a total collapse of the radiant tube. For this reason, when a crack is detected, it is necessary to close the tube resulting in the furnace losing of heating capacity and efficiency. Due to the high temperature and the use of a protective atmosphere the tubes cannot be replaced with the furnace in operation and it is necessary to stop the completely the line and cold down the furnace. This operation requires the line stoppage during at least two or three days, so the replacement of the damaged radiant tubes is done during the yearly standstill, and only in case of extreme need, a specific stoppage for changing tubes is planned. So, the limited opportunities of changing the radiant tubes makes very important its reliability. Other limiting factor of the productivity is the maximum working temperature of the radiant tubes. As the thermal exchange is mainly due to radiation, the higher temperature achieved by the tube, the higher is the exchange. In case of metallic radiant tubes, try to work at temperatures higher to 1000ºC lead to damage them and so on, diminish the capacity of the furnace. In case of the current ceramic radiant tubes it is possible to achieve temperatures around 1200º C without thermal affection, but these materials are fragile and do not resist the impact of the strip in case of breakages inside the furnace, problem which could happen from time to time. The value of the present project is to develop a material merging the best of both materials, with good thermo-mechanical properties and able to work at high temperatures, which will make possible to increase the lifetime of the radiant tubes. The proposed material should also be ready for the next transformation by the change of the traditional fuels to the new energy sources.

Ceramic Matrix Composites Market: The market size for ceramic matrix composites is estimated to be USD 9,4 billion in 2019 and is projected to reach USD 23,3 billion by 2029 [150]. The Ceramic Matrix Composites market is expected to grow at a CAGR of 9 - 10% during the forecast period (2019-2026). The ceramic matrix composites market was dominated by the oxide/oxide ceramic matrix composites segment, in terms of volume. This is due to the high demand for oxide/oxide ceramic matrix composites in aerospace, industrial, and automotive applications. Oxide/oxide ceramic matrix composites are considered as the material of choice in the aerospace, industrial, and automotive industries due to their low weight and high-temperature resistance properties. It is also widely used in industrial applications. They are less expensive than SiC_f/SiC ceramic matrix composites.

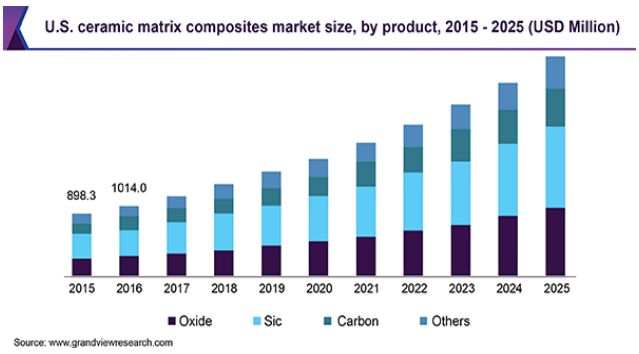


Figure 2.4 *- US ceramic matrix composites market per product 2015-2025*

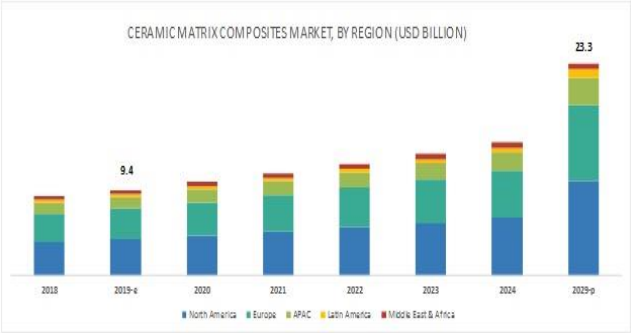


Figure 2.5 *- Ceramic matrix composites market by region.*

North America is expected to be the largest market for ceramic matrix composites, owning to increasing demand from aerospace & defense and automotive end-use industries in the region. The market growth is attributed to the increased demand for ceramic matrix composites from the aerospace & defense, and energy & power end-user industries.

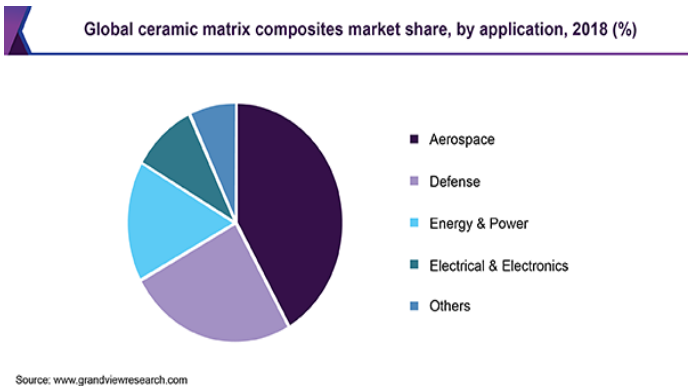


Figure 2.6 *- Global ceramic matrix composites market per application in 2018*

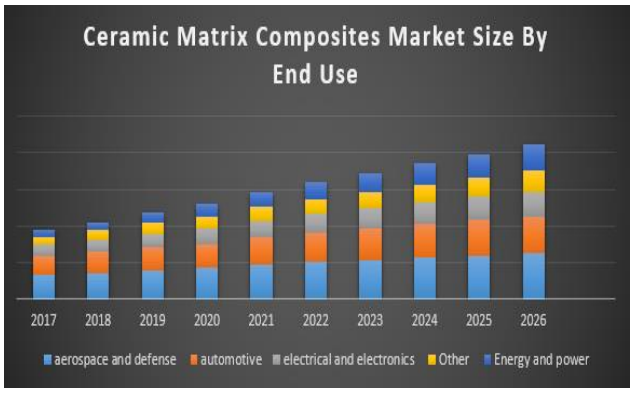


Figure 2.7 *- Global ceramic matrix composites market share by end use from 2017 to 2026*

In the *automotive sector*, the growing need to reduce weight and improve efficiency is driving the demand for ceramic matrix composites. This, in turn, will help counter the effect of growing fuel prices and reduce carbon emission. Manufacturers of luxury and sports car engines prefer to use ceramic matrix composites in their engine components as they are strong, lighter, and can withstand very high temperatures. They are expected to increasingly replace conventional alloys and metal components in automotive engines as conventional alloys need extensive dedicated cooling air while ceramic matrix components can operate with little or no cooling. The increasing use of CMCs in the *aerospace & defense industry* is mainly due to its properties such as high damage tolerance, fracture toughness, and high temperature, wear & corrosion resistance. It is projected that ceramic matrix composites will replace titanium metal- and other composites-based components in aircraft structures. Ceramic matrix composites are used in aerospace gas turbine engines, nose caps, exhaust nozzles, and missile propulsion systems in the aerospace & defense end-use industry. *Energy & power sectors*: the risk associated with introducing ceramic matrix composites in new applications is considered to be a big roadblock for the market growth. Market players are anticipating high growth in the ceramic matrix composites market in the coming years due to their use in newer applications. For instance, Japanese researchers [151] are considering it for building next-generation nuclear reactors. These composites are increasingly finding applications in the medical and diagnostic industry for making implants and diagnostic kits. However, **growing environmental regulations and high production cost associated with CMCs are the primary factors hindering the growth of global ceramic matrix composites market**. In effect, the cost of ceramic matrix composites has been a concern for many industries over the last decade. Due to the high manufacturing cost and time associated with these composites, gaining customer confidence is a major challenge for the market players. The prices of ceramic matrix composites are controlled, to a large extent, by the costs of ceramic fibers. The prices of carbon/carbon ceramic matrix composites are comparatively lower than silicon carbide/silicon carbide and oxide/oxide ceramic matrix composites due to the large-scale production of carbon fibers in the last few years. On the other hand, the production of alumina and silicon carbide fibers is still very low. Another factor contributing to high prices of ceramic matrix composites is the manufacturing of the matrix; the production of a matrix needs expensive batch processes at high temperatures in a controlled atmosphere. Also, ceramic matrix components involve densification costs and machining costs, along with numerous inspection steps if the component is to be used commercially or where the risk of failure may endanger lives. Many industries consider ceramic matrix composites for a limited number of premium applications owing to the high costs of raw materials, fabrication, and inspection compared to monolithic ceramics and other conventional materials. This high cost is a major

restraint to the growth of the market. **CEM-WAVE will represent an innovative solution to this current problem. In fact, the CEM-WAVE novel proposed process, while maintaining the highly performant properties of CMCs, will extremely reduce the costs and time for making components, thus making CMCs truly competitive and sustainable alternatives for process industries in energy-intensive sectors such as steelmaking.** In fact, since the investment cost for the construction of a conventional CVI plant system compared to the MW-CVI is almost unchanged what affects the final price is the energy saving deriving from the shorter working hours to produce an equal amount of CMC.
Currently there are two main factors that influence CMC cost: the expensive raw materials (in particular for the SiC fibers) and the processing technology. CEM-WAVE project aims to abate production costs, with a sustainable process, lowering the energy consumption and processing time. This goal, joined with the current birth of European ceramic fibers producers (https://www.bjsceramics.com/bjs-ceramics-2/products/) will certainly open new market opportunities enabling a wider usage of CMCs not only limited to energy intensive sectors, leading to a further reduction in the overall cost of these materials.

2.1.3 Environmental impact

Global population growth and tremendous economic development has brought us to the crossroads of long-term sustainability and risk of irreversible changes in the ecosystem. Energy efficient and eco-friendly technologies and systems are critically needed for further growth and sustainable development. Expanding global industrialization will continue to put pressure on energy resources, heightening concerns about energy efficiency. At the same time, environmental regulations' mandates to reduce emissions will continue to be an incentive for developing cleaner, more efficient combustion cycles.While ceramic matrix composites were originally developed to overcome problems associated with the brittle nature of monolithic ceramics, today the composites can be tailored for customized purposes and offer energy efficient and eco-friendly applications, including aerospace, ground transportation, and power generation systems.In fact, ceramic matrix composites combine reinforcing ceramic phases with a ceramic matrix to create materials with new and superior properties (toughness, electrical conductivity, thermal conductivity, thermal expansion, high-temperature stability, high thermal shock resistance, high hardness, high corrosion resistance, light weight, non-magnetic and non-conductive properties and versatility). For the processing industry, related benefits of using ceramic composites include increased energy efficiency, increased productivity, increased resistance to different substances and conditions and regulatory compliance. CEM-WAVE will pursue the reduction of energy consumption and CO_2 emissions both in the production process of CMCs developed in the project, and in steelmaking industrial processes, where the CMCs components -produced in CEM-WAVE- will be used in the radiant tube furnaces. CEM-WAVE consortium is confident that there will be substantial reductions in CO_2 emissions; the quantity and percentage of reduction is therefore both one of our KPIs and an expected impact. The reduction of CO_2 of at least 20% is really a KPI of our process. The full LCA, LCC and TA performed in CEM-WAVE definitively will show the benefits of proposed technologies in terms of **using optimal amounts of raw materials, energy consumption and other energy saving features**. Moreover, the foreseen spreading of CMCs in other sectors can allow the use of **more durable materials**, thus immediately **decreasing generation of waste**. Another indirect good environmental impact consists of the **development of materials for more efficient engines**, allowing a reduction of fuel consumption and thus reducing not renewable resources consumption and related pollution/greenhouse effects.

2.1.4 Barriers, obstacles and framework conditions

The consortium has adequately examined all the known factors that may prevent CEM-WAVE from achieving its goals. The success of the project and the exploitation of the results depend on many external factors. The primary risks/barriers related to project implementation are covered in section 3.2 of the Technical Annex. The longer-term barriers and obstacles towards achieving impact relate to moving at the project end from TRL5 into widespread market replication. These barriers can be considered in the following areas:
1. Post-project financing. A key component of WP9 will be to prepare the case for post project financing from commercial or public funding sources to ensure the completion of the journey over the Innovation Gap and through the notorious Valley of Death. This will be supported by Life Cycle Costing (LCC) activities undertaken in WP8 and the project management activities in WP10. **KNEIA is experienced in securing commercial investment from long term sources of finance for innovative technologies**. This experience has contributed to the structure of the proposed plan of work. It has already identified potential post-project financing sources that could be pursued depending on the project results and the risks that may arise (both technical and economic) which will be tackled in the activities carried out in WP10.

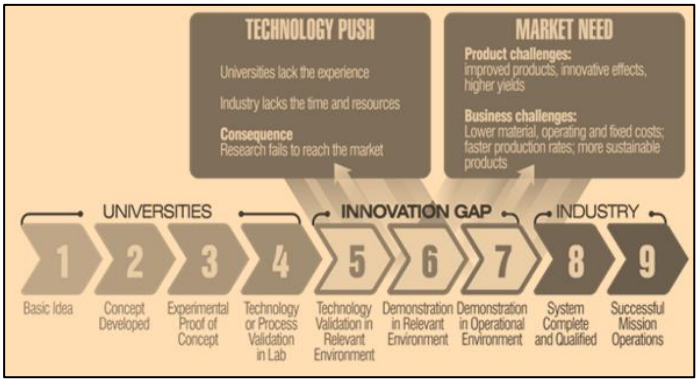


Figure 2.8 *- Innovation Gap within the TRL scale (source IRIS).*

2. Technical barriers. Typical key barriers to the broad application of ceramic matrix composites include the lack of specifications, databases, attachment concepts, in-service repair methodology, high cost, and scale-up. The technical risks and mitigation strategies covering the project delivery have been analysed and are covered in section 3.2. Any scale-up and industrialization of a process will involve a

certain level of risk. However, in addition there are risks that competitive actions may result in IP being protected that would limit the proposed innovation strategies. It is important that, in obtaining any post-project financing, there is the "Freedom to Operate" and that there are no intellectual property (IP) conflicts. Initial IP searches undertaken in the preparation of this proposal (see section 1.4) have shown that there is Freedom to Operate. However, throughout the project, the partnership will employ due diligence in ensuring that IP risks do not become barriers to progress.

3. Political and regulatory environment. The Kyoto Protocol and various European directives have set binding targets on countries to reduce their GHG emissions. Europe depends on imports for a significant proportion of its energy demand, especially for transport. Increasingly these import supplies are through state owned enterprises and employed as a foreign policy tool. The ambition of this project is to contribute to a more energy efficient processing and manufacturing industry in Europe. All partners from CEM-WAVE consortium are committed to fully comply with all applicable regulations and other legal requirements to conduct their research in a highly ethical and legal manner. Main regulation related to CEM-WAVE materials will be identified and reviewed, updated and addressed in WP8: among them: Environmental Management, UNEP specifications Guidelines for Social Life Cycle Assessment of Products, etc. A special attention will be paid to the expected "Proposal to support zero carbon steelmaking processes by 2030", announced in the recently published "The European Green Deal" as an action to be undertaken before the end of 2020.

2.2 Measures to maximise impact

A) Dissemination and exploitation of results

The project results cannot be only oriented towards the excellence in the product itself. The needs and possibilities of the final users have to be considered in order to make real the economical payback and revenues of the public and private investment. Thus, excellence in dissemination and exploitation is of outmost importance to maximise the impacts obtained in the project.

Dissemination plan

Our dissemination plan encompasses a comprehensive series of actions **to broadcast the non-confidential results that are generated during the project to the widest and most relevant audience possible.** The goal is to heighten awareness about the results, to contribute to knowledge building in Europe, and to ensure that as many stakeholders as possible have access to these results. This includes: the relevant industrial sectors (ceramic matrix composites, steelmaking and manufacturing), standardisation bodies, retailers, market actors (environmental consultancies, etc.), policy makers, researchers and the academic community, mass media (TV, radio and the written press) and multipliers for addressing the general public, as well as other targeted stakeholders (other H2020, EC and national R&D initiatives, ESIF stakeholders, A.SPIRE). The most appropriate dissemination tools and channels will be selected for targeting each and the expected impacts of our dissemination activities are also defined. The consortium will actively disseminate project results with a multi-faceted approach. The partners have an excellent track record in scientific publications, conference presentations and a strong history of engagement with ceramic matrix composites and steelmaking professionals, scientists, regulators, managers, commercial organisations, and the citizens. All research results will be disseminated to the scientific community, using multiple approaches: via scientific journals, conferences, the project website and regular newsletters. This section includes the preliminary version of the Plan for the Dissemination and Exploitation of the Project's (PDER) results, which will be enlarged and updated during the execution of the project. The purpose of the PDER is to provide a formal planning document for using and disseminating knowledge throughout the project. The plan will facilitate the common understanding of the aims of the dissemination activities, and will assure the dissemination does not interfere with the IPR management but is at its service. To protect the exploitation potential, a **confidentiality policy will underpin our dissemination activities**. The consortium experience in collaborative projects points towards the most effective default position being that all information is confidential unless otherwise agreed as non-confidential. The dissemination strategy will reflect the broad expertise of the consortium and the diverse nature of the project results and related work packages. Considering the profile of each target audience, the most appropriate dissemination tools and channels will be selected for targeting each and the expected impacts of our dissemination activities are also defined. Within the framework of WP9, the partners will build on and execute this dissemination plan following a process of analysing: **Why** – the purpose of the dissemination action; **What** will be communicated – the message; **To whom** – the audience; **How** – the method; **When** – the timing. Performance indicators will be established as part of the upgraded plan, such as number of citations (for publications), number of hits on web, number of downloads (for electronic documents/materials), number of industry consultations (from tradeshows/exhibitions), number of policy-maker engagements, etc.The dissemination of the results of CEM-WAVE will be an important action, once identified valuable intellectual property has been protected, in order to reach the widest possible impact to facilitate the take-up of the new technologies. For this reason, a comprehensive **Communication and Dissemination Plan** as part of the **Plan for the Dissemination and Exploitation of the Project's results** will be developed under the supervision of the Dissemination, Communication and Exploitation Manager (Cristina Barragan from KNEIA, leader of the Communication and Dissemination Team), to ensure a high visibility of the funded actions and help to maximise the impact of the CEM-WAVE results. Dissemination and Communication activities will be defined by the Valorization Committee. Concerning the dissemination plan, it aims at divulging, in accordance with IPR restrictions, the main innovative aspects, which evolve during the development of the project. In line with the above considerations all partners will be involved in the definition of the dissemination strategy, which will be included in the detailed dissemination plans that will be issued at the end of month 18 of the project. These plans will give clear measures and priorities of how to spread the solutions elaborated by the project consortium to the end users. Our dissemination plan will be complemented by a **professional communication campaign** (discussed below). This will contribute to efficient

coordination and dissemination of the project outputs across the relevant European and international communities to promote the project. Moreover, the dissemination strategy will also consider how it can best be used **to maximise the future exploitation of the results of the research**. **CEM-WAVE will count on several tools and channels for the dissemination of the project results summarized in the following table:**

Table 2.2 *- Dissemination channels*

Type of dissemination	***Target audience***	***Estimated amount***
Leaflets	Industry, academia, policy-makers	2
Press releases	All stakeholders	2
Public website	All stakeholders	1
Newsletter	Industry, academia, policy-makers	3
News via social media (LinkedIn, Facebook, Twitter, Youtube, SlideShare)	All stakeholders	8
Articles in technical papers	Industry	3-5
Articles at market oriented and economic journals	Industry, consumers	3-5
Peer reviewed scientific papers	Academia	8-12
Presentations at scientific conferences	Academia	6-9
Presentations at technical conferences and exhibitions	Industry	10-12
Webinar	Academia and industry	1
Final symposium	Academia and industry	1

The dissemination activities will be carried out through publications in specialized **open access journals** of ceramic matrix composites, materials, steelmaking and manufacturing sectors. Targeted journals include: ***Journal of the European Ceramic Society, Ceramic International, Materials, International journal of applied ceramic technology, International Journal of Impact Engineering, Materials & Design, Materials Science and Engineering, New Carbon Materials, Journal of Composite Materials, International Journal of Heat and Mass Transfer, Ceramics in Modern Technologies,*** among the others. All these publications reach a wide spectrum both of the scientific, public administration and technical community. Priority will be given to journals that allow Golden and Green Open Access options. Consortium members, except where it is specifically agreed otherwise, hold copyright to the publication which will be published under the 'Creative Commons Share Alike 3.0 Unported License'. The CEM-WAVE partners will therefore submit technical papers based on the project results in these journals. Potential scientific and technical market-oriented publications have been already pre-defined:

Table 2.3 *- Potential scientific and technical publications*

Topic	**Estimated month**	**Target groups**
Ceramic fibre coatings based on ceramic precursors	M24	Scientific community
Porous fibre preforms with adapted properties for use in microwave processes	M24	Scientific community
Joining of oxide-oxide CMCs by pressure-less methods	M24	Scientific community
Joining of CMCs to dissimilar materials and their characterization	M24	Scientific community
Monitoring and modelling of the processing of CMC	M24	CMC converters directly and other processing industries in order to give more visibility to our solutions
Optimized ceramic inlays for radiant combustion tubes	M24	Scientific/engineering community
Thermal properties of CMCs for radiant combustion tubes	M24	Scientific/engineering community
Image pattern recognition as a tool for the modelling of ceramic matrix composites processing	M24	Scientific/engineering community
Image-based microscale modelling of the dielectric behaviour of ceramic-matrix composites	M28	Scientific/engineering community
Multiscale, multiphysics and AI-powered modelling of ceramic matrix composites manufacturing	M32	Scientific/engineering community
Corrosion resistance of ceramic matrix composites for radiant tubes with biogas combustion	M28	Scientific/engineering community
Oxide fibre preform impregnation; vacuum impregnation versus injection-based techniques	M24	Scientific community
Processing of oxide-oxide composites via microwave enhanced chemical vapour infiltration	M24	Scientific community

The effect of process variables on the property characteristics for ox-ox composites produced by MW-CVI	M24	Scientific community
Joining of monolithic and composite SiC: thermomechanical and microstrctural characterisations	M24	Scientific community
Joining of monolithic and composite SiC: thermophysical characterisations	M24	Scientific community
Water corrosion of monolithic and composite SiC in H2/CH4 mixtures	M24	Scientific community

The publication and presentation activity of CEM-WAVE focuses on interdisciplinary dissemination, which highlights the achievements that were possible thanks to the critical mass of experts who get together in the context of CEM-WAVE. Following Horizon 2020's open access policy, the consortium will ensure that peer-reviewed scientific publications resulting from the CEM-WAVE project will be deposited in repositories and made open, meaning free of charge online access for all the potential users. Even though it is believed that Open Access can also boost the visibility of European research, and in particular offer small and medium-sized enterprises (SMEs) access to the latest research for utilisation, it has to be kept in mind that Open Access will not interfere with Consortium decisions to exploit project research outcomes commercially or by patent process. Therefore, the project will pay attention to the legitimate interests of the partners in terms of data protection and IPR issues related to commercial impacts of the project outcomes. CEM-WAVE will develop and implement open access to the project research results according to the following main conditions: 1) All peer reviewed research articles will be deposited into an on-line repository; 2) To make the best effort to have open access to the articles with a delay (embargo) of 6-12 months as maximum.

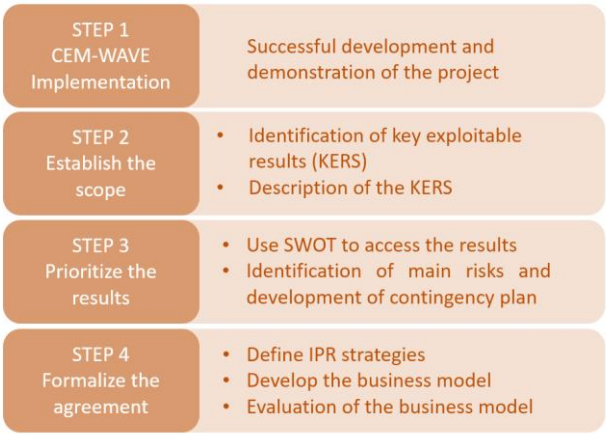
An important feature of our dissemination strategy is the **"CEM-WAVE Roadshow"**: an agenda of events that will be mapped for the duration of the project to maximise resources and to bring mutual benefits to events being organised by third parties. CEM-WAVE will schedule the trade fairs, exhibitions, workshops, seminars, conferences, round tables, etc. of relevance in Europe and internationally that CEM-WAVE consortium members can attend. Our target is to have CEM-WAVE present in **at least 18 events in Europe and internationally** over the course of the project. A philosophy of 'not wasting' any dissemination opportunities will be instilled in the consortium and partners will be encouraged to capitalise on the relevant marketing and dissemination events they are attending or organising to plan a dissemination activity for CEM-WAVE. Finally, CEM-WAVE will gratefully take advantage of specific dissemination events organized by the European Commission to present the project and its results as for example **A.SPIRE, Environment and Circular Economy Info Days** Workshops, European Researchers Nights as well as Science Festivals. A preliminary selection of relevant conferences/events includes: ***International Ceramics Congress (CIMTEC), International HT-CMC conference (HT-CMC 11), International CMC conference, Hannover Messe, JEC, International Symposium on Dynamic Response and Failure of Composite Materials (DRAF), International Conference And Exposition On Advanced Ceramics And Composites, Teratec Forum, SRA (Society for Risk Analysis), European Ceramic Society conference (ECerS), ICEAF VII, Ceramitec, R2B,*** *etc.*

CEM-WAVE project dissemination activities will link with the broader dissemination strategies of the ***A.SPIRE Association (of which 3 of the 13 partners are associated: ArcelorMittal, ENEA, Fraunhofer)***: to boost the impact of our project, CEM-WAVE will follow the guidelines of the CSA project called **SPRING** (https://www.spire2030.eu/spring), funded by the SPIRE programme, destined to support the sustainability potential and the education, training, dissemination and exploitation activities for the enhancement of impact of SPIRE projects. In addition, CEM-WAVE will coordinate its dissemination efforts also with the ***European Ceramic Society*** stakeholder panel, and other relevant industrial association and standardisation bodies at pan-European level. CertiMaC, third party of ENEA participating to the CEM-WAVE project, is also **member of UNI (National Standardisation Body)** and of ECTP (**European Construction Technology Platform**). We will aim to couple CEM-WAVE with meeting and events of these initiatives, to contribute to efficient coordination and dissemination of the project outputs across the relevant European and international communities to promote the project results One of the partners, POLITO, is member of the KMM virtual institute (KMM-VIN) (http://www.kmm-vin.eu/), which gathers more than 60 industrial and academic partners from all over EU: dissemination and communication activities such as project description on the KMM Newsletters, dedicated KMM workshops, etc. are foreseen. CEM-WAVE will also promote the coordination of common dissemination events with other SPIRE PPP funded projects. In CEM-WAVE, the plan is to organize a final conference event in month 42 to release the fully evaluated CEM-WAVE project results to all stakeholders; the event will be organised by UNIPI. UNIPI will organise a webinar to present intermediate results at M24. The webinar approach will allow reaching a wider audience limiting the costs of organisation. Given the partiality of results to be presented at M24 the audience could take advantage to a remote dissemination event limiting the costs for participation while being duly informed on the latest innovation development in the sector. CEM-WAVE will launch a dedicated website immediately after project kick-off (M4 the latest), populating it with regular non-confidential updates on the progress and a real-time "scoreboard" for our indicators. Each of the consortium members will leverage their own websites and other online presence (e.g. social media, project blogs, newsletters and on-line platforms) to disseminate information about the project and increase visibility of the project's website. All consortium members will leverage their PR networks to publish press releases stating the project objectives and collaboration agreements in each participant state to drive early adoption and interest. CEM-WAVE will also use social media and dissemination channels to maintain the momentum until preliminary results can be published. In addition, the project will gratefully make use of any dissemination route offered by A.SPIRE and the European Commission, including publication of CEM-WAVE brief on CORDIS and similar websites. The project will also develop a comprehensive strategy for activities with the general public as

part of WP9. This strategy will include public engagement, press releases, communication campaigns and publication of articles in newspapers and popular magazines; all to be used to communicate the potential impact of CEM-WAVE results for the economy, the environment and the society at large. Television is considered a very valuable channel in this sense, for this reason local channels will be contacted to explore the possibility to participate in dissemination programs. “Futuris” in the “Euronews” channel, will also be considered for a wider coverage. The project will take benefit of the EurekAlerts! news-release distribution platform hosting news releases produced by universities, journal publishers, medical centres, government agencies, corporations, and other organizations engaged in all disciplines of scientific research. Public and non-academic stakeholders will be invited in the press releases, in order to raise the attention in the territory of the participating organizations. Both KNEIA and UNIPI will make available the support of their communication units, specialised in maximising the media and public outreach as well as in building targeted dissemination tools for research and innovation results.

B) Business Plan and Exploitation of the results

CEM-WAVE will validate (from a technical, environmental and socio-economical point of view) an innovative CMC production process to produce high quality CMCs for future applications in steelmaking and other energy-intensive industrial sectors. As such, our project outputs are highly relevant for future market exploitation. A preliminary exploitation strategy and business case has been conceived considering the contribution of the results to patents, to new or existing products, services, infrastructures, knowledge bases and standards, to ensure the project innovations are delivered to the market. Our intention is to realise all the exploitation pathways for the results:



a) **Commercial exploitation-** the exploitation route of CEM-WAVE will be twofold: a) the industrial use of the project knowledge and technology developments, for the **creation of new processes to deliver high competitiveness and growth impacts** (CEM-WAVE use case will be the use of the developed materials in radiant tube furnaces in the steelmaking process); b) the production of the material itself in industrial setting, thus improving the material’s production business opportunities.
b) **Further research-** providing the knowledge and skills for future spin-out research and development projects that will have a high impact for raising the European knowledge base to prepare CMCs for steelmaking and manufacturing products and equipment, while promoting these industrial sectors transition to the green economy;
c) **Contribution to standards-** the project results will contribute towards a regulation and standard development in the field of sustainable development of CMCs.

The CEM-WAVE project is a market- and exploitation- oriented research and development initiative, our proposed Exploitation Strategy is a 4-Step approach, as summarised aside. During the implementation of the project, the most promising results will be identified (a preliminary list of potential KERs is listed in Table 2.3.). The expected impacts can only be realised through a robust post-project exploitation strategy. Table 2.3 gives a summary description of the exploitable knowledge, products or technologies, and results from CEM-WAVE project:

Table 2.4 *- CEM-WAVE Key exploitable results*

KERs	Proposed IPR tool	WP	Owners	Potential/Targeted end users
Fibre preforms both oxide and non-oxide CMCs	Know-how and patents to be licensed to companies	1	FHF	Textile industry, CMC manufactures, plant engineering and construction manufacturers, Aerospace/Aeronautics sector
Fibre coating technology both for oxide and non-oxide CMCs	Know-how and patents to be licensed to companies	1	FHF	CMC manufactures, ceramic industry (Refractory, Catalysis, Filtration)
Component coating process and coatings both for oxide and non-oxide CMCs	Know-how and patents to be licensed to companies	6	FHF	CMC manufactures, ceramic industry (Refractory, Catalysis, Filtration)
Pressure-less joining process both for ox-ox and non-oxide CMCs, easily transferable to the industrial scale; wider use of CMCs in several applications, not only limited to steel manufacturing companies	Patent, licensing to interested companies	6	POLITO	CMCs radiant tubes manufacturers Aerospace/Aeronautics Companies
Monitoring system for CMC processing	Patent, licensing to interested companies	3	IRIS	Companies converting the new CMC materials

Designed ceramic inlays for radiant burner tubes	Know-how and patents to be licensed to companies	3-6	CNRS & other partners of LCTS: U. Bordeaux, Safran, CEA	Industries that make use of tubular burners
Environmental assessment of the composite materials, designed for high temperature applications.	Patent	8	R-Tech	Metal industries, aerospace companies, consumers
Fully functional tube for use in radiant tube furnaces	Know-how and Patenting	7	AMIII	AMIII itself
The combination of High-Power Solid-State Microwave Generators at different frequencies	Know-how and patents for license to companies and other organisations	2, 3	FM	Manufacturer of high-temperature furnaces for high-tech products from the ceramic industry, manufacturer of CVD systems.
High temperature IR-Measuring Systems for the detection and control of the homogeneity of the microwave fields	Know-how and patents for license to companies and other organisations	2, 3	FM	Manufacturer of high-temperature furnaces for high-tech products from the ceramic industry, manufacturer of CVD systems.
Enhanced manufacture of oxide CMCs via the use of microwave-heated chemical vapour infiltration	Know-how and patents for license to companies and other organisations	3	UoB	CMC manufacturers, aerospace sector, chemical industry, steel and other materials industries
Preform impregnation technology	Know-how and patents for license to companies and other organisations	1	UoB	CMC manufacturers and other ceramic manufacturing companies and organisations
Manufacture of flat plates and open end tubes from oxide-oxide composites	Know-how and patents for license to companies and other organisations	3	UoB	CMC manufacturers, aerospace sector, chemical industry, steel and other materials industries, other ceramic manufacturing companies and organisations
Potential for obtaining a fibre / matrix interphase coating that can act as both a microwave absorber and yield good composite properties for a weak interface ox-ox composite	Know-how and patents for license to companies and other organisations	3	UoB	CMC manufacturers, aerospace sector, chemical industry, steel and other materials industries, other ceramic manufacturing companies and organisations
Design and manufacture of CVI system to UoB	Know-how and patents to protect ATL's designs and allow sale of equipment to ed users.	2	ATL	CMC manufacturers and other ceramic manufacturing companies and organisations serving aerospace and nuclear industries
Upgrade of CVI system at UniPi	Know-how and patents to protect ATL's designs and allow sale of equipment to end users.	3	ATL	CMC manufacturers and other ceramic manufacturing companies and organisations serving aerospace and nuclear industries
Corrosion data for monolithic and composite SiC in combustion environments	Know-how for license to companies and other organisations	4	ENEA	Radiant tubes producers; steel, glass, concrete and ceramic industry
Composite SiC for thermomechanical application up to 1600°C	Know-how for license to companies and other organisations	6	ENEA	CMC manufacturers, aerospace sector, chemical industry, steel, glass, concrete and ceramic industry
Optimization and automation of the operating conditions of single-mode and multimode reactors for microwave-assisted processing	Know-how and patents to be licensed to companies	2,3	IPCF-CNR	Manufacturers of high-temperature furnaces for high-tech products from the ceramic industry, manufacturers of MW-assisted processes systems

In-operando analysis of the dielectric properties and the infiltration stage of the sample. Monitoring of the structural properties of the CMC in use	Know-how and patents to be licensed to companies	2,7	IPCF-CNR	Manufacturers of high-temperature furnaces for high-tech products from the ceramic industry, manufacturer of MW-assisted processes systems

Partners will be involved in identifying and addressing the target groups for CEM-WAVE and also future target groups. A set of measurable indicators will be established to account for the progress during the implementation of CEM-WAVE and to measure the effect/impact after the project ends (in the medium-long run). Additionally, in order to promote the adoption of the project's solutions after its termination, the final exploitation plan will include a socio-economic impact study that will be performed towards the end of the project. The partners will analyse and validate the primary and secondary market potential, and structure a market penetration & development plan accordingly. Preliminary market analyses in CMC and steelmaking sector has been performed and is described in 2.1. Cooperation with regulatory bodies and third-party sales and distribution licensees in the field of ceramic matrix composites, steelmaking and manufacturing will be implemented. This process has already begun through the concept development and pre-project market validation work carried out by the partners within their existing customer bases. The final Exploitation and Business Plan, as part of the PEDR, will be released in month 42. It will be drafted during the project course, according to IPR restrictions and information resulting from:

- The knowledge in the CMC field;
- Evaluation of possible industrial outcome for the research results obtained during this project;
- Estimation of the economic impact of the processes developed under the project;
- Observation of market trends and positioning of project results;
- Market analysis (actual market needs and size in Europe).

CEM-WAVE exploitation team will periodically review current and future industrial CMCs materials to estimate evolving opportunities and threats for the product's positioning on the market and to ensure that exploitation strategies are up to date.

Exploitation Interest and strategy of the Consortium Members: The partners have indicated their exploitation intentions for CEM-WAVE exploitable results. The expected results of individual partners and the intended method of the exploitation are summarized as follows (Table 2.5):

Table 2.5 - *Exploitation strategy*

Partner	Exploitation strategy per partner
UNIPI	Expertise that is gained throughout the project will be used in future projects and future R&D activities; scientific knowledge will be exploited in scientific publications and academic didactical activity; possible participation to initiatives related to a trademark with the consortium.
IPCF-CNR	Expertise gained from the Project for future projects and commissioned R&D activities; scientific publications. Know How from results obtained from the project and technology transfer.
FM	Know How from results obtained from the project for developing and manufacturing High Temperature Microwave Ovens under certain atmospheric conditions for the composite and ceramic industry.
FHF	Exploitation of results in scientific publications and know-how use in future projects. Expansion of the technical portfolio of the institute in the field of ceramic fibre preforms and EBC coatings. Patent of new findings and technology transfer to industry.
UOB	Expertise gained throughout the Project will be used in future projects and R&D activities; scientific knowledge will be exploited in scientific publications and academic teaching activity. Furthermore, UOB will use the project results and expertise gained during the project to expand its portfolio of services offered to industry. Patent of results obtained from the project.
ATL	Use expertise gained from project to broaden portfolio of available ceramic processing equipment to a global market. Market new types of equipment directly to ATL's large existing customer base and to potential new customers through attendance at conferences and exhibitions.
ENEA	Expertise gained from the Project for future projects and commissioned R&D activities; scientific publications. Know How from results obtained from the project and technology transfer.
CNRS	Exploitation of results in scientific publications and know-how use in future projects. Training and consultancy on multiscale MW-CVI modeling. Patent on ceramic inlays and licensing.
POLITO	Expertise gained throughout the Project will be used in future projects and R&D activities; scientific knowledge will be exploited in scientific publications and academic teaching activity. Furthermore, POLITO will use the project results and expertise gained during the project to expand its portfolio of services offered to industry. Patent of results obtained from the project.
IRIS	Further development (productisation and customisation service) of the CMC monitoring systems and software embedding processing optimisation models. Commercialisation thereof. Use of the knowledge generated in further projects related with process control, as well as of the fact new sensors will have been integrated (IR camera, dielectric sensors) in our platform PATBox® wherever relevant for B2B activities.

AMIII	Experience from results in the tests to plan future furnaces and define annealing process limits.
R-Tech	Expertise gained throughout the Project will possibly used as a standard and replicated in future research projects and R&D activities; scientific knowledge will be exploited through dissemination in scientific publications and participating in national and international conferences and exhibitions. The project results and expertise gained during the project will expand the portfolio of services offered to the targeted industries.
KNEIA	As dissemination and IP valorisation expert, KNEIA is interested in take full advantage of the knowledge to be developed in terms of market mapping, and providing full services of technology valorisation in relation with future exploitation of the developed solution.

Foreseen business approach: The exploitation strategy will start with a comprehensive market analysis of the CMC market, which will be conducted among the wide network of CEM-WAVE partners and their close contacts with the CMC community. Several inputs (market latency, market potential, adoption rate or uptake, etc.) from the market analysis will be used for the consecutive analysis, in which an economical evaluation using a "cost-benefit" model will be performed. Consequently, several alternative strategies will be prepared and prioritized choices will be made so as to propose the most appropriate business model, which will hopefully maximize future growth and return on investment for the products developed by CEM-WAVE. Different partners have identified the strong potential of the CEM-WAVE products for the development of a spin-off company, and have expressed their interest to take advantage of it, so specific consideration to this will be made in the Exploitation and Business Plan. During the development of the project, this possibility will be considered at every step taken related to the protection of IPR. Various partners in the team have an excellent background in identifying commercial opportunities in the application areas of CEM-WAVE. Another exploitation option to be explored will be entrusting sales and marketing of the new products developed to an existing international Ceramic composite producer/supplier, since it offers an existing infrastructure. This option must be carefully evaluated and the returns quantified. Whether this or other exploitation strategies that would allow boosting the commercialization of the CEM-WAVE products are selected, IPR management will be considered for assuring the conciliation of the rights and eventual compensations of the IP owners. The project will make use also of different tools to achieve good exploitation of the results including PESTLE-analysis, Porter 5 forces analysis and SWOT analysis. Exploitation plans, outline financial arrangements and any follow-up will be developed further during the project.

CEM-WAVE is intended as a strategic project from the industrial point of view, considering that industry and market needs are the main drivers of development and integration of the proposed Microwave assisted Chemical Vapour Infiltration (MW-CVI) processing technologies. The CEM-WAVE Consortium therefore will address the need of new technical solutions for European companies in the field to stay competitive in the market of CMCs, requiring such products at significantly lower cost: this novel proposed process, while maintaining properties of the currently used CMCs, will extremely reduce the costs for making components, thus lowering costs of CMC components that can access other market sectors such as energy-intensive process industries (e.g. steelmaking), and able to resist to future harsher conditions. Compared with the common expectation of a RIA (generally bringing technology from TRL 3 to TRL5), CEM-WAVE is expected to have a strong impact on the real economy through the deployment of its solutions in a relatively shorter than average time-frame. This is the case since CEM-WAVE is more market-oriented than most RIAs out of a deliberate choice. This approach has been suggested by the results of the preliminary market analysis, by the experience in the CMCs sector of multiple partners and by the appealing conditions of the material under study that, despite being developed in order to be used in future processes where high properties of CMC will be a requirement, can already positively impact current processes in terms of environmental sustainability, energy efficiency, lifetime of equipment. It has namely been estimated that it is most likely best to pursue commercialisation immediately, rather than apply for a second round of public support to research and innovation (for instance through an IA), because the experience of the partners suggests that the ideal time-frame needed to reach high TRLs and ultimately the market is shorter than the one allowed by EU and national funded projects (However, the possibility of a Fast Track to Innovation project to develop specific solutions will be considered as well). The Business Model Canvas [152] methodology from A. Osterwalder has been used in order to define the initial business plan below, hence to clearly define: value proposition; key partners, activities and resources; customer relationships; cost structures, revenue streams, value-delivery channels (Table 2.7). The Canvas in the next table summarizes the business models for the exploitation of our project results.

Table 2.7 *- Business Model Canvas*

Key partners	**Key activities**	**Value propositions**	**Customer relationship**	**Customer segments**
- CEM-WAVE Consortium members - Manufacturing sector - Steelmaking sector - CMCs production specialists and Industry - Scientific community	- Environmental consulting services on steelmaking equipment and CMCs materials and CEM-WAVE manufacturing technologies - Cost evaluation of CEM-WAVE materials and processing - Economic feasibility analysis of CEM-WAVE and market application	- Increased understanding of environmental and cost impacts and their transferences - Decision support on environmental impact for product/process	- Effective dissemination and marketing activities - Organisation of workshops, seminars and training sessions - Online platform for support to clients - Customer-based exploitation strategies - Strategic partnerships	- CMCs industry and composites producers - Steelmaking companies - Manufacturing industries - CMCs research agents

- Investors (CMC, steelmaking and manufacturing sectors)	- Tailored Business strategy: Exploitable Results and Technological Implementation Plan **Key resources** -CMCs solutions -Technical and financial capability -Effective dissemination and marketing plan, presence in the EU R&D community -Client portfolio in the EU market -Databases on: environment, society and market -Continuous R&D activities	- CEM-WAVE systems cost and performance data - Social and socio-economic benefits information - Education and Training - Contribution to International Standards for CMCs products - Better management of team members and cash flow - Overall customer benefit	**Channels** -Website -Academic journals and key magazines -Periodic newsletter -Direct contact with key stakeholders (meetings, email, etc.) -Presence at key international conferences -Communication and Dissemination Plan (as part of this initiative within WP9)	- Engineering and consultancy companies - Entrepreneurs, Spin-off, Star-up
Cost structure - Staff costs - Raw materials - Design and prototyping and adaptation of equipment - Software and database requirements - Travelling expenses for dissemination		**Revenue Streams** - IP Licensing and Royalties by agreement - Direct sales of the products - Services (Operation and maintenance) - Advertises		

The business plan has three main goals:

- to enable aligning CEM-WAVE solutions with the existing market needs (therefore, it has been deemed preferable to draft a preliminary version already in the proposal phase);
- to identify a model(s) for an economically feasible exploitation of the different CEM-WAVE results;
- to favour the identification of a joint solution for joint ownership and exploitation of project results, according to the individual business strategies, in strict correlation with the CEM-WAVE PEDR.

The business plan has been created through the following 4 activities:

A. <u>Eco system analysis and mapping.</u>

The value chain has been preliminarily studied and mapped; the actors and the micro/macro environment variables has been identified; a strong representation of the full value chain, from the sensors producers to the final users of the CMCs materials via the materials developers and IT solutions providers has been selected for participation in the CEM-WAVE consortium. PESTLE and Porter methodologies will be applied for the EU market; such activities, once completed, will be important in order to define the competitive environment in which partners will move to offer the added value of CEM-WAVE products and services. A market analysis has been conducted and it will be completed (WP9) for the CEM-WAVE new products/technologies, focused on EU. The preliminary market analysis results set against the characteristics of the CEM-WAVE solutions (adaptability, low cost, safety enhancement, etc.), have suggested to already identify both initial and succeeding target markets. This is reflected in the sector choices for market deployment: (i) steelmaking industry; (ii) composites producers; (iii) Sensor and AI modeling: Development of new multi-sensing units and AI modelling.

B. <u>Measurement of the global benefit</u>;

Building on the preliminary analysis, a full mapping of the most promising applications of CEM-WAVE solutions will be performed, and an approach to provide the relevant stakeholders with value selected; it will be a hybrid between the (long established) 'project-based system and 'pre-designed (scale) based system' – often described as the 'taxi' and 'bus' systems. Business models using the taxi system create value by interacting with specific clients to solve specific problems (for example consulting firms – such as McKinsey, large law firms, movie makers). Those that utilize the bus system add value by producing 'one-size-fits-all' solutions in a repetitive manner from a standardized format. CEM-WAVE systems will be <u>hybrid</u> in the sense that the business model will be tailored for specific uses and developed in close collaboration with clients and all stakeholders, but at the same time it will target markets whereby, within each one and at times across them, there are ample possibilities of replications, and on the basis of modular building stones (CMCs, sensor types, AI algorithms, etc.).

C. <u>Shaping a business model for sustainable eco-system, on CEM-WAVE product-services offering</u>

For each target market a preliminary business model has been drafted and it will be refined in task T9.3, starting from eco-system analysis and mapping, hence allowing to define how to share the added value deriving from CEM-WAVE products amongst the existing (CEM-WAVE Partners) and new actors (e.g.: Stakeholders involved in the Multi-Stakeholders Platform).

The business models are in each case based on the general CEM-WAVE solution but tailored on the needs and specifications of the target markets listed above. This will include also a full consideration of environmental and economic aspects. The LCA will determine the impact of CEM-WAVE solutions with regard to the environment, with a strong focus on their ability to reduce price, extend equipment lifetime and increase energy efficiency (thus reducing energy-related emissions); the intention being both

maximising the environmental benefits and using their rigorously quantified estimating as a marketing tool. With regard to the economic aspects, the focus will be on the reduction of CMCs cost to users and producers.

D. Definition & Implementation of a disrupting, yet reliable, business plan for the Partners.

Finally, based on the business models adopted, business plan for the project results will be designed, mixing products and services (Task 9.3). This activity will focus on detailing the elements needed to deliver the value proposition to the customer and to achieve the business goals for the new CEM-WAVE processes, products and services. It will include financial, operational and marketing information. With regard to the financial aspects, when bringing an innovation to the market, a most important factor to be considered is the financing of the investment; in the case of CEM-WAVE the up-front investment is not deemed to be large, while the operational capital needed is sizable as most activities are labour-intensive and extensive use of testing facilities is required. Most importantly, the business plan will also define the best approach, or approaches, to commercialisation from a legal and contractual point of view; the most promising routes are considered as of now in-house exploitation by the industrial partners, with the additional possibilities of creating a dedicated company for exploitation (possibly in joint-venture with a large CMCs producer or user). Indeed, an integral part of the CEM-WAVE Business Plan is the Intellectual Property and Knowledge Management described in following paragraph and closely integrated in the CEM-WAVE PEDR.

AMIII USE CASE

AMIII use case is of fundamental importance because, among the energy intensive industries identified in the A.SPIRE consortium, steelmaking is one of the most representative sector in which is needed the reduction of the environmental impact. Obviously, the results obtained in CEM-WAVE can be successfully transferred to other high energy intensive industries where resistance to high temperatures is the key property. The radiant tubes built with the materials developed in this project should substitute the current metallic tubes (casted and Inconel ones). The price of these radiant tubes varies depending on the price of the Nickel, but it could be considered that a target price for a W shape casted radiant tube is 8 k€ and 12k€ in case of Inconel.The average lifetime of these tubes depends on several factors, but could be ranged between four and eight years, despite higher lifetime could be achieved. These values can be used for estimating a range for the cost for the tubes developed with the new material, taking a target lifetime of at least 12 years. An installation cost of 1k€ is also included in this evaluation. The higher cost of the Inconel is compensated by its efficiency: both materials have similar thermal conductivity, but the thickness is case of Inconel is lower (standard 5 mm) than in casted tubes (standard 8 mm) which results in a higher temperature in the surface with the same burner power. As the radiation is the power four of the temperature, an increase of 50-60º C could result in a 20% of improvement in the efficiency in the hottest point. As the temperature distribution in not homogenous in these tubes, a more realistic value could be 5-10% of improvement, so an average of 7% is taken in the calculations. To make the CMC-based components market competitive with respect to current Inconel employed materials, the target cost of the tube, including the improved efficiency factor and the expected increased lifetime of 12 years, should be in the range of 18-36 k€. Other factors that could further reduce CMC costs could be the possibility to achieve higher annealing temperatures or new chemistries to be processed, but it is difficult to evaluate the economic impact of this improvement because it depends on the market demand. Therefore, the exploitation strategy will be defined by the final cost of the radiant tubes and its application to higher temperature cycles. A competitive cost will make a wider deployment to almost all the industrial lines possible, meanwhile a higher cost, but with the possibility of opening new product capabilities, will be limited to the top lines (technologically adapted to the new products).

Table 2.8 - *Estimated cast and Inconel materials from current radiant tube furnaces.*

Material	Estimated cost	Lifetime	Cost (k€/y)	Cost* efficiency (k€/y)	Cost 12 years
Casted	8 k€	4 years	2.25	2.25	26 k€
		8 years	1.12	1.12	13 k€
Inconel	12 k€	4 years	3.25	3.02	36 k€
		8 years	1.62	1.50	18 k€

Intellectual Property, knowledge protection and regulatory issues

Strategy for the management and protection of knowledge and intellectual property rights: To pave the way forward for commercial exploitation and dissemination of the project results, the consortium will implement an integral strategy for the effective **management of the knowledge.** The management of the IPR is not expected to present any especial complexity; the selection of the Dissemination Communication and Exploitation (DCE) Manager (KNEIA) has been carefully made and the coordinator will also contribute to the management of IPR if necessary, based on its resources and competences related with the different exploitation routes envisaged. The project results will be continuously assessed in terms of **novelty, patentability and protection** as part of an effective knowledge management strategy. For the results obtained, ownership, protection and use will be considered in detail. Ownership of patents will be clearly defined and governed by the Consortium Agreement that partners will co-sign. Results shall be owned by the participant generating them. To avoid doubt or confusion over ownership, a protocol of procedures will be defined, which will be followed by each partner to clearly show the development of the generation of knowledge or results. In cases results are generated jointly (i.e. where the separate parts of some result cannot be attributed to different participants), it will be **jointly owned**. The allocation and the terms of exercising this joint ownership will be agreed in **Joint Ownership Agreements**. However, an overriding principle will be that the **participating industry partners are always provided with the access that they need to the results to exploit them in accordance with the needs and strategy of their business.** As such, access rights to

results for implementing the project will be granted on a royalty-free basis. Access rights for use purposes will be granted either **royalty-free**, or on **fair and reasonable conditions** (these conditions will be established in **Exploitation Agreements** that will be negotiated between the requester and granter). Different **potential business models** are considered, that could be conditioned by intellectual property laws of each country.

Table 2.9 - *Knowledge, results, property and exploitation strategy of CEM-WAVE partners.*

Partner	Knowledge/Competences/ Background	Results to which the partner will contribute	Exploitation Interests
FHF	Novel ceramic fibre preforms and technology to handle highly stiff and brittle ceramic fibres. Adjustment of appropriate open porosity to be infiltrated with slurries and by MW-CVI processes in order to obtain dense and high strength matrices.	Ceramic fibre based preforms with adapted porosity for oxide and non-oxide CMCs	Technology transfer by licensing to enterprises with the priority to project partners
FHF	Functional fibre coatings to ensure high MW absorption and to contribute to fibre-matrix bonding and even control bonding (weak to strong) depending on the microstructural requirements. Technology for the coating of ceramic fibres in a continuous and cost-efficient roll-to-roll-process.	Fibre coatings for oxide and non-oxide CMCs	Technology transfer by licensing to enterprises with the priority to project partners
FHF	Development of EBC coating systems with closed porosity and with well adapted CTE to CMC substrates to withstand severe environments with increased lifetime.Technology for the application of the slurry based EBC systems onto tubes.	Component coatings	Technology transfer by licensing to enterprises with the priority to project partners
POLITO	The pressure-less joining process is addressed to all the components made of CMCs which will benefit from the development of a new joining technology, considering the following further aspects: 1. Reproducibility 2. Bonding performance 3. Risk mitigation 4. Savings in effort The most important customer segment is the steel industry, but some of the innovations in joining process might be useful for other markets (i.e aerospace).	Joining process and joining materials developed for oxide-oxide and non-oxide CMCs	Licensing to companies, priority project partners
POLITO	Development of new exploitable manufacturing technologies based on joining of CMCs to metals to enable new design solutions for heating element components, with improved reliability and in-service performance.	Joining process and joining materials developed for dissimilar materials (i.e CMCs to metals)	Licensing to companies, priority project partners
CNRS	Optimized ceramic inlays	Combustor efficiency optimization	Technology transfer by licensing to enterprises with priority given to project partners
R-Tech	Risk Assessment, Environmental impact assessment	Risk management of the project. Assessing the carbon footprint of the chosen materials and the processes.	Consultancy to companies.

FM	Knowledge and IP related to the Development and manufacturing of technically demanding High-Power Microwave Generators 900-930 MHz and 2.4-2.5 GHz, technology for combination such frequencies and IR-Systems for microwave power control and matching	Microwave Technology. Control Unit (regulation of the microwave oven)	Development and manufacturing of technically demanding High Temperature Microwave Ovens under certain atmospheric conditions (normal, vacuum, overpressure) for the ceramic and composite industry
ATL	Knowledge and IP relating to the design, manufacture, operation, and maintenance of microwave enhanced chemical vapour deposition (MW-CVI) equipment.	Optimised design of MW-CVI equipment to carry out the following: 1. Ceramic fibre based preforms with controlled fibre loadings and porosity for oxide CMCs. 2. Oxide CMCs with controlled matrices in terms of composition, microstructure and macrostructure. 3. Fibre coatings for oxide and non-oxide CMCs	Manufacture for sale and support of MW-CVI equipment for the manufacture of oxide and non-oxide CMCs in aerospace and other markets.
UoB	Knowledge and IP related to the construction of oxide fibre preforms and the ability to impregnate them with oxide powder particles via slurries with a view to controlling the resultant fibre loading and porosity, and hence gas permeability. Knowledge of the effect of the process variables on the resultant preform properties and their consequences on the ability to obtain controlled porosity and hence properties in the final oxide-oxide composite.	Ceramic fibre based preforms with controlled fibre loadings and porosity for oxide CMCs.	Technology transfer by licensing to companies and other organisations with priority going to project partners.
UoB	Knowledge and IP related to the ability to fill the porosity in the above preforms with oxide matrices via the process of microwave enhanced chemical vapour infiltration (MW-CVI). Knowledge of the effect of the process variables on the resultant preform properties and their consequences on the ability to obtain controlled porosity and hence properties in the final oxide-oxide composite.	Oxide CMCs with controlled matrices in terms of composition, microstructure and macrostructure.	Technology transfer by licensing to companies and other organisations with priority going to project partners.
UoB	Knowledge and IP related to the effect of different interfacial coatings on the fibres on a) the microwave absorption characteristics and b) the consequent mechanical properties of the composites.	Fibre coatings for oxide and non-oxide CMCs	Technology transfer by licensing to companies and other organisations with priority going to project partners.
UNIPI, ENEA	Knowledge and IP related to the ability to fill the porosity in SiC preforms with SiC matrices via the process of CVI and microwave enhanced chemical vapour infiltration (MW-CVI).	Non-Oxide CMCs with controlled microstructure and macrostructure.	Technology transfer by licensing with priority going to project partners.
ENEA	Knowledge and IP related to the joining of monolithic and composite SiC, for optimising the radiant tube in term of production costs.	Methodology for joining SiC-based materials, for use up to 1600°C	Technology transfer and/or licensing
ENEA/ CertiMaC	Knowledge about the thermophysical and thermomechanical characteristics for a joining mixture suitable for monolithic and composite SiC	Data for thermomechanical and thermophysical modelling	Technology transfer and/or licensing

CNRS/ ENEA/ UoB	Knowledge (microstructural and thermomechanical changes) about the recession in relevant combustion environments (H_2 and H_2/CH_4 mixtures)	Kinetics data for modelling and reliability assessment	Technology transfer and/or licensing
IPCF-CNR	Knowledge and IP related to the design of single-mode and multimode MW reactor chambers, development of techniques for improving energetic efficiency and heating homogeneity, automation of the operating conditions, in-operando analysis of the dielectric properties and of the infiltration stage, monitoring of the structural properties of the produced CMC when in use	MW power coupling and distribution, dielectric and structural properties	Technology transfer by licensing to companies and other organisations with priority going to project partners

Data Management

A variety of research data will be generated/collected in the project including (but not limited to): components characteristics data, processes and product data, customer survey data, etc. A detailed Data Management Plan (DMP) will be prepared by M6 within the activities of WP10 (Coordination and Management). Access will be via the shared area of the project website. A key challenge is therefore to ensure the impartiality and accuracy of any data collected as data can become biased towards an individual or organisation's objectives and secondary data may be misinterpreted if it was collected for a different purpose. To mitigate these weaknesses, we will triangulate between sources where possible, and look for result confirmation across different data/information sources. The project coordinator in collaboration with the project Steering Committee, after the General Assembly gives a precise mandate, will take all the appropriate measures to make the data openly available and usable for third parties for study, teaching and research purposes via licensing or other means. Primary focus in data sharing will be on the data underlying prospective scientific publications ensuring the validation of results presented in publications. At the beginning of the research project the research consortium will decide and agree on the tasks, roles, responsibilities and rights relating to data collection, dataset management and data use. During the project, the data will be used by the research consortium members while after the end of project all requests for further use of data will be considered carefully and whenever possible approved by the principal investigator. Permission for data use will be granted providing there are no IPR or confidentiality issues involved or any direct overlap of research questions with the primary research. Ownership of datasets will belong to the project consortium. Datasets' variables and value names will be constructed following general data processing principles (F.A.I.R. data, i.e. Findable, Accessible, Interoperable and Re-usable data). The common metadata schema Dublin Core will be used to store the data enhancing an easier data discovery and re-use. Possibility for follow-up utilization of the data will be ensured by documenting data collection methods as well as the contents of the datasets. The datasets generated will be archived at the premises of UNIPI for data security reasons and in addition archived in a common and – if feasible – open data repository depending on the decision of the Management Committee. UNIPI is responsible for curating, preserving, disseminating and deleting the datasets in its possession. Retention time for curated datasets is the same as other project materials at UNIPI, by default twenty years. CEM-WAVE will develop and implement open access to the project research results as explained in section 2.2., making the best effort to deposit all peer reviewed research article into online repositories and have open access to the articles with a delay (embargo) of 6-12 months as maximum.

Standardisation as essential tool to reach the market

To ensure long term sustainability and widest possible use of CEM-WAVE outcomes, the project will interact with the appropriate bodies to contribute to standardization as needed during the course of the project and following through the participation of respective consortia members. In particular actions will be directed to Cen, CENELEC, ETSI, amongst others. The standardization and fulfilment of regulations of CEM-WAVE products will be scrutinized and assessed. It is worth to mention that CertiMaC, third party of ENEA participating to the CEM-WAVE project, offers a wide range of services related to R&D, characterisation and **certification f**or traditional and innovative materials and components. CertiMaC is also recognised by the European Commission and by the Italian Ministry of Infrastructure and Transport as **Notified Body no. 2685** in accordance with the Construction Product Regulation - CPR (EU) 305/'11. CertiMaC is also **member of UNI (National Standardisation Body)** and of ECTP (European Construction Technology Platform). The direct involvement of CertiMaC in the CEM-WAVE project will facilitate the future standardization and certification of our new materials.

C) Communication activities

Communication activities are considered to be an important aspect of this project, to ensure that we effectively promote our project and its outputs, as well as its benefits and impacts to the stakeholders that we want to reach. As one of the keys to successfully gaining impact **is through our ability to secure stakeholder 'buy-in'**, i.e. the cooperation or positive participation of our key stakeholders, they will be the target of our communication activities. It is relevant that citizens are aware of the developments and the societal benefits, especially in the context of the positive impacts for CMCs development and sustainability, and the improvement in the production and processes, the economic growth and jobs related to the project implementation, etc. Moreover,

it is important to communicate the societal benefits as the project is benefiting from tax-payers' contributions. Through our communication activities we have an opportunity to ensure that citizens are informed and feel like they are getting value for money.

Table 2.10 *- Communication table*

Target Audience	Communication Objectives	Major channels that will be used
Industry value chains and Scientific Community	To generate industry and scientific community awareness of the project aims and expected impacts and to stimulate interest in future adoption and investment in the project results. The consortium will promote the distribution of the research results, especially to industrial CMC manufacturers and end-users. Prototypical components will be shown on exhibitions in different branches with high industrial participation like the international tradeshow for composites, e.g. the JEC in Paris.	Professional social media channels (thought leadership, blogs, business/case studies (documented around the performance, feasibility and benefits of our product prototypes and their pilot processes). Articles in technical magazines and journals. Participation in tradeshows, conferences and exhibitions CEM-WAVE Workshops and demo days.
R&D specialists	Report about new CMCs materials and their properties	Publications in journals. Conferences with relevant theme as oral and poster presentations.
CMCs industry, Composite Manufacturers, Material Matrix Manufacturers, Ceramic matrix composite distributors and suppliers	Report about new CMCs materials and their properties in relevant applications. To demonstrate the improved capability and efficiency of microwave CVI equipment over conventional CVI that we build currently.	Publications in specialized magazines. Conferences with relevant theme as oral and poster presentations.
Standardisation and regulatory bodies	Report about new CMCs materials and their properties in relevant applications	Publications in journals. Conferences with relevant theme as oral and poster presentations
Steelmaking Industry	To demonstrate the benefits of the new CMCs materials as well as their increased sustainability throughout the value chain	Articles in specialized journals and magazines. Sectorial conferences. Posters/Oral Presentations.
CMC converters	To demonstrate the benefits of being able to monitor the process to optimise the resulting materials	Articles in specialized journals and magazines. Sectorial conferences. Posters/Oral Presentations.
Consumers and citizens	To ensure that consumers and citizens are aware of the project developments and the associated societal benefits, especially in the context of the positive impacts for environmental sustainability (CO_2 emissions), not to mention the favorable impacts for growth and jobs, etc.	Mass media (press releases) Social media (facebook, twitter)

Through *WP9,* we will plan out our communication activities and develop the necessary communication tools (project website, leaflet, posters, social media accounts, etc.) to guarantee that communication activities will be effectively intertwined with the project dissemination effort. Stakeholders and general public engagement and support are crucial to increase the impacts of the project in terms of industrial competitiveness, environmental protection and a transition to the green manufacturing economy in Europe.
To maximise the impact of CEM-WAVE communication activities the communication plan will follow these steps:

1. **Identify and list all possible stakeholders** - the target audiences of our dissemination activities will be used as the starting point (in addition to any other stakeholders that might be considered important). Up to now an important stakeholder on the ceramic market (Sacmi) already manifested its interest about CEM-WAVE results (endorsement letter attached in Annex 4).
2. **Organise our stakeholders** - following a stakeholder analysis, the target audience of our communication activities will be categorised into "communities of support" for helping CEM-WAVE to realise its ambition, goals, objectives and impacts.
3. **Development of the CEM-WAVE brand** - in order to help us in engaging with our stakeholders the principles behind the CEM-WAVE brand (visually identified through its logo) will be further elaborated and documented.
4. **Design and implementation of effective communication campaigns (and supporting tools and materials) -** our aim is to generate understanding within the different communities of support as to what CEM-WAVE is seeking to achieve. We will ensure regular communication to our stakeholders through these campaigns, which will involve the use of emails, face-to-face meetings (in the case of policy-makers), leaflets, posters, the website and social media, etc. A press and media relations campaign will be implemented throughout the project to ensure media interest and coverage. This will include direct briefings for selected key publications whilst providing them with additional materials to ensure that high levels of coverage are achieved. The impacts of these campaigns will be regularly assessed to enable any modifications or improvements to be made to ensure our

communication efforts are effective.

***Table 2.11** - Target groups: The project will develop a specific communication strategy towards each of the stakeholders of CEM-WAVE:*

Stakeholder	CEM-WAVE Benefits and Communication strategy
Steelmaking industries	*Production and use of innovative, sustainable equipment/products for the steelmaking sector: contacted through online forums, mailing, events and congresses, advertising in sector related magazines. Knowledge of the environmental impact of different materials. Risk identification on time will have positive impact to the existing businesses.*
CMCs industry, Composite Manufacturers, Material Matrix Manufacturers, Ceramic matrix composite distributors and suppliers	*Articles in specialized journals and magazines. Sectorial conferences. Posters/Oral Presentations.*
CMC converters	*Articles in specialized journals and magazines. Sectorial conferences. Posters/Oral Presentations.*
Standardisation community	*Project information and dissemination of project results.*
Academia	*Articles in specialized journals and magazines. Sectorial conferences. Posters/Oral Presentations.*
General public	*Information about CEM-WAVE and its achievements. Public website, datasheets creation, marketing campaigns*

3. Implementation

3.1 Work plan — Work packages, deliverables and milestones

3.1.1 Overall structure and Pert Chart

A competitive work plan has been devised in order to deliver on the objectives of the CEM-WAVE project to ensure that it fulfils its goals and vision to catalyse the activities to produce novel materials capable to resist to fluctuation and extreme future conditions and to provide the requirements of the CEM-WAVE use case. KPI have been used as the starting point for planning the work that needs to be implemented within the project. All the actions that are required to meet the objectives of this Research and Innovation Action have been collectively analysed by the interdisciplinary consortium. It was agreed how best to distribute the workload among the various experts by making best use of their relevant fields of knowledge, capacity, infrastructure, resources. This bottom-up planning process has greatly facilitated the task of arriving at a clear, comprehensive and realistic work plan (see Gantt Chart and Pert Diagram below), whereby all partners have ownership of the work plan and its approach. Of a total of 10 WPs aimed at accomplishing the broad scope of this SPIRE topic as previously justified, our R&D work is split into 6 WPs, connected with the one in which a use case will be analyzed, each with clear tasks that will enable the project's objectives to be fulfilled through a coherent structure that will also facilitate project management (WP10), sustainability study (WP8) and the further dissemination and exploitation of the results (WP9).

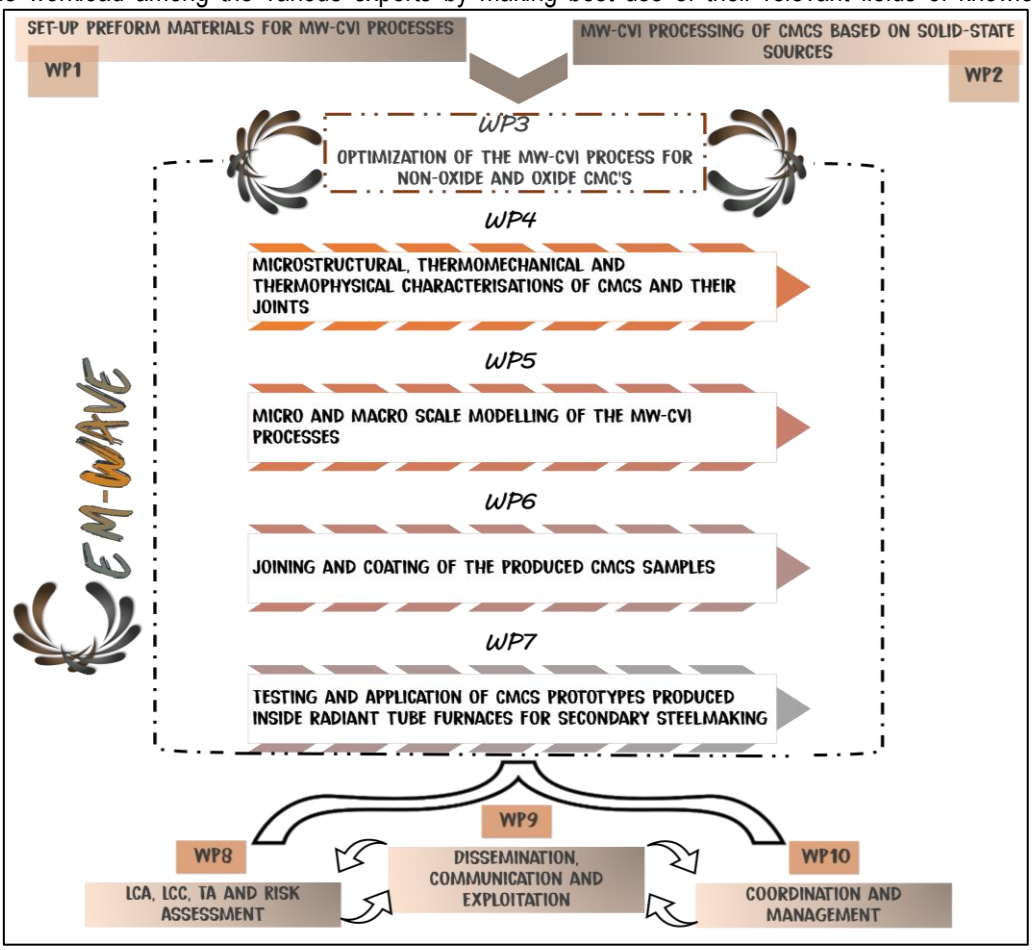
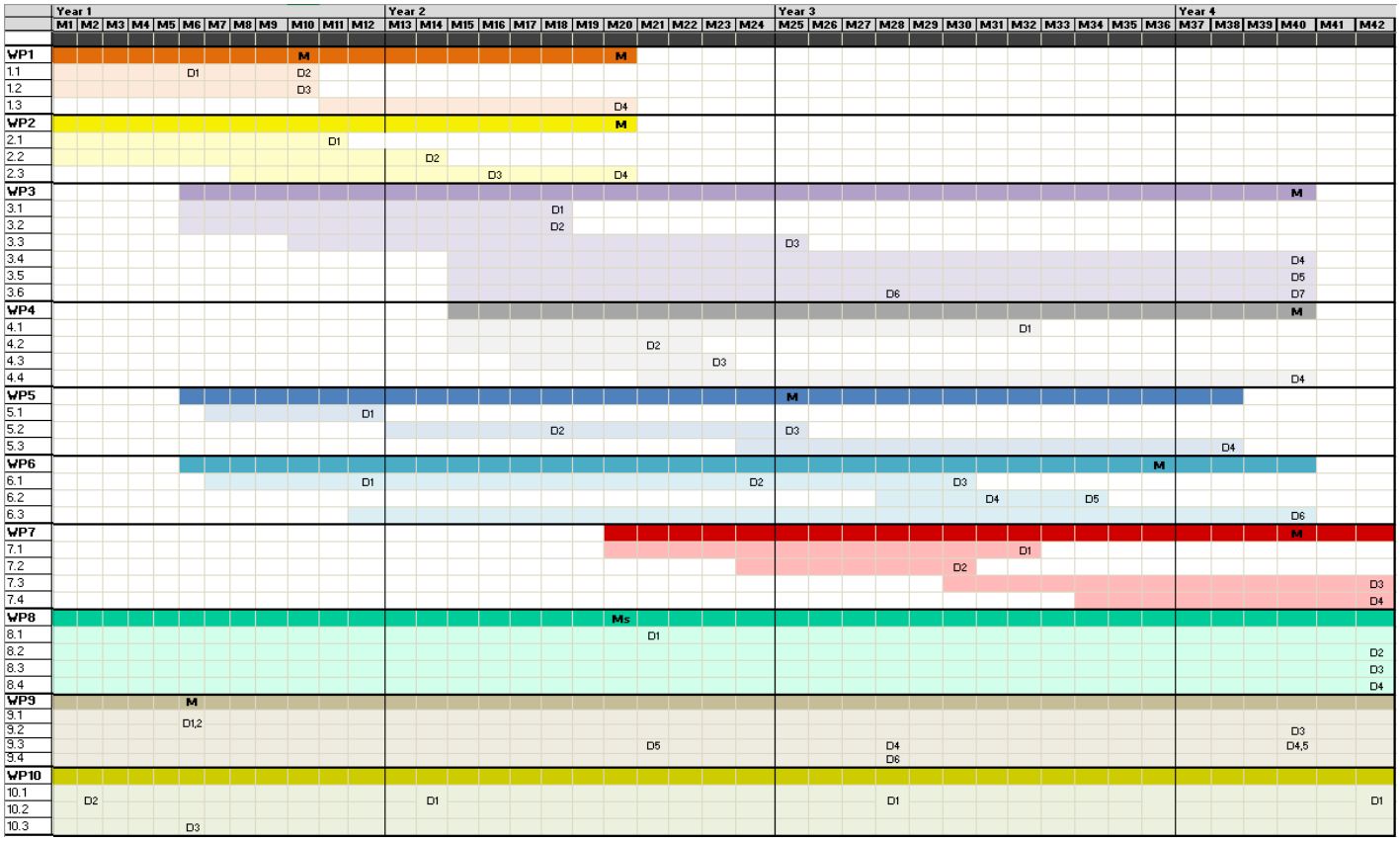


***Figure 3.1** – CEM-WAVE Pert Chart.*

The first research phase (WP1 and WP2) deals with the preparation of the fiber coating and improve the knowledge of the solid-state MW technology. CEM-WAVE heart is characterized by WP3, in which oxide and non-oxide CMCs will be produced wity the implemented MW-CVI technique. All around there are WPs of support about the WP3 activity: AI assisted modeling (WP5), Characterization also in relevant environment (WP4) and joining R&D on CMCs (WP6). Contemporary, the CMC gradually designed, produced and joined will be tested by CEM-WAVE use case work package (WP7). As described before the work plan is encased in a robust environmental and thermoeconomical analysis (WP8), in a fundamental project exploitation, dissemination and communication plan (WP9) and in well-established and efficient project management structure and procedures (WP10). The work plan is linked to the objectives set out in Section 1.1 and the consortium is confident that this work plan is robust to ensure that the project's S&T and innovation goals are fulfilled.

3.1.2 Gantt chart



3.1.3 Detailed work packages description

Table 3.1a *- List of WPs*

WP No.	WP Title	Lead Part. No.	Leader Short Name	Person Months	Start Month	End month
WP1	Set-up preform materials for MW-CVI processes	4	FHF	28	1	20
WP2	MW-CVI processing of CMCs based on solid-state sources	2	IPCF-CNR	69,5	1	20
WP3	Optimization of the MW-CVI process for non-oxide and oxide CMCs	1	UNIPI	118,3	6	40
WP4	Microstructural, thermomechanical and thermophysical characterisations of CMCs and their joints	6	ENEA	75,6	15	40
WP5	Micro and Macro scale modeling of the MW-CVI processes	7	CNRS	25,1	6	38
WP6	Joining and Coating of the produced CMC samples	9	POLITO	36,02	6	40
WP7	Validation of CMC in steelmaking relevant environment	11	AMIII	69,5	20	42
WP8	LCA, LCC, TA & Risk assessment	12	R-Tech	30	1	42
WP9	Dissemination, Communication and Exploitation	13	KNEIA	62,1	1	42
WP10	Coordination and Management	1	UNIPI	35,3	1	42
			Total months	549,42		

Table 3.1b *- Work package description*

WP No.	WP1					Lead beneficiary					FHF		
WP title	**Set-up preform materials for MW-CVI processes**												
Part. No.	1	2	3	4	5	6	7	8	9	10	11	12	13
Short name	UNIPI	IPCF-CNR	FM	FHF	UoB	ATL	ENEA	CNRS	POLITO	IRIS	AMIII	R-TECH	KNEIA
Person Months	5,5	4	0	**13,5**	1	0	1	0	1	0	2	0	0
Start Month	1					End Month					20		

Objectives

- Development of oxide and non-oxide preforms to be processed in WP3 by microwave CVI assisted densification. Moreover, the preform will be used for joining experiments in WP6
- Different fibre preform architectures will be required. The preform thickness will be between 3 mm to 9 mm
- Different fibre lay-up architectures and fibre orientations will be investigated to achieve an appropriate porosity
- For the primary stabilisation of the fibre preforms, binder systems will be developed
- Total porosity and the pore size distribution of the preforms will need to be adjusted

Description of work

Firstly, the technical and specific requirements for applications of both oxide and non-oxide CMCs will be defined in more detail by AM. This includes especially the sample geometries and maximum service temperature. At the beginning of the process chain for the CMC production, both oxide and non-oxide ceramic fibres need to be coated to adjust the interfacial domain between fibre and matrix for the production of Weak Interface Composites, WICs (Task 1.1 & Task 1.2), e.g. non-oxide and oxide CMCs with low matrix porosity. Note that uncoated preforms will be required for the Weak Matrix Composites, WMCs, e.g. for the oxide composites with high matrix porosity. At FHF, ceramic fibres will be coated continuously in a roll-to-roll process by a wet-chemical route. The technology can provide coated fibres in sufficient quantity and in a reasonable time. The coated fibres will be delivered to Task 1.3, where preforms with selected geometries will be developed by techniques such as filament winding or prepreg technology based on 2D fabrics. Three manufacturing loops will be needed to achieve the most suitable preform quality. Commercially available SiC fibres like 2nd or 3rd generation SiC fibres (e.g. UBE-ZMI, UBE-Tyranno SA, NGS-Hi Nicalon, NGS Hi-Nicalon type S) as well as oxide fibres (e.g. 3M-Nextel 610 & 720) will be necessary to allow manufacturing and application temperatures up to 1200°C. The preforms can either be infiltrated first by MW-CVI or be joined first and then infiltrated

Task 1.1 Coating set-up of non-oxide fibres for CMC's production (M1-10)

Non-oxide ceramic fibres will be coated in a continuous roll-to-roll process at FHF. The coating parameters will be optimized to achieve homogeneous coatings with a high degree of coverage and minimized fibre bridging. The latter will be of great importance to have flexible tows after application of the coatings and, in turn, flexible tows are required in Task 1.4 to develop a stable process for the preparation of the preforms by filament winding techniques. An interface system based on BN and PyC with additional layers of SiC or Si_3N_4 on top of BN will be applied on the fibres. The coated fibres will be characterized by various methods such as scanning electron microscopy, X-ray diffraction, chemical analysis, tensile tow testing, etc. in order to evaluate the quality of the coatings and potential fibre damage due to the coating process. The coated fibres will be transferred to Task 1.4 for the preparation of preforms.

Leader: FHF *Participants*: UNIPI, ENEA, IPCF-CNR, POLITO

Task 1.2 Coating set-up of oxide fibres for CMC's production (M1-10)

At FHF, oxide ceramic fibres will be coated by a continuous wet-chemical process for the WICs as described in Task 1.1. Uncoated fibres will also be made available for the work on WMCs. An interface system based on oxides, such as yttria-silicates and phosphates ($Me^{3+}PO^{4-}$) as well as other microwave absorbing coatings, will be developed. To avoid damaging of the fibres due to thermal loads, it will be of great importance to design the coating precursors and the process in such a way that the maximum temperature of pyrolysis of the coatings does not exceed 1200°C. The coated fibres will be characterized using the same methodology used for the non-oxide fibers characterisation in Task 1.1 and then delivered to Task 1.3 for the preparation of preforms.

Leader: FHF *Participants*: UoB, ENEA, IPCF-CNR, POLITO

Task 1.3 Preparation of preforms with selected geometries (M11-20)

According to the specific requirements of AMIII about the preform geometries, at FHF, different kinds of commercially available ceramic fibres will be used to manufacture preforms. Ceramic slurries will be developed to build up the primary ceramic matrix. These slurries containing ceramic particles can be either on a water or a resin basis. The matrix should keep the preform in shape and must exhibit a minimum mechanical strength for the further handling, transportation and MW-CVI processing after thorough drying. Both oxide as well as non-oxide CMCs will be based on continuous ceramic fibres with a certain surface design or coating applied in Task 1.1 and Task 1.2. Ceramic fibre rowings, especially when they have a thicker coating, are very stiff and therefore highly sensitive to classical textile manufacturing. Therefore, the textile processing parameters must be investigated. Since fibre undulation in 2D fabrics often causes fibre breaking during the crimping, the preparation of such preforms will be primarily carried out by using a crimping free fibre winding technique. The diameter of the preforms will be varied by the winding core geometry. The primary size of the cores defines the final geometry of the samples. By using this

method, tubes up to about 500 mm in length and up to 300 mm in diameter can be realized. Wall thicknesses can be also varied from 1-10 mm. In order to minimize tension stresses within the structure, the winding angle of the fibres can be varied from about 30° to 90°. After removing from the textile machine, the preforms will be stabilized by a drying and a curing process. To stabilize the fibre preforms thermally and to adjust the porosity, a high temperature treatment process up to 1150°C or beyond must be accomplished. A suitable pyrolysis programme for SiC_f/SiC and an appropriate sintering process for oxide CMC will be carried out. During their heat treatment, the thermal stability of the selected fibre types has to be taken into consideration. All processing parameters will be investigated and developed by using uncoated ceramic fibres in the 1st loop. In the 2nd loop, coated fibres will be used for all of the WIC non-oxide and oxide composites; uncoated fibres will be used for the WMC oxide composites. In the 3rd loop, tubular preforms with the best suited porosity to the MW-CVI processes and high homogeneity will be manufactured. The processed preforms will be finally checked by non-destructive computed tomography method to detect defects such as macropores, delaminations or cracks within the microstructure. These possible defects will be minimized by iterative process steps.

Leader: FHF *Participants*: UNIPI, UoB, ENEA, POLITO, AMIII

Deliverables

D1.1 (M6): Design freeze for coating process for non-oxide and oxide CMCs (FHF)
D1.2 (M10): Report on coating set-up of non-oxide fibres for CMC's production (FHF)
D1.3 (M10): Report on coating set-up of oxide fibres for CMC's production (FHF)
D1.4 (M20): Report on prepared preforms with different and selected geometries (FHF)

WP No.	WP2					**Lead beneficiary**					IPCF-CNR		
WP title	**MW-CVI processing of CMCs based on solid-state sources**												
Part. No.	1	2	3	4	5	6	7	8	9	10	11	12	13
Short name	UNIPI	IPCF-CNR	FM	FHF	UoB	ATL	ENEA	CNRS	POLITO	IRIS	AMIII	R-TECH	KNEIA
Person Months	5,5	**27**	10	0	1	23	0	0	0	3	0	0	0
Start Month	1					**End Month**					20		

Objectives

- Determination of the dielectric properties of the materials at the temperatures of the infiltration process
- Identification of the optimal conditions for the efficient microwave heating of the samples
- Integration of solid-state sources in the existing MW-CVI reactors
- Definition of an automatic procedure for the tuning of the solid-state sources to the optimal MW frequency

Description of work

The activity of WP2 will start with the determination of the dielectric properties of the oxide and non-oxide samples of interest and, on the basis of these properties, will continue with the identification of the most efficient operational conditions of the MW-CVI reactors. The existing MW-CVI reactors will then be upgraded with the passage from magnetron sources to tunable solid-state sources. The final part of the activity of WP2 will focus on the definition of automatic procedures capable to operate the reactors in the most efficient conditions.

Task 2.1 High temperature dielectric characterization of selected non-oxide and oxide fibres (M1-11)

The complex dielectric permittivity of the non-oxide and oxide fibres of interest will be determined as a function of the temperature up to 1200°C and at a frequency around 2.45 GHz by means of measurement system available at the IPCF-CNR. The measurements will be conducted either in standard atmosphere or in vacuum according to the sample characteristics. Samples composed of different fibres, deniers, etc. and including additives designed to improve the mechanical or dielectric characteristics, will be investigated. The dependence of the dielectric properties as a function of the infiltration level will be investigated as well.

Leader: IPCF-CNR *Participants*: UNIPI, UoB, FM

Task 2.2 Optimization of the electromagnetic heating process inside the MW-CVI reactors (M1-14)

The electromagnetic heating process of the sample inside the MW-CVI reactors will be investigated by means of rigorous numerical modelling based on the software Comsol Multiphysics. The aim of this Task will be to conduct simulations of the reactors in their operational conditions, solving simultaneously and in a self-consistent way the coupled equations governing the electromagnetic and the thermal problems on the basis of the thermal power delivered to the reactor, the dielectric and thermal properties of the sample determined in Tasks 2.1, and the temperature of the reactor walls. The main outcome of the modelling will be the temperature profile in the sample region, as a function of the electromagnetic field frequency, of the balance between the electromagnetic losses in the sample and that in the chamber walls, and of the injected powers. The optimization of the operational conditions of the reactor will be conducted with the goal of improving the uniformity of the temperature distribution in the sample and the efficiency of the microwave heating process. The main control parameter will be the microwave frequency, which will be adjustable thanks to the use of solid-state sources. The numerical modelling will be conducted on the high-performance computing cluster available at IPCF-CNR.

Leader: IPCF-CNR *Participants*: UNIPI, UoB, FM, IRIS

Task 2.3 Integration of solid-state sources and automation of the MW-CVI reactors (M8-20)

The objective of this task is twofold: first, the integration of new, fully tunable solid-state sources in the existing MW-CVI reactors; second, the automation of the resulting system. The integration of the solid-state sources with the reactor chamber will be implemented by means of a coupling system capable of providing high coupling levels (of the order of -10 dB or better) in the typical operational conditions of the reactor loaded with a sample. In order to achieve this result, a suitable coupling structure will be chosen and employed in combination with an efficient commercial stub tuner. Once the solid-state sources have been integrated into the MW-CVI reactors, a procedure will be developed with the aim of automating the tuning of the solid-state source emission. Program routines will be prepared to handle a 'probe and pump' technique, in which the source will first be swept at low power in order to identify the optimal working frequency (the 'probe'); then, a high-power emission of a given duration will be setup at the optimal working frequency (the 'pump'). At the end of this pumping, a new probe sweep will be setup in order to identify the new optimal working frequency, which, in general, changes due to the sample heating. The variation of the optimal working frequency along the infiltration process will provide an indication of the progress of the sample densification.

Leader: FM *Participants*: UNIPI, UoB, IPCF-CNR

Deliverables

D2.1 (M11): Dielectric properties of the samples of interest (IPCF-CNR)
D2.2 (M14): Optimisation of the operational conditions of the MW-CVI reactors (IPCF-CNR)
D2.3 (M16): Integration of the solid-state sources in the MW-CVI reactors (FM)
D2.4 (M20): Automation of the MW-CVI reactors (IPCF-CNR)

WP No.	WP3					**Lead beneficiary**					UNIPI		
WP title	**Optimization of the MW-CVI process for non-oxide and oxide CMCs**												
Part. No.	1	2	3	4	5	6	7	8	9	10	11	12	13
Short name	UNIPI	IPCF-CNR	FM	FHF	UoB	ATL	ENEA	CNRS	POLITO	IRIS	AMIII	R-TECH	KNEIA
Person Months	**41**	14	5	1	16,1	6	0	11,2	2	21	0	0	0
Start Month	6					**End Month**					40		

Objectives

- Screening of the best operating parameters for oxide fibre and CMC production
- Screening of the best operating parameters for non-oxide fibre and CMC production
- Developing new sensors for tuning, in real-time, the processing parameters for CMC production
- Pilot lab-scale production of oxide and non-oxide CMC samples with the new solid-state microwave technology
- Assessment of the amount of time-reduction for oxide and non-oxide CMC production
- Obtaining CMC products with properties at least comparable to conventionally heated CVI processes
- Improving thermal efficiency by introducing a new ceramic inlay

Description of work

For both oxide and non-oxide CMC production, screening of the best operating conditions will be undertaken in two separate pilot plants: the oxide CMCs will be produced by UoB whilst the non-oxide by UNIPI. In parallel, IRIS will select suitable sensors and the related software platform that will be integrated in both pilot plants; these will enable the monitoring, in real-time, of the main process conditions and will greatly assist the optimisation process of the best operating conditions. In parallel, new solid-state microwave generators will be installed in both systems; this will overcome many of the current issues faced when using microwave heating technology. Samples of different geometries, from WP1, will be infiltrated to produce a range of samples for thermo-mechanical characterization (WP4) and the investigation of different joining techniques (WP6). The improvement in thermal efficiency is a key-factor in achieving the primary project goal, hence a new ceramic inlay will be introduced into the CMCs to further improve the energy transfer and hence improve the thermal efficiency of the processing. It should be noted that microwave CVI has already been shown to be at least an order of magnitude faster than conventional CVI and so the goal of the new ceramic inlays is to further enhance this process.

Task 3.1 Research and tuning of the operating parameters for MW-CVI non-oxide CMC production (M6-18)

On the basis of the previous experience at UNIPI on a pilot plant for non-oxide CMC production, the parameters to be tuned and optimized with the implementation of a new energy microwave source (solid-state technology) will be:

- *Operative pressure:* plasma formation must be avoided (which occurs at too low a pressure) whilst, at the same time, guaranteeing that SiC infiltration starts at the sample core without the inclusion of surface porosity;
- *Infiltration temperature:* this must ensure that a stoichiometric SiC deposit is achieved with the correct crystallinity and that undesired secondary reactions are avoided whilst simultaneously avoiding sample damage due to the exothermic nature of the main reaction, which can be triggered if the process temperature is too high;
- *Reagents ratio:* this must be optimised in order to obtain the best results in terms of reaction efficiency (g/h SiC deposition);
- *Microwave power:* the goal is to use the optimal power level from a thermo-economic analysis perspective whilst, at the same time, guaranteeing uniform heating of the sample for the entire duration of the infiltration process. It will be

also important to find the best compromise between the MW power and operative pressure in the required gaseous environment to achieve the optimum infiltration temperature whilst avoiding plasma formation.

Once determined, these operating parameters will be adjusted to the different geometries and sizes of fibre preforms coming from WP1. This research activity will be carried out in the UNIPI pilot plant with the contribution of microwave optimization of IPCF-CNR and FM know-how. The data achieved in this task will be transmitted to WP8 in order to transfer useful information for LCA, LCC, TA and Risk Assessment.

Leader: UNIPI *Participants*: IPCF-CNR, FM, ATL

Task 3.2 Research and Tuning of the operating parameters for MW-CVI oxide CMC production (M6-18)

Oxide-oxide CMCs will be produced by MW-CVI exploting the same benefits expected from the MW heating described in Task 3.1, with the one exception that, unlike silicon carbide fibres, alumina fibres do not absorb readily MW from room temperature. Two approaches will be followed; (i) the introduction of "fugitive additives", which will be utilised for the manufacture of Weak Matrix Composites (WMCs), that will absorb MW helping the first heating of the preforms up to 600°C – 800°C (depending on the chemical composition of the oxide fibres); (ii) application of a coating to the precursor oxide fibres that will fulfil two functions, a) it will provide the required weak interface, for the manufacture of Weak Interface oxide Composites (WICs), and b) it will provide the required microwave absorption characteristics. On the basis of the previous experience at UoB on a lab scale MW-CVI facility used to date for non-oxide CMC production, the parameters to be tuned and optimized with the implementation of a new energy microwave source (solid-state technology) will be fundamentally the same as those being investigated at UNIPI for the non-oxide composite work, i.e.:

- Operative pressure: plasma formation must be avoided (which occurs at too low a pressure) whilst, at the same time, guaranteeing that alumina infiltration starts at the sample core without the inclusion of surface porosity;
- Infiltration temperature: this must ensure that a stoichiometric Al2O3 deposit is achieved (though it is far less likely to be non-stoichiometric than for SiC deposits) with the correct crystallinity and that undesired secondary reactions are avoided whilst simultaneously avoiding sample damage due to the use of excessive process temperatures;
- Reagents ratio: this must be optimised in order to obtain the best results in terms of reaction efficiency (g/h Al2O3 deposition);
- Microwave power: the goal is to use the optimal power level from a thermo-economic analysis perspective whilst, at the same time, guaranteeing uniform heating of the sample for the entire duration of the infiltration process. It will be also important to find the best compromise between the MW power and operative pressure in the required gaseous environment to achieve the optimum infiltration temperature whilst avoiding plasma formation.

Once determined, these operating parameters will be adjusted to the different geometries and sizes of fibre preforms coming from WP1. This research activity will be carried out in the UoB microwave CVI facility with the contribution of microwave optimization of IPCF-CNR and FM know-how. The data achieved in this task will be transmitted to WP8 in order to transfer useful information for LCA, LCC, TA and Risk Assessment.

Leader: UoB *Participants*: IPCF-CNR, FM, ATL

Task 3.3 Development of advanced process monitoring tools (M10-25)

IRIS will define the specifications for monitoring the CMC processing together with UoB, UNIPI, and IPCF-CNR. Suitable tools for measuring, in-situ, temperature and dielectric properties of the CMC at infiltration conditions will be setup and the customised interface to the process designed and manufactured. They will be integrated in the respective processes. The temperature of the CMC sample will be monitored during the infiltration process by means of a thermal camera operating at the temperatures of interest, around 1000°C. The camera will acquire the sample temperature profile looking through a small pipe vacuum sealed by means of a quartz window. The dielectric properties of the sample will be monitored along the processing by means of the optimal working frequency determined in Task 2.3. Indeed, this frequency is defined by the geometry of the reactor chamber and by the shape, size, and dielectric properties of the sample. By means of a rigorous numerical modelling, the inverse problem of finding the dielectric properties corresponding to a given optimal working frequency can be accurately solved. The software to acquire, store and process the data from the monitoring tools will be developed by IRIS on the basis of its PATBox platform. A design of experiments will be prepared in line with tasks 3.4 and 3.5 to validate and calibrate the monitoring. Data driven models will correlate the materials characterisation information during process and process parameters to the final offline quality control of the materials from WP4 with inputs from multiscale numerical models from WP5 as applicable. After an initial training period, these models will be embedded in a decision support tool that will serve 2 complementary purposes: i) initial process set up depending on a target product characteristic and ii) real process control to adjust the process parameters to the optimisation targets based on the data monitored in real time.

Leader: IRIS *Participants*: UoB, UNIPI, FM

Task 3.4 Production of infiltrated non-oxide CMC samples (M15-40)

In the UNIPI pilot plant with the new microwave solid-state source developed in WP2, CMCs will be produced with quality comparable to the materials obtained with traditional CVI process heating but in a much faster time.

SiC_f /SiC disks with a diameter of 100 mm and a thickness of 10 mm will be made in a suitable number to be characterized from both a thermal and mechanical point of view in relevant environments in WP4. Some of the samples produced will be sent to WP6 to act as the basis for the work on joining, whilst separately a specific ceramic inlay will be developed by sintering to

be applied to the CMC preform through joining in task 6.4 with a view to obtaining a further improvement in thermal efficiency. The objective to be reached for the SiC_f/ SiC samples must be at least comparable with the properties of CMCs obtained using conventionally heated CVI (density 2.4 g/cm^3) but in an infiltration time (of ~50 hours), which is considerably less than the current benchmark (~1000 h) thus reducing manufacturing costs significantly and so allowing these materials to be used for new applications.
Leader: UNIPI *Participants*: IPCF-CNR, FHF

Task 3.5 Production of infiltrated oxide CMCs samples (M15-40)
In the UoB lab-scale MW-CVI facility with the new microwave solid-state source developed in WP2, CMCs will be produced with quality comparable to the materials obtained via conventional green forming followed by sintering but with potentially significantly enhanced properties. Al_2O_{3f} / Al_2O_3, Al_2O_{3f} / Al_2O_3 – ZrO_2 and Al_2O_{3f} / YAG – ZrO_2 disks with a diameter of 50 mm and a thickness up to 10 mm will initially be made in a suitable number to be characterized from both a thermal and mechanical point of view in relevant environments in WP4. Subsequently the size of the best samples produced will be increased to 100 mm diameter with similar thicknesses. Some of the samples produced will be sent to WP6 to act as the basis for the work on joining, whilst separately a specific ceramic inlay will be developed by sintering to be applied to the CMC preform through joining in task 6.4 with a view to obtaining an improvement in thermal efficiency. The objective to be reached for the Al_2O_3f / Al_2O_3 samples must be to achieve properties that are at least comparable with the properties of CMCs obtained using conventional green forming and sintering (density 3.4-3.5 g/cm^3), however, since the processing temperature will be significantly lower (<1000°C vs 1200-1300°C) it is anticipated that there will be less strength loss in the alumina fibres and hence the properties of the MW-CVI CMCs will be superior. It is currently unknown how the manufacturing costs will compare, but assessing this will be a goal of the Task.
Leader: UoB *Participants*: IPCF-CNR, FHF

Task 3.6 Design and processing of ceramic inlays for increased thermal efficiency of CMC samples (M15-40)
The thermal efficiency of the tube can be greatly enhanced by the insertion of ceramic inlays that can have the effect of increasing the gas turbulence, the gas/solid thermal contact surface and the radiative heat transfer throughout the tube section. The task will consist in designing optimal inlays in terms of increased heat transfer efficiency without increasing too much the pressure head necessary to have the gases flowing through the pipes. The optimization will be based on computations of heat and mass transfer in 3D structures designed by CAD. Then, ceramic inlays based on the obtained design will be produced.
Leader: CNRS *Participants*: UNIPI, UoB, POLITO

Deliverables
D3.1 (M18): Optimisation of operating parameters for MW-CVI non-oxide CMC production (UNIPI)
D3.2 (M18): Optimisation of operating parameters for MW-CVI oxide CMC production (UoB)
D3.3 (M25): Report on advanced process monitoring tools (IRIS)
D3.4 (M40): Report on production of infiltrated non-oxide CMC samples (UNIPI)
D3.5 (M40): Report on production of infiltrated oxide CMC samples (UoB)
D3.6 (M28): Optimal design of ceramic inlays (CNRS)
D3.7 (M40): Manufactured ceramic inlays (CNRS)

WP No.	WP4					**Lead beneficiary**					ENEA		
WP title	**Microstructural, thermomechanical and thermophysical characterisations of CMCs and their joints**												
Part. No.	1	2	3	4	5	6	7	8	9	10	11	12	13
Short name	UNIPI	IPCF-CNR	FM	FHF	UoB	ATL	ENEA	CNRS	POLITO	IRIS	AMIII	R-TECH	KNEIA
Person Months	39,5	6	0	0	2,6	0	**12**	7,5	12	0	1	0	0
Start Month	15					**End Month**					40		

Objectives
- The materials, derived from WP3, will be characterised both during development and after it, the latter with the aim of fulfilling the industrial application requirements.
- The materials will have to withstand ageing in the expected working conditions. Considering for example, the materials that will find applications as heating elements (e.g. the SiC-based composites).
- Testing of the high temperature emissivity, thermal conductivity and other physico-chemical, microstructural and mechanical characterisation useful for the project.

Description of work
In WP4, physico-chemical, microstructural and mechanical characterisation will be undertaken to show that the new CMC materials that are being developed will perform better than currently used materials, particularly in terms of superior thermomechanical properties. On the basis of this increased performance, the impact of the research programme on the European steel industry, amongst others, will be estimated. Relevant characteristics will be agreed together with steel producer end-user ArcellorMittal (AM). Ceramic oxides have some known limitations when used for high-temperature radiant-heating element applications. These limitations are linked to: (1) low high-temperature bending strength, (2) limited impact resistance at high temperature and (3) limited creep, wear and fatigue resistance. Classical solutions for improving some of these

characteristics is to move from monolithic oxide to non-oxide ceramics, and from monolithic to long-fibre reinforced oxide ceramics, or to long-fibre reinforced non-oxide ceramics. SiC_f/SiC is expected to be an optimised solution for radiant tubes production, however hydrothermal corrosion has to be carefully considered, since methane and hydrogen combustion might induce "recession" that is the condition in which SiC is continuously consumed. EBC could be applied to solve this problem, however it is impossible to produce good EBCs inside a SiC-based radiant tube. WP4 would investigate this type of corrosion to find the threshold between passive and active oxidation, that is the temperature not to be exceeded to avoid SiC consumption.
In WP4, the measure of thermomechanical, optical and thermophysical properties will be carried out for both ox/ox, SiC_f/SiC and joints (WP6), in the case of SiC_f/SiC only in the relevant passive oxidation temperature range. Ageing tests will be performed into static and dynamic conditions (burner rigs).

Task 4.1 Preliminary water corrosion tests and thermomechanical characterisation on CMC samples and their joints (M15-32)
In this task, CNRS will establish the passive-to-active threshold temperature regarding water corrosion, in the range 800-1300°C, in the expected working conditions (simulating biomethane, hydrogen, or biomethane/hydrogen mixtures combustion). This input is necessary to optimise the materials (and in particular SiC based CMC) for their application as radiant tubes, with an internal burner. Room and high temperature mechanical characterisation of CMCs samples will be performed by ENEA and POLITO in accordance with the international standards, e.g. ASTM C1359 & UNI EN 12789. In particular, tensile, 4-point bending, C-ring and flexural creep strength up to 1500°C will be measured, whilst impact tests at high temperature will be performed by CNRS. In this task, POLITO will be mainly involved with the mechanical characterisation of the joined samples produced in WP6. AMIII will contribute by giving technical information about the expected working condition of the radiating elements in their industrial facilities.
Leader: ENEA *Participants:* CNRS, AMIII, POLITO

Task 4.2 Measurement of thermal conductivity and emissivity (M15-21)
Starting from the materials designed in the previous WPs (ox/ox, SiC_f/SiC and the best performing joining solution), a thorough thermo-physical characterization of the solutions developed will be undertaken in this task. ENEA Linked 3rd party: CertiMaC will assess thermal conductivity and specific heat by means of both traditional experimental methods (e.g. heat flow meter) and innovative methods (e.g. Laser Flash Analysis - LFA method) up to 1250°C. This will allow the selection of the best solutions in terms of thermal behaviour with reference to the application. The employment of proper post-analysis methods will enable the evaluation of the different effects such as two-dimensional heat flow, heat dispersion from the sample, effects of radiation for temperatures above 500°C and of heat exchange by convection from the sample to the flow of ambient air around it. The combination between thermal conductivity and specific heat will enable the determination of a fundamental parameter – thermal diffusivity (units: mm^2/s) – a material-specific property that characterises unsteady heat conduction and describes how quickly a material reacts to a change in temperature, thus playing a strategic role in qualifying components / materials behaviour in the framework of thermal exchange problems. Moreover, the emissivity, which is the parameter representing the ability to absorb, transmit and emit infrared energy, will be measured at high temperatures by CNRS by means of spectroradiometers and/or spectrophotometers. AMIII will contribute by giving technical information about thermal characteristics of currently used materials for state-of-the-art heating elements.
Leader: CNRS *Participants:* ENEA Linked 3rd party: CertiMaC, AMIII

Task 4.3 Microstructural, physico-chemical and dielectric characterization of CMC samples and their joints (M18-23)
High temperature thermal conductivity and emissivity are expected to be strongly dependent on microstructure, which, in turn, is dependent on the composition and the CMC production process. Microstructure characterisation will be mainly performed using SEM-EDS, TEM and XRD investigation, which will compare the current traditional materials and the new CMCs. High temperature dielectric characterisation will be conducted by IPCF-CNR, both in support of the MW-CVI process development (Task2.1) and in the assessment of the finite CMC products. POLITO will investigate CMCs and joined CMCs by micro-CT and will perform the microstructural characterization (SEM, EDS, XRD) of the joined samples and post-mortem analysis of mechanically tested joints, thus giving a continuous and fruitful feedback to WP6 to improve joint performances.
Leader: ENEA *Participants:* ENEA Linked 3rd party: CertiMaC, IPCF-CNR, POLITO, UoB, UNIPI

Task 4.4 Measurements of wear resistance, thermal efficiency and accelerated ageing tests (M21-40)
Wear resistance is a relevant parameter for the application and will be evaluated by ENEA at room temperature, using an abrasimeter (and recrystallized alumina as abrasive agent).
In order to estimate the thermal efficiency of the designed solutions FE modelling activity will be carried out by ENEA Linked 3rd party: CertiMaC on the basis of the expected component geometries (AMIII), experimental thermal conductivity and emissivity data. This modellisation activity will be carried out in connection with Task 5.3. The parameters will be employed as inputs for the thermal modelling of the components, which will be carried out under the effective working conditions, in steady state operation and during the worst application scenario in order to investigate in a predictive way the whole behaviour of the innovative new materials in comparison to the current traditional ones in use today. The results of the simulations will be employed to evaluate the heat transfer ability and to improve thermal efficiency of the selected materials.

Accelerated ageing tests will be performed on the selected CMCs in the expected working conditions, followed by microstructural, thermomechanical and thermophysical characterisations (Tasks 4.1-4.3). Micro X-ray computed tomography, µCT (POLITO), will permit the correlation of the changes at the macro level (e.g. in the mechanical characteristics) to microstructural changes. Another relevant parameter that could well change in a critical way upon ageing is the fibre-matrix interface, which will need a specific, TEM-based study; this will be undertaken by UoB using their TEM, STEM and dual-beam FIB. Accelerating ageing tests are the only way to simulate several years of degradation and to ensure that the best materials are proposed for the application in steel production. Lower maintenance costs and energy savings are expected in steel production, as a consequence of improved characteristics of CMCs compared to currently used materials. This represents a relevant part of the project impact, since it affects sustainability of mass production. In order to make these evaluations quantitative, WP4 will collect inputs for subsequent analysis in WP8.
Leader: ENEA Linked 3rd party: CertiMaC *Participants:* CNRS, ENEA, AMIII, POLITO, UoB

Deliverables
D4.1 (M32): Water corrosion tests and thermomechanical characterisation on CMC samples and their joints (ENEA)
D4.2 (M21): Measurement of thermal conductivity and emissivity (ENEA Linked 3rd party: CertiMaC)
D4.3 (M23): Microstructural, physico-chemical characterisation of CMC samples and their joints (POLITO)
D4.4 (M40): Measurements of wear resistance, thermal efficiency and accelerated ageing tests (ENEA)

WP No.	WP5					**Lead beneficiary**					CNRS		
WP title	**Micro and Macro scale modelling of the MW-CVI processes**												
Part. No.	1	2	3	4	5	6	7	8	9	10	11	12	13
Short name	UNIPI	IPCF-CNR	FM	FHF	UoB	ATL	ENEA	CNRS	POLITO	IRIS	AMIII	R-TECH	KNEIA
Person Months	2,5	0	0	0	1	0	0	**18,6**	0	0	0	0	0
Start Month	6					**End Month**					38		

Objectives
- Acquire sufficient data on the material reinforcement architecture at various scales
- Prepare computational meshes for image-based simulations
- Perform two changes of scales to obtain effective geometrical and (tensorial) transfer laws vs infiltration degree
- Perform local infiltration simulations for identification and optimization
- Produce a full simulation of part infiltration by MW-CVI
- Determine optimal conditions for infiltration time, material quality and energy consumption

Description of work
The objective of WP5 is to achieve a multi-scale, multi-physics modelling approach to the MW-CVI technique. Three scales have to be addressed distinctly: the intra-bundle, fibre scale (tens of microns); the geometrical REV scale (a few millimetres) and the part scale. All scales are related and require specific simulations.

Task 5.1. Development of new Image-based numerical modeling using AI methodologies for the MW-CVI process (M6-12)
This task addresses the preparation of realistic numerical simulation domains from the acquisition of X-ray Computerized Micro-Tomographic scans and Scanning Electron Microscope micrographs. The first step will consist of acquiring sufficient image data through X-ray CMT and SEM imaging. Then, image processing tools using Artificial Intelligence will be adapted and used to detect the precise arrangement of the reinforcement architecture in the CMT scans. The result will be a segmented image with an additional field containing all local orientations of the fibre tows. Next, 3D images will be produced from the segmented data and converted into meshes suited for the numerical resolution of the heat, mass and charge transfer model.
Leader: CNRS *Participants:* UNIPI, UoB

Task 5.2 Micro-scale modeling of MW-CVI processes (M13-25)
MW-CVI process modeling requires the solution of conservation equations for mass, momentum, heat and electrical charge. These equations themselves require laws providing the evolution of the local values of geometrical and transfer properties. They are: (i) internal surface area and/or average pore diameter, (ii) ordinary diffusion, Knudsen diffusion, Darcian permeability tensors, (iii) heat conductivity tensor, (iv) tensorial dielectric constant and electrical conductivity (gathered in a complex electrical conductivity). All these properties will be obtained by homogenization (or volume averaging) techniques on elementary computational domains called Representative Elementary Volumes (REV). First, the techniques will be applied to the bundles and will provide effective bundle coefficients as a function of the local fibre density and of the local degree of infiltration. Local infiltration will also be simulated in order (i) to obtain, by comparison with actual experiments, a confirmation of the diffusion/reaction rate ratio, and (ii) to determine optimal infiltration conditions by varying this ratio. Secondly, the same type of computation will be performed again, but at a larger scale: the computational domains will span several tows to that a full material REV be considered.
Leader: CNRS *Participants:* UNIPI, UoB

Task 5.3 Macro-scale modeling of MW-CVI processes (M24-38)

This task will solve mass, heat, momentum, species and electrical charge balances needed for the simulation of the MW-CVI process, in a computational domain that spans the whole part to be infiltrated. All geometry and transfer coefficients will be fed into this model in order to accurately represent the resulting infiltration degree achieved throughout the whole part by the end of the processing run. The model will be run under several conditions in order to estimate the impact of parameter variations on the final quality of the CMC, as assessed by: its pore volume fraction, thermal conductivity and leak tightness, and, for the process efficiency, in terms of total infiltration time, injected microwave power and quantity of source gases used per kg of produced material.
Leader: CNRS *Participants:* UNIPI, UoB

Deliverables
D5.1 (M12): 2D and 3D images at various scales of the fibrous reinforcement and/or fully infiltrated composites (CNRS)
D5.2 (M18): Suitable computational meshes for all computations (CNRS)
D5.3 (M25): Effective laws for geometrical and transfer properties as a function of infiltration degree at mesoscale (CNRS)
D5.4 (M38): Effective laws for geometrical and transfer properties as a function of infiltration degree at macroscale (CNRS)

WP No.	WP6					**Lead beneficiary**					POLITO		
WP title	**Joining and Coating of the produced CMC samples**												
Part. No.	1	2	3	4	5	6	7	8	9	10	11	12	13
Short name	UNIPI	IPCF-CNR	FM	FHF	UoB	ATL	ENEA	CNRS	POLITO	IRIS	AMIII	R-TECH	KNEIA
Person Months	1	0	0	6,02	1	0	2	1,5	**24**	0	0	0	0
Start Month	6					**End Month**					40		

Objectives
- Develop and test novel joints for oxide and non-oxide CMCs
- Develop and test novel coatings for oxide and non-oxide CMCs

Description of work
The WP is structured with a view to: defining the technical requirements for the joints & coatings that will be needed; identifying the joining & coating materials & processes according to the requirements; procuring the materials needed; narrowing the list of materials and processes identified based on the results of the preliminary characterization and resulting performance. The outcomes of this WP are central to the whole project: suitable joints & coatings will be proposed and tested in close collaboration with WP3 and WP4.

Task 6.1 Developing and testing novel joints for oxide and non-oxide CMCs (M6-30)
The joining materials & technologies will be selected according to the technical requirements; once identified and procured, the materials & processes will be narrowed on the basis of the results obtained during the preliminary characterization and the resulting laboratory-scale performance. Special attention will be paid to pressure-less joining technologies in order to provide industry-friendly solutions; glass-ceramics, pre-ceramic polymers and brazing alloys with suitable compositions will be designed and tested as joining materials; the CMC surfaces will also be suitably prepared, to improve joint strength and resulting performance; mechanical machining, plasma structuring and selective matrix removal of the composite matrix and surface modification by chemical reaction will all be studied and used with appropriate joining materials. Laboratory scale samples of CMCs joined to themselves and/or to other ceramics will be prepared and tested at room and working temperature at J-TECH@POLITO laboratory (POLITO). ENEA will contribute to this task by testing a pressure-less joining method for non-oxide CMCs based on the use of a high temperature resistant brazing mixture. The most promising results will be transferred to WP7 for testing under fluctuating and extreme conditions and in industrial setting. Continuous feedback between this task and WP7 will be of paramount importance for the success of this task.
Leader: POLITO *Participants:* UNIPI, UoB, ENEA

Task 6.2 Joining of the ceramic inlays to the CMC samples (oxide and non-oxide) (M28-34)
This task will focus on joining of CMCs to monolithic ceramics by using the materials & techniques developed in Task 6.1 and that are considered as the most promising. Laboratory scale joined components will be prepared and tested at room and working temperature at J-TECH@POLITO laboratory (POLITO). The ENEA brazing method will be also evaluated and eventually tested for non-oxide CMCs. The most promising results will be transferred to WP7 for test under fluctuating and extreme conditions and in industrial setting. Continuous feedback between this task and WP7 will be of paramount importance for the success of this task.
Leader: POLITO *Participants:* UNIPI, UoB, ENEA, CNRS

Task 6.3 Deposition of selected coatings for CMC samples (M12-40)
The coating materials & technologies will be selected according to the technical requirements; once identified and procured, the materials & processes will be narrowed, on the basis of the results obtained during the preliminary characterization and the resulting laboratory-scale performance. Special attention will be paid to PVD, EPD and slurry based coating technologies in order to provide industry-friendly solutions: ceramics, pre-ceramic polymers and functionally graded alloys/structures with suitable composition will be designed and tested as coating materials; the CMC surfaces will also be suitably prepared, to improve coating adhesion and performance: mechanical machining, plasma structuring and selective matrix removal of the

composite matrix and surface modification by chemical reaction will all be studied and used with appropriate coating materials, in tight contact with Task 6.1. Laboratory scale samples of coated CMCs will be prepared and tested at room and working temperature at J-TECH@POLITO laboratory (POLITO); the most promising results will be transferred to WP7 for pre-prototype test under fluctuating and extreme conditions and in industrial setting. Continuous feedback between this task and WP7 will be of paramount importance for the success of this task. At FHF, EBCs will be deposited on selected oxide and/or non-oxide CMC samples via a slurry coating technique. The formulation of the slurries will be developed and the rheological properties will be adjusted for the coating process. The EBCs will be multilayer systems comprising at least one bond coat and one top coat. The EBC will be designed to minimise thermal stresses by adjusting the coefficient of thermal expansion to the CMC substrates. The bond coat needs to be chemically compatible with the underlying CMC substrate. For non-oxide CMCs, FHF has great experience in mullite- and Zr-titanate-based bond coat systems. As top coats, rare earth silicates will be considered. The top coats will be designed to raise the gas-tightness, which is an important requirement for good corrosion protection. This can be achieved by minimizing cracks in the top coat and by designing the microstructure of the coatings with closed porosity. The overall objective is to provide an EBC system that protects the underlying CMCs from corrosion during use.
Leader: FHF *Participants:* POLITO, UNIPI, UoB, ENEA

Deliverables
D6.1 (M12): Report on novel joining materials and techniques for oxide and non-oxide CMCs (POLITO)
D6.2 (M24): Report on characterisation results for novel joints for oxide and non-oxide CMCs at laboratory scale (POLITO)
D6.3 (M30): Final report on characterisation results for novel joints for oxide and non-oxide CMCs (POLITO)
D6.4 (M31): Report on joining of the ceramic inlays to the oxide and non-oxide CMC samples (POLITO)
D6.5 (M34): Final report on joining of the ceramic inlays to the CMC samples (CNRS)
D6.6 (M40): Report on EBC coatings on CMC samples and joint materials (FHF)

WP No.	WP7					**Lead beneficiary**					AMIII		
WP title	**Validation of CMC in steelmaking relevant environment**												
Part. No.	1	2	3	4	5	6	7	8	9	10	11	12	13
Short name	UNIPI	IPCF-CNR	FM	FHF	UoB	ATL	ENEA	CNRS	POLITO	IRIS	AMIII	R-TECH	KNEIA
Person Months	7	6	4,5	0	1	0	3	5	2	14	**27**	0	0
Start Month	20					**End Month**					42		

Objectives
- Testing of oxide and non-oxide CMCs samples produced for future fluctuating and extreme conditions
- Compare their efficiency with current materials employed in radiant tube furnaces
- Monitor the evolution and variation of material properties during the aging test

Description of work
The work of this WP will consist on testing the CMC in simulated relevant harsh environments. a support will be designed for the insertion of the CMCs prototypes and a set of tests will be carried out in the radiant tubes pilot lab of AMIII. The data gathered during these tests will make possible to evaluate the performance of the CMCs in both future possible scenarios and in current processes.

Task 7.1 Evaluation of CMCs samples in relevant environment with future fuel mixtures (M20-32)
To use biogas mixed with clean hydrogen as fuel for combustion in the next future could make sense if materials used are insensitive to main impurities present in biogas (sulfur, siloxane, halogens) and to phenomena caused by the presence of hydrogen. In order to assess CMCs behavior in such working conditions, experiments will be performed on chosen CMCs samples under a similar thermal flux produced by a combustion flame chemically modified to be representative of biogas/hydrogen mixtures. This thermal load will be carried out by means of thermal spray processes such as flame spray or High velocity oxygen fuel spray from a biogas/hydrogen/oxygen mixture to which additives representing the main impurities will be added. Using high velocity flame could be an interesting way to accelerate recession phenomena in order to simulate long duration test and compared solution with low recession rate.
In a first step, working conditions defined by AMIII will be reproduced by means of thermal spray processes and controlled using calorimeter. In a second step, a comparison between CMCs sample before and after thermal loading will be performed at the lab to estimate recession effect. Microstructural and composition studies on aged samples will be performed.
Leader: CNRS *Participants:* AMIII, ENEA, ENEA Linked 3rd party: CertiMaC

Task 7.2 Technical Evaluation of the CMCs applied to the business case (M24-30)
The technical feasibility of using CMCs in the future for the construction of the radiant tubes depends on several factors:
- Resistance to the thermal fatigue. A radiant tube suffers fluctuations in the temperature which can limits its lifetime. The CMC should resist several heating & cooling cycles without damages
- Resistance to the high temperatures. The alloys currently used for the construction of the radiant tubes are limited to work at temperatures below 1000º C. The CMC must resist higher working temperatures.
- Mechanical resistance. The CMC must present a higher strength than current ceramic materials used for the radiant tubes, which are fragile, and it does not resist mechanical impacts.

- The CMC should present a high heat transfer coefficient and have an overall good thermal efficiency

With the data gathered during the trials it will be evaluated if the CMC fulfill all these requirements.
Leader: AMIII *Participants:* ENEA, ENEA Linked 3rd party: CertiMaC, CNRS, POLITO

Task 7.3 Validation of CMCs samples in current CEM-WAVE use case conditions (M30-42)
The samples of CMC will be tested in the radiant tubes pilot furnace of AMIII. This pilot plant is in the facilities that AMIII R&D has inside the ArcelorMittal plant of Gijón (Spain). The pilot plant was designed for testing commercial equipment or developing new one. The samples will be tested in this facility because it can simulate the industrial working conditions. A special support will be designed and constructed to carry out the tests on the CMC samples, because the foreseen diameter and length of the samples are smaller than the radiant tubes' ones. This support will make possible to place the sample inside the furnace chamber and to circulate the combustion gases though it, keeping the gas tightness of the whole device. Once the sample is ready, a test campaign will be carried out in which the working conditions will be varied. Different furnace start-stop cycles will be carried out, combining different heating ramps and target temperatures. The composition of the gas used as fuel will also be varied to evaluate possible effects on the material in case of using alternative fuels to the currently used natural gas.
Leader: AMIII *Participants:* UNIPI, UoB, ENEA, ENEA Linked 3rd party: CertiMaC, CNRS, POLITO

Task 7.4 On line health monitoring of CMCs properties during use (M34-42)
Building on the CMC manufacturing process monitoring strategies developed in WP3, IRIS will tailor the process analytical technology software platform to additionally perform structural monitoring on the walls of the ceramic composite tubes samples intended for use in highly demanding processes. The structural monitoring of the walls of the CMC tubes will focus on the detection of possible cracks and on the degradation of the tube walls. Two combined strategies will be implemented. The first one, based on an IR camera, will monitor the temperature distribution on the surface of the tube, whose alteration during a prolonged use can indicate the presence of a local damage (crack) or a diffused component degradation (reduction in the tube density). A second approach will be based on the electrical resistance of the tube, determined along its length by means of graphite electrode placed at the beginning and the end of the tube, which is sensitive to both the presence of extended cracks, which increase the electrical resistance, and to the density and thickness of the tube walls. The inputs of the two sensors will be correlated in order to minimize the risk of 'false positives'.
Data driven models will correlate the real time data from the IR and electrical sensors with inputs from characterisation in accelerated ageing conditions provided by WP4 and multiscale numerical models from WP5 where applicable. The platform will control the sensors, map their data on a 3D model and apply algorithms to assess the structural health along use (i.e. probability of failure) and recommend optimum maintenance for the CMC user.
Leader: IRIS *Participants:* IPCF-CNR, FM

Deliverables
D7.1 (M32): Report on evaluation of CMC samples lifetime in relevant environment with future fuel mixtures (CNRS)
D7.2 (M30): Technical feasibility of CMC application to the CEM-WAVE use case (AMIII)
D7.3 (M42): Report on validation of CMC samples in current CEM-WAVE use case conditions (AMIII)
D7.4 (M42): Report on CMCs health monitoring during use (IRIS)

WP No.	WP8					**Lead beneficiary**					R-TECH		
WP title	**LCA, LCC, TA & Risk assessment**												
Part. No.	1	2	3	4	5	6	7	8	9	10	11	12	13
Short name	UNIPI	IPCF-CNR	FM	FHF	UoB	ATL	ENEA	CNRS	POLITO	IRIS	AMIII	R-TECH	KNEIA
Person Months	2	1	0,5	0,5	0,5	1	0,5	0,5	0,5	1	1	**19**	1,5
Start Month	1					**End Month**					42		

Objectives
- Assess the environmental impact of CEM-WAVE processes and products
- Model the sustainability of CEM-WAVE including different scenarios
- Risk Assessment of CEM-WAVE processes
- Steer and optimize the economic sustainability of CEM-WAVE processes and products
- Assess of energy and exergy impacts comparing the new developed CMCs production with conventional ones

Description of work
WP8 will investigate in depth the specifications and requirements to allow the environmental, energetical, cost sustainability and risk analysis of CEM-WAVE processes and products. LCA, LCC, TA and RA will be carried out following technical and regulatory requirements.

Task 8.1 Life Cycle Assessment (LCA) (M1-42)
The environmental assessment of the CMCs will be carried out using Life Cycle Assessment (LCA) software suite GaBi following the internationally recognized methodology ISO 14040/14044 and compare the performance with current materials available in the market. Following the LCA methodological framework: Goal and scope definition, Life Cycle Inventory, Life Cycle Impact Assessment, Life Cycle Interpretation, R-Tech will compare the physical and mechanical characteristics of the CMCs, manufacturing processes, and the end-user specifications. The LCA will be developed in two stages; (i) At the material

level with the characteristic information of the material from the scientific partners, (ii) At protocol level with the end-user specific information. The required information includes the functional units with the main inputs/outputs of energy, materials and water in the system boundary. Additionally, the energy consumption, the carbon footprint, the water footprint and the inorganic release of compounds will be assessed.
Leader: R-TECH *Participants:* All partners

Task 8.2 Life Cycle Costing (LCC) (M1-M42)
LCC is an analytical methodology for calculating the total cost of a material, technology and product adopted in the project. This task will provide the cost profiles of the analysed systems and design of materials, and facilitate the choice of the technological route for the production that have a higher economic benefit for the end-users. Similar to LCA, the LCC will be developed in two stages; (i) Devoted to the cost analysis at the material level, (ii) Developed at a product prototype level in collaboration with the end-users, providing a set of economic indicators for further optimization.
Leader: R-TECH *Participants:* All partners

Task 8.3 Thermo-economic analysis and Energy optimization (TA) (M1-42)
The main objective of this task is to assist the optimisation of the energy and costs that hugely influence LCA (Task 8.1) and LCC (Task 8.2). The possible alternatives and criteria (cost and energy) will be defined for different processes in collaboration with involved partners and the Multi-Criteria Decision Assistant (MCDA) tool will be used for evaluating the best alternative in the given scenarios.
Leader: R-TECH *Participants:* All partners

Task 8.4 Risk assessment (Technological and non-technological barriers) (M1-42)
The risk management will be carried out following the ISO 31000/31050, considering the main steps; (i) risk identification, (ii) risk analysis, (iii) risk assessment and (iv) risk evaluation, together with monitoring and communication. The risk identification will be an integral part, updating regularly throughout the project through workshops and surveys. The identified risks, as well as the early warnings for possible emerging risks will be constantly monitored by developing an effective monitoring strategy in collaboration with the whole consortium. The identification of the risks of the project will cover the areas, such as (i) technological risks (specific materials and process) (ii) Market risks (market adoption and acceptance), (iii) IPR issues and exploitation (patents and scientific knowledge management) (iv) Environment, regulations and safety (challenges in Policy, Regulation and Law according).
Leader: R-TECH *Participants:* All partners

Deliverables
D8.1 (M21): Mid-term LCA for CEM-WAVE MW-CVI oxide and non-oxide processes (R-TECH)
D8.2 (M42): Final report on LCA, LCC for CEM-WAVE products (R-TECH)
D8.3 (M42): Report on thermo-economic analysis and energy optimisation (R-TECH)
D8.4 (M42): Report on technological and non-technological barriers (Risk assessment) (R-TECH)

WP No.	WP9					**Lead beneficiary**					KNEIA		
WP title	**Dissemination, Communication and Exploitation**												
Part. No.	1	2	3	4	5	6	7	8	9	10	11	12	13
Short name	UNIPI	IPCF-CNR	FM	FHF	UoB	ATL	ENEA	CNRS	POLITO	IRIS	AMIII	R-TECH	KNEIA
Person Months	9	1	1	1	1,4	1	1,5	1	2	2	1	1,2	**39**
Start Month	1					**End Month**					42		

Objectives
- Enable potential future exploitation of the project results by disseminating them to the relevant stakeholders
- Ensure that the findings of the project are widely communicated to the scientific community and the general public
- Document undertaken and proposed dissemination and communication activities
- Ensure the project results reach the relevant stakeholders who will use and implement them
- Guarantee IPR management according to Guidelines of the European Commission within the Consortium Agreement

Description of work
In this WP, KNEIA will lead the activities of market strategy, dissemination and exploitation of CEM-WAVE products and results. The WP9 workplan is, in fact, encased in a robust project exploitation, dissemination and communication plan that involves a Multi-Stakeholders Platform and the scientific community through workshops and promotional material. Up to now one important stakeholder on the ceramic market (Sacmi) manifested its interest about CEM-WAVE results.

Task 9.1 CEM-WAVE clustering (M1-42)
This task will ensure that CEM-WAVE is networked with all relevant national and international projects, initiatives and networks in EU and internationally, and that the results and impact of the project are effectively fully disseminated to all relevant audiences: industry, policy, research and academic communities. A dissemination plan will be developed, building on the preliminary plan provided in Section 2.2, and the following activities among others will be executed and reported:

- Publishing at least 6 scientific articles in journals and conferences (subject to the IP protection publication policy of the consortium);
- Organizing special sessions and scientific workshops as a part of large conferences with high impact;
- Using the project and its results in teaching, in course materials and as a subject for theses in universities;
- Targeted placement of articles and features in relevant trade and industry focused publications;
- Organisation of the "CEM-WAVE Agenda" at events - an agenda of relevant European and international events that will be planned for the 36-month duration of the project: trade fairs, exhibitions, workshops, seminars, conferences, round tables, etc., that the consortium is already planning to attend or organising as part of their ongoing activities. It is expected CEM-WAVE will capitalise on these existing events (target: 18 events), as well as the organisation of dedicated CEM-WAVE workshops. 2 workshops are envisaged, one in the second year of the project and a Final CEM-WAVE workshop/conference that will be organised in Italy to be able to showcase the project results and impacts at the end of the project; other dissemination and networking opportunities will be the planned demo and training events.
- Networking with other projects is also very important: it will ensure knowledge interchange and reuse between previous and future projects. Importantly, and in line with the call text, we will ensure a close interaction with the other proposals selected for funding under the same or closer topics. The networking plan will take relevant projects within international stakeholders and other EU / national projects into account (some of them have been identified in Section 1.3). It will also focus on networking between consortium partners and their respective areas. Most of the project partners are members of relevant networks, groups and associations -such as the SPIRE Consortium, SUSCHEM, etc. These memberships will be instrumental to promote cooperation and integration with CEM-WAVE. EU resources and supports will be used where applicable throughout the project to support the networking, dissemination and standardisation activities.

The progress of this task will be reported in the Interim and Final Plans for the dissemination and exploitation of the project's results.

Leader: KNEIA *Participants:* All partners

Task 9.2 Communication plan (M1-42)

Subtask 9.2.1. Communication Plan and reporting on communication activities

As discussed in Section 2.2, we will draw up a Communication Plan that will be effective in engaging with our targeted stakeholders during the project in order to engage with them, mobilize their interests and secure their buy-in and support for the project. This plan will be closely linked with the project dissemination activities and it will be following the steps of a) identifying all possible stakeholders; b) organize the stakeholders into "communities of support" for helping us to present our ambition, goals, objectives and impacts; c) develop the CEM-WAVE brand as a powerful marketing and communications tool; and d) design and implement an effective communication campaign. In addition to planning and designing our Communication campaign, communication reports at project interim and at the end of the action (D9.6 and D9.9 respectively) will document the communication activities that have been implemented and all associated communication materials will be annexed.

Subtask 9.2.2 Project website

The project website (D9.1) architecture will be carefully planned to gather the different target audiences who, from the central landing page, will be able to navigate easily to dedicated sub-sites depending on their particular area of interest. The website will serve as a powerful communication tool to all our target audiences (it will also house a restricted access area for partners to store information and facilitate file sharing and internal communication). The public site will present the project brand and will provide well-presented non-confidential project information including the project concept, the partners, core objectives, related literature, easy to understand infographics presenting the strategic, competitive and economic benefits of the project, information about Horizon 2020, the 'CEM-WAVE Green Channel' of open access publications, links to other relevant websites, networks, projects, communities, conferences, etc. The platform will also enable dissemination of the non-confidential results of the project (public reports and deliverables, publications as part of our green open access, etc.). It will also house the on-line training tools and educational materials and will incorporate videos (official project video and eventual videos produced by other partners to describe part of the process). At the end of the project, the relevant public information and results will be available at this site with the aim to maintain the site as a central virtual hub for our post-project activities for at least 10 years.

Subtask 9.2.3 Communication and promotional materials

Project leaflets, posters, project video (D9.3), newsletters and promotional materials (gadgets for tradeshows, etc.) will also be designed in order to support project communication and branding efforts. These materials will be reported and Annexed to the Communication Reports (D9.6 and D9.9).

Subtask 9.2.4 Social / digital media channels

This subtask will capitalise on the tremendous potential of reaching some of our targeted audiences through social media channels: a social media grid will be drawn up; it will plot the most appropriate social media channel/sites for targeting specific audiences. Appropriate pages will be designed for each channel and a content maintenance plan will be put in place for managing the streams of information across these channels to secure and maintain followers. Social and digital media will be particularly powerful in helping to create 'communities of support' for the project. Community creation will be achieved via tools such as: Google hangouts, Reddit, Pinterest and other social media networks such as Facebook, Twitter, LinkedIn and YouTube. The project will create interest on social media and digital platforms by using visual media, videos, animations, icons

and info-graphic imagery, and importantly, mobile enabled content and richer content experiences for users of any of the digital platforms.
Leader: KNEIA *Participants:* All partners

Task 9.3 Exploitation of results and Business model development (M1-42)
For each of the key exploitable results, a business plan will be drawn up. A draft of these Business Plans will be delivered at M24 (D9.4) and final Plans in M36 (D9.7). Built on the information already provided in Section 2.2, the Business Plans will be structured as follows:

- The planned key exploitable results and their expected key applications: development status, facts and figures that facilitate the assessment of the potential impact, differences from existing competing products/services, main IPR issues to be clarified in the consortium agreement;
- The exploitation team: partners involved in the exploitation activities providing information on their previous experience and business interests, complementing external resources if not in the consortium;
- The market: the potential geographical and economic size of the market, market trends, main competitors and competitive advantage;
- The business model: marketing strategy, main clients; following inputs from business models in terms of ceramic matrix composites and manufacturing technology market supply;
- The financial projections: sales forecasts, investment needed for covering the costs, additional funding that will be identified through e.g. ESIF, national programmes, private funding sources etc., complementary or parallel projects;
- The commercialization roadmap: the steps planned before the product is ready for the market (i.e. proof of concept, prototyping, demonstrations of technological performance and cost effectiveness field trials, pilots, validation and standardization issues, regulatory requirements).

Leader: KNEIA *Participants:* All partners

Task 9.4 Multi-Stakeholders Platform (M1-42)
This task will be directed towards community building and stakeholder engagement through the creation of an ad hoc online platform, which will constitute both a stakeholder meeting point and a knowledge sharing place. The project will develop a Multi-Stakeholders Platform (MSP) to ensure the engagement, inclusion and active participation of all relevant stakeholders and their diverse perspectives. The stakeholders involved will foresee the engagement of representatives of research centers, economic and social actors: private companies engaged in several strategic EU applications (furnace industry for steel production, aerospace and energy fields), together with legislators, regulators standardisation bodies and industrial participants. The MSP will also be an instrument for knowledge sharing and engagement with other successful projects under the Horizon 2020 call LC-SPIRE-08-2020, and other H2020 and future Horizon Europe projects. The role of this platform will be to collect the information provided by industry stakeholders identified during the project in order to define common development campaigns aimed to unlock the potentials of CEM-WAVE strategies in different EU Countries. This platform will be nourished through reports, publications, press briefings and newsletters on project results and other relevant information. In particular, the MSP will define a set of relevant topics influencing the take-off of CEM-WAVE technologies and solutions. Among them, a special focus will be dedicated to: standardization issues, supply chain facilitating measures and sustainability criteria development for CEM-WAVE products. The involvement of stakeholders and the valuable inter - communication management strategy will be a crucial aspect to be carried out by all project partners. The active collaboration of CEM-WAVE identified actors will be the basis to achieve the final target. An online Business Impact Forum will be established, as part of the MSP, gathering a wide range of stakeholders with a potential interest in implementing the project outcomes and where a return on investment is feasible. This forum will contribute to better focus our business planning, strategies and post-project exploitation routes.
Leader: KNEIA *Participants:* All partners

Deliverables
D9.1 (M6): Project Website (KNEIA)
D9.2 (M6): Communication Plan and Report, to be updated every reporting period (KNEIA)
D9.3 (M40): Project video (KNEIA)
D9.4 (M28 and M40): Business Plans - following robust business plan templates and guidelines (KNEIA)
D9.5 (M21 and M40): Plan for the dissemination and exploitation of the project's results (KNEIA)
D9.6 (M28): Multi-Stakeholder Platform (KNEIA)

WP No.	WP10					**Lead beneficiary**					UNIPI		
WP title	**Coordination and Management**												
Part. No.	1	2	3	4	5	6	7	8	9	10	11	12	13
Short name	UNIPI	IPCF-CNR	FM	FHF	UoB	ATL	ENEA	CNRS	POLITO	IRIS	AMIII	R-TECH	KNEIA
Person Months	**23**	1	1	1	0,8	1	0,5	1	1	2	1	1	1
Start Month	1					**End Month**					42		
Objectives													

- Coordinate and supervise the project research and innovation activities to be carried out according to the work plan, also to monitor and ensure quality and timing of project deliverables;
- Carry out the overall administrative and financial management and reporting of the project;
- Manage contacts with the EU Commission and establish effective internal and external communication;
- Prevent and resolve possible conflicts, ensuring that risk management is an integral part of the project management strategy.

Description of work

WP10 will: coordinate and monitor research activities, administrative and financial management; coordinate the reporting and deliverables production and will address the resolution of possible conflicts. In addition, it will continuously ensure contacts with the EC and compliance with the procedures described in the Grant Agreement and in the Consortium Agreement through participation in EC project reviews and regular internal review of periodic reports. UNIPI will be in charge of the administrative coordination of the CEM-WAVE project and lead this WP. The WP leaders (FHF (WP1), IPCF-CNR (WP2), UNIPI (WP3), ENEA (WP4), CNRS (WP5), POLITO (WP6), AM (WP7), STW (WP8), KNEIA (WP9), UNIPI (WP10)) will support the coordination and project management, *e.g.* organize and host meetings and will participate in the Steering Committee (SC). The Innovation Management Team (IMT) leader and the IPR Manager as well as the members of the Communication and Dissemination Team (CDT) will be among the staff member of institutions of the General Assembly as mentioned in section 3.2.1.

Task 10.1 Coordination, project meetings and reporting (M1-42)

Subtask 10.1.1 Establishment of the management guidelines

UNIPI will establish the Management Guidelines that sets out all reporting requirements, deliverable sign-off procedures, templates for all project documents, meeting schedules, and participants' roles and responsibilities. It will allow setting up common rules that will be applied by all.

Subtask 10.1.2 Coordination and reporting

The Project Coordinator (CO) will organize, implement, and follow-up a kick-off meeting and face to face meetings (two per year). The CO, assisted by the Project Management Team (PMT) and the hosting Partner, will convene and manage the project meetings. The progress and results achieved by the project will be reported in the periodic activity reports required by the Funding Agency under the supervision of the coordinators and WP leaders. If necessary, recovery actions will be identified and activities revised accordingly.

Subtask 10.1.3 Organization of Kick-off and periodic meetings

The CO will organize, implement and follow-up a kick-off meeting and yearly face to face meetings (two per year). The CO, assisted by the PMT and the hosting Partner, will convene and manage the project meetings. The task will include preparation of related documentation (*e.g.,* agendas, invitations, documents and reports, minutes, and action lists, *etc.*) and securing location of meeting places/catering. The kick-off meeting will be held early (possibly by Month 1) to familiarize all partners with the structure and operation of the consortium, their detailed activities for the following period. The periodic General Assembly and Steering Committee (SC) meetings will regularly bring members of project together to provide guidance on the project implementation. Additionally, they will represent the opportunity to solve any issues impeding the progress of the project and to approve the appropriate contingency plans. Additional technical meetings through Skype, teleconference and other web-based means, as well as technical visits will be convened when and where necessary.

Subtask 10.1.4 Communication and exchange with the European Commission

Communication and exchange with the EC will be guaranteed by the Project coordinator, who will represent the only interface with it. She will act on behalf of and in agreement with the members of the Consortium and will report to the SC and the General Assembly any request by the Agency. The coordinator will inform and dialogue with the EC in relation to any opportunity, risk and problem, which may arise internally and during the project implementation.

Subtask 10.1.5 Operational management of project boards

The objectives and fields of action of each Governing body are described in section 3.2.

Leader: UNIPI *Participants:* All partners

Task 10.2 Overall legal, Financial and Administrative management (M1-42)

The CO, assisted by the PMT, will supervise and guarantee coherence among project activities and procedures described in the Grant Agreement and the Consortium Agreement. The CO will lead discussion on amendments and revisions of these Agreements if necessary.

Leader: UNIPI *Participants:* All partners

Task 10.3 Coordination of Knowledge and Innovation management activities (M1-42)

The CO, assisted by the Project Management Team, will manage the EC contribution and distribute funds among partners according to the rules established in the Grant Agreement and the Consortium Agreement. Financial statements, relevant certificates (if necessary) and project reports will be collected and submitted to the EC. The PMT will guarantee assistance to individual partners on administrative and financial issues. The CO, in cooperation with the SC, will carefully review the results of the project and make them available to the competent project boards listed above and in section 3.2. Results will be communicated and shared in accordance with Grant Agreement and Consortium Agreement provisions, the Plan for

Exploitation and Dissemination of Results ensuring their complementarity. Actions under this task will be carried out in close cooperation with the IPR Manager, paying attention to monitoring, management and exploitation of those. In terms of Data management, the task will include the definition and put into practice of activities related with guaranteeing the accuracy, standardization and security of data, but also protection of scientific data, commercialisation and Intellectual Property Rights (IPR). This task will be integrated in the execution of the scientific/technical tasks, and the results included in the deliverables of the technical WPs, except aspects in relation with commercialization and IPR, that will be tackled in the Plan for Communication, Dissemination and Exploitation. *Leader:* UNIPI *Participants:* All partners
Deliverables **D10.1 (M14, M28, M42):** CEM-WAVE interim and final reports (UNIPI) **D10.2 (M2):** CEM-WAVE Project Management Guide (UNIPI) **D10.3 (M6):** Data Management Plan (UNIPI)

Table 3.1 c *- List of Deliverables*

Del. No.	Deliverable Name	WP No.	Leader	Type	Diss. level	Del. date (month)
D1.1	Design freeze for coating process for non-oxide and oxide CMCs	WP1	FHF	R	CO	6
D1.2	Report on coating set-up of non-oxide fibres for CMC's production	WP1	FHF	R	CO	10
D1.3	Report on coating set-up of oxide fibres for CMC's production	WP1	FHF	R	CO	10
D1.4	Report on prepared preforms with different and selected geometries	WP1	FHF	R	CO	20
D2.1	Dielectric properties of the samples of interest	WP2	IPCF-CNR	R	CO	11
D2.2	Optimisation of the operational conditions of the MW-CVI reactors	WP2	IPCF-CNR	R	CO	14
D2.3	Integration of the solid-state sources in the MW-CVI reactors	WP2	FM	R	CO	16
D2.4	Automation of the MW-CVI reactors	WP2	IPCF-CNR	R	CO	20
D3.1	Optimisation of operating parameters for MW-CVI non-oxide CMC production	WP3	UNIPI	R	CO	18
D3.2	Optimisation of operating parameters for MW-CVI oxide CMC production	WP3	UoB	R	CO	18
D3.3	Report on advanced process monitoring tools	WP3	IRIS	R	CO	25
D3.4	Report on production of infiltrated non-oxide CMC samples	WP3	UNIPI	R	CO	40
D3.5	Report on production of infiltrated oxide CMC samples	WP3	UoB	R	CO	40
D3.6	Optimal design of ceramic inlays	WP3	CNRS	R	CO	28
D3.7	Manufactured ceramic inlays	WP3	CNRS	R	CO	40
D4.1	Water corrosion tests and thermomechanical characterisation on CMC samples and their joints	WP4	ENEA	R	CO	32
D4.2	Measurement of thermal conductivity and emissivity	WP4	ENEA Linked 3rd party: CertiMaC	R	CO	21
D4.3	Microstructural, physico-chemical characterisation of CMC samples and their joints	WP4	POLITO	R	CO	23
D4.4	Measurements of wear resistance, thermal efficiency and accelerated ageing tests	WP4	ENEA	R	CO	40
D5.1	2D and 3D images at various scales of the fibrous reinforcement and/or fully infiltrated composites	WP5	CNRS	R	CO	12
D5.2	Suitable computational meshes for all computations	WP5	CNRS	R	CO	18
D5.3	Effective laws for geometrical and transfer properties as a function of infiltration degree at mesoscale	WP5	CNRS	R	CO	25
D5.4	Effective laws for geometrical and transfer properties as a function of infiltration degree at macroscale	WP5	CNRS	R	CO	38
D6.1	Report on novel joining materials and techniques for oxide AND non-oxide CMCs	WP6	POLITO	R	CO	12
D6.2	Report on characterisation results for novel joints for oxide and non-oxide CMCs at laboratory scale	WP6	POLITO	R	CO	24
D6.3	Final report on characterisation results for novel joints for oxide and non-oxide CMCs	WP6	POLITO	R	CO	30
D6.4	Report on joining of the ceramic inlays to the oxide and non-oxide CMC samples	WP6	POLITO	R	CO	31
D6.5	Final report on joining of the ceramic inlays to the CMC samples	WP6	CNRS	R	CO	34
D6.6	Report on EBC coatings on CMC samples and joint materials	WP6	FHF	R	CO	40
D7.1	Report on evaluation of CMC samples lifetime in relevant environment with future fuel mixtures	WP7	CNRS	R	CO	32

D7.2	Technical feasibility of CMC application to the CEM-WAVE use case	WP7	AMIII	R	CO	30
D7.3	Report on validation of CMC samples in current CEM-WAVE use case conditions	WP7	AMIII	R	CO	42
D7.4	Report on CMCs health monitoring during use	WP7	IRIS	R	CO	42
D8.1	Mid-term LCA for CEM-WAVE MW-CVI oxide and non-oxide processes	WP8	R-TECH	R	CO	21
D8.2	Final report on LCA, LCC for CEM-WAVE products	WP8	R-TECH	R	CO	42
D8.3	Report on thermo-economic analysis and energy optimisation	WP8	R-TECH	R	CO	42
D8.4	Report on technological and non-technological barriers (Risk assessment)	WP8	R-TECH	R	CO	42
D9.1	Project Website	WP9	KNEIA	DEC	PU	6
D9.2	Communication Plan and Report to be updated every reporting period	WP9	KNEIA	R	CO	6
D9.3	Project video	WP9	KNEIA	DEC	PU	40
D9.4	Business Plans - following robust business plan templates and guidelines	WP9	KNEIA	R	CO	28 & 40
D9.5	Plan for the dissemination and exploitation of the project's results	WP9	KNEIA	R	CO	21 & 40
D9.6	Multi-Stakeholder Platform	WP9	KNEIA	DEC	PU	28
D10.1	CEM-WAVE interim and final reports	WP10	UNIPI	R	CO	14, 28 & 42
D10.2	CEM-WAVE Project Management Guide	WP10	UNIPI	R	CO	2
D10.3	Data Management Plan	WP10	UNIPI	R	CO	6

3.2 Management structure and procedures

3.2.1. Project management structure

CEM-WAVE is a research project that will run for 42 months, bringing 13 partners from 5 countries in close collaboration. The dimension of the consortium as well as the multidisciplinary and cross-functional nature of the project requires a hierarchical and robust management structure in order to ensure a smooth workflow.

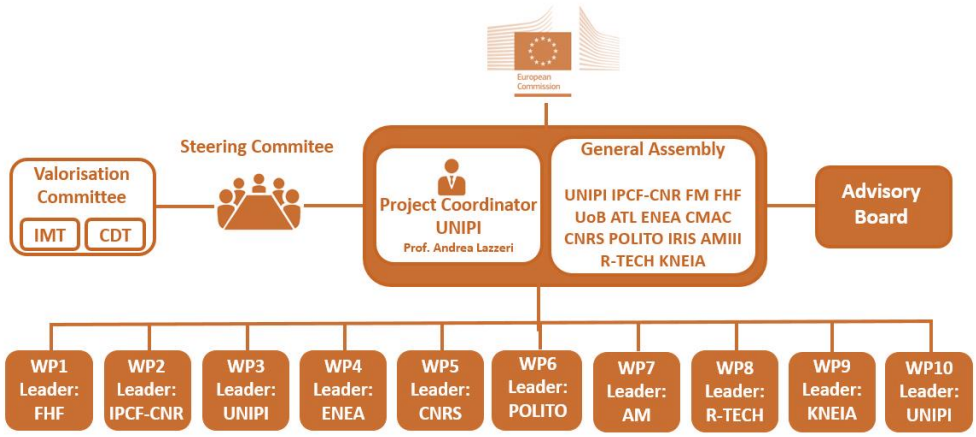


Figure 3.2 *- CEM-WAVE project management structure.*

The **Project Management Team** will lead the administrative and financial issues, providing support to the **Project coordinator** and the **Steering Committee**, which will handle the day to day procession of tasks. Fundamental decisions will be taken by the **General Assembly** formed by one representative of each consortium member. The **Valorization Committee** will be specifically in charge of boosting the market uptake while ensure dissemination does not jeopardize IP protection and exploitation interests.

Project Coordinator (CO)

Prof. Andrea Lazzeri from UNIPI will be the Project Coordinator. He will take up the administrative/financial and technical/scientific management of the project. As CO he will act as the contact point of the project with the European Commission, and will be responsible of resolving any possible administrative/financial, contractual or technical/scientific issue or misunderstanding amongst the beneficiaries. His responsibilities will include:

- ✓ Monitoring of the project proper implementation;
- ✓ Act as intermediary for all communications between the beneficiaries and the EU;
- ✓ Submit any information, deliverables and reports to the EC previously gathered from the consortium members;
- ✓ Ensure all payments are made to the beneficiaries according to the GA and the CA, and organize the project finances;
- ✓ Organize and chair the GA and the Steering Committee meetings;
- ✓ Control the overall project schedule and objectives maintenance, establish and monitor progress indicators and identify and manage associated risk;
- ✓ Maintain regular contact with the Steering Committee members and ensure the execution of decisions taken by the General Assembly;
- ✓ Report the project progress and risks to the EC and the General Assembly;

The project coordinator will be assisted by the Project Management Team, formed by personnel of UNIPI. This team will support also the Steering Committee. They are responsible for the day-to-day management of the H2020 project. They are responsible for the organization of meetings, coordination of the reporting, serving as helpdesk for queries by the consortium and any other task defined in the related WP (WP10).

Steering Committee (SC)

The SC consists of the Work-package leaders (WPL) and forms the communication and decision platform concerning scientific and technical demands of the work plan chaired by the CO. The SC will provide excellence to the scientific management of the project but also reflect both an interdisciplinary and inter-sectoral approach to the overall research objectives of the project. The SC will provide strong support to the CO for the overview of the project and for monitoring progress, making strategic decisions relating both to the research and to its dissemination, and for ensuring effective communication among the participating institutions. The SC will meet at least monthly by electronic means. The responsibilities of the SC include:

- ✓ Monitoring progress against the project management plan;
- ✓ Reviewing and approving changes made to project resource plan, schedules, scope, goals, cost estimates, etc.;
- ✓ Making strategic decisions regarding the prioritization of project deliverables and approving interim deliverables;
- ✓ Reviewing and approving the project development strategy;
- ✓ Reviewing and suggesting solutions for the issues critical to project success;
- ✓ Resolving conflicts between stakeholder groups.

Work-package Leaders (WPL)

In addition to its membership to the SC, WPL have the responsibility for the coordination of the technical work of the group associated with the related WP. In this role, each WPL will be assisted by all participants of that WP. The WPL will have the following tasks and responsibilities:

- ✓ Monitoring of the progress of the technical work on a day-to-day basis in the work package, including scientific managing and reviewing process, identifying risks and proposing appropriate actions to the CO and the SC;
- ✓ Submit and interim semestral reports to the CO regarding task progress;
- ✓ Preparation of deliverables and transmission of documents to the CO and the SC;
- ✓ Transmission of deliverables of the partners within the work package to the CO;
- ✓ Monitor the execution and submission of deliverables in the WP to the CO;
- ✓ Work together with the Valorization Committee to enhance dissemination and exploitation of results;
- ✓ Decide on scientific and technical aspects according to the WP;
- ✓ Promote communication amongst other WPL.

General Assembly

The General Assembly includes one representative from each partner and is chaired by the CO. The assembly meets every six months as a regular basis, although extraordinary meetings can be required by the CO. By default, any decision can be taken by a majority of 2/3 of the voters, and to vote there should be at least 2/3 of the members present or represented. The decisions made by the General Assembly are mainly on proposal by the SC. The General Assembly decides on high-level matters relating modifications and amendments of the work plan and the work packages, acceptance of new parties as well as the exclusion of parties, budget related aspects, modification of the Consortium Agreement and premature completion/termination of the project

Valorisation Committee

The Valorisation Committee is the body in charge of making the consortium to take the best advantage in terms of market of the project and its results. The committee is constituted of two distinct teams, namely the **Innovation Management Team (IMT)** and the **Communication and Dissemination Team (CDT).**

IMT - The innovation in the CEM-WAVE project will be driven by end-user needs to fully address the expectations of the industry clients. The IMT is constituted of the consortium representatives having a deep knowledge of the targeted markets and the specific environment of the industrial sectors concerned. ArcelorMittal and FM will be the pillars of Innovation Management Team in the CEM-WAVE project, although the complete composition will be defined at an early stage of the project. The IMT will manage the implementation of the innovation cycle by pulling R&D activities towards marketable products and technologies and the creation of new business opportunities. The IMT will guide R&D activities by giving feedbacks to ensure that the innovation cycle leads to successful commercial applications. Key exploitable results (KERs) will be identified by the IMT and key strategic issues, such as IPR protection mechanisms, IPR sharing and routes to exploitation will be discussed extensively during meetings of the IMT. The IMT will support partners who have generated KERs to prepare business plan canvas and subsequent business plan draft. The IMT will work under the lead of Dr. Ciro Avolio of KNEIA who is the designated IMT leader in the CEM-WAVE project. He is the right person to stimulate innovation with and for industrial partners within the consortium. The project will have also an IPR Manager from UNIPI. With an extensive experience in IP management and exploitation, the manager will assist the partners in relation with the protection of the developed knowledge (patent search, patentability study, writing, IP valuation and contractual aspects)

CDT- The Communication and Dissemination Team will coordinate all actions related to dissemination of project information and communication of project progress and results towards the stakeholder groups as well as to the general public. It will be responsible to agree upon the relevant strategy, monitor and evaluate the performed dissemination activities and their impact and

adjust the established strategy accordingly. Although all partners will actively participate in dissemination and communication activities, it has been decided that members of CDT will be UNIPI (as Coordinator), POLITO (representing research/academic partners), FM (representing industry), and KNEIA (as WP9 leader). The CDT will be led by Cristina Barragan from KNEIA. She will be responsible for leading the consortium in planning and implementing the project Communication and Dissemination activities, as well as ensuring the project reaches out to different audiences in the most effective way, and will also ensure the confidentiality of the results is safeguarded until the IP is protected.

Advisory Board (AB). This body consists of a voluntary group of experts who will engage with the project on ad hoc basis according to the specific needs of CEM-WAVE consortium and on yearly basis at project meetings. This body has no voting rights and a confidentiality agreement will be signed with all its members. The composition of AB is set to cover the prominent experts both in academia and in industrial sectors, and its final composition and role will be detailed in the Consortium Agreement. The preliminary composition is: Byungil Yoon (DACC CARBON Co., Ltd., South Korea), Alexey Kulik (Soft-Impact Russia), Alberto Ortona (SUPSI, Switzerland), Roland Weiss (Consultant for CMCs, Germany).

3.2.2. Meetings and communication

The Steering Committee will meet at least monthly as a regular basis by electronic means, in order to ensure the appropriate follow-up of the project. The Valorisation Committee will meet every three months as a regular basis by electronic means, where aspects in relation with the dissemination, and identification and management of Key Exploitable Results will be discussed. The GA will be convoked every 6 months, although extraordinary meetings can be proposed by the CO in case critical situations arise that might require it. During the meetings of the GA, the Steering Committee and the Valorization Committee will present their activities, advances and conclusions.

3.2.3 Decision-making

The CO is responsible for submitting relevant decisions in scientific and technical terms, including risk management, to the evaluation of the SC. Decisions will be taken by a simple majority. In case of strategic decisions concerning actions to solve conflicts, mitigate critical risks to project implementation, make amendments to the grant agreement, etc., the SC –chaired by the CO- will require the decision of the GA. In the GA by default, any decision can be taken by a majority of 2/3 of the voters, and to vote there should be at least 2/3 of the members present or represented. In case of this situation, the CO has a quality vote.

3.2.4 Appropriateness of organisational structure and decision-making processes

The foreseen organisational structure and decision-making processes take into account the geographic distances between the partners of the consortium and their respective competencies and expectations, the size of the consortium as well as the representation of each participating institution. The organisational structure and decision-making processes are tailored to achieve the research ambition and expected innovation impact of the project outcomes. Responsibility of each body is clearly outlined, and the internal following procedures will ensure each member understands his/her responsibility and the context in terms of process and follow up appropriately pushes for a strong and committed collaboration between the members. The CO will take advantage of an experienced Project Management Team, and the Steering Committee will perform a consensual strategic management of the project. End-users pulled technologies/products will be useful to ensure the management of the innovation and related Intellectual Property Rights (IPRs) by the Valorization Committee in its efforts in successfully pushing CEM-WAVE outcomes to the market. The functions and bodies at all levels are occupied by the most appropriately skilled and experienced persons in order to efficiently address S&T and innovation objectives of the CEM-WAVE project.

3.2.5 Innovation management

A key objective of the CEM-WAVE project is the steady translation of exploitable scientific and technological results into market-oriented products. The approach of the Innovation Cycle by the CEM-WAVE consortium consists in a very pragmatic Innovation Management essentially driven by commercial actors from steel and ceramic composites industrial sectors. Scientists and engineers will closely interact with industrial partners who perfectly know the market needs and volume potential, industrial and technological constraints and opportunities as well as regulatory standards to match the demand. The Valorization Committee and its related procedures is designed to do not leave to change the identification of exploitable results, but boost on an organized, conscious, professional and committed manner the focus on exploitation and market uptake of the research efforts.

Table 3.2a *- List of milestones*

MS no.	Milestone name	WP(s)	Due date	Means of verification
MS1	Communication Plan	WP9	6	A robust communication plan to support the recognition of CEM-WAVE as a reference in the CMCs market place will be available and to secure stakeholder engagement with the project.
MS2	Coating set-up of high degree coverage of non-oxide fibers	WP1	10	The most suitable coating materials and techniques for non-oxide fibers will be selected for the subsequent preforms production to be used in the MW-CVI plant (WP3).

MS3	Delivery of improved preforms	WP1	20	Selected geometries (plates and tubes) of oxide and non-oxide preforms will be send for the infiltration in the MW-CVI plants (WP3).
MS4	Feasibility of solid-state technology for MW-CVI process	WP2	20	The new solid-state technology for oxide and non-oxide MW-CVI processes will be available.
MS5	Mid-Term LCA	WP8	20	LCA, and LCC analysis will be performed. The mid-term results will be used to evaluate the sustainability of the innovative CEM-WAVE processes and applications.
MS6	Mid-Term Risk Assesment	WP8	20	Risk analysis evaluating technological and non-technological barriers will be developed to guarantee the feasibility of CEM-WAVE objectives
MS7	Numerical Solution giving complete Production of a set of optimal processing parameters allowing reduction of time and improving efficiency	WP5	25	A complete simulation of MW-CVI will be carried out to evaluate the optimized operative conditions to decrease infiltration and increase efficiency
MS8	Joining of oxide and non-oxide CMCs achieved	WP6	36	CMC/CMC and ceramic/CMC joining technique will be available to produce pieces with complex geometry
MS9	Production of non-oxide and oxide samples	WP3	40	Oxide and non-oxide CMCs samples will be available for physical and mechanical characterization and for the testing in relevant and future environment (WP 4-7).
MS10	Validation of CMCs properties in relevant environment and for future extreme and fluctuating conditions	WP4-7	40	CMCs with tailored properties and high lifetime for relevant user case environment and for future extreme and fluctuating conditions will be available.

The management structure has been conceived to minimize risks and facilitate the establishment of mitigation measures, as well as quickly address any problem that may arise. The main elements of our risk management strategy are based in 7 key aspects:

1) Early identification of risks: The WP Leaders are responsible for analyzing the possible risk sources as part of the regular review of critical risks for implementation list, during the monthly WP team meetings.
2) Efficient communication: The continuous follow up of the tasks will be guaranteed from the monthly interaction amongst WP teams and WP Leaders, and subsequent progress teleconferences with the CO.
3) Prioritization of risks: The risk tracking will be organized considering its severity (defined as the likelihood and consequence).
4) Full understanding of reason and impact of each risk: An initial analysis of each risk has been performed in order to define its impact and likelihood, but this analysis will be reviewed as the project evolves –in coincidence with the consortium meetings as default- in order to not underestimate possible deviations.
5) Track risks and associated tasks: definition of a list of critical risks for implementation, as the basis for a systematic review.
6) Develop preventive tasks for each risk: definition of risk-mitigation measures in the critical risks for implementation table.
7) Establish responses to each risk: as the elements required for the materialization of a risk arise, the SC will put into practice without delay the corrective measures required for overcoming the deviations.

The table below compiles a list of any significant risk identified at this stage of the project preparation and the associated contingency plans. Risks in Table 3.2b are related to the overall project and are then complemented by WP-specific risks.

Table 3.2b *- Critical risks for implementation*

Description of risk (level of likelihood: Low / Medium / High)	WP(s)	Proposed risk-mitigation measures
Execution Risks – Partner Related		
Partner leaving the project. Allocated and possibly key-part of the work cannot be performed, delaying the project schedule. Likelihood: Low; Severity: Medium.	All	If possible, the affected tasks can be allocated to another partner(s). Otherwise, the flexible management structure allows quick inclusion of new partners in the consortium if necessary.
Key-person left or is temporarily not available or part of work allocated to one partner cannot be performed on time and with expected quality. Likelihood: Low; Severity: Medium.	All	Consortium partners are involved in the related areas with more than one staff member, ensuring an immediate substitution. Furthermore, the project as a whole has technical excellence in related disciplines spread across the partners, providing

		additional substitution possibilities or an appropriate shift of resources and responsibilities.
Execution Risks – Planning Problems		
Resources are underestimated. Part of the planned work cannot be completed in accordance with plan. Likelihood: Medium; Severity: High.	All	Such a situation is identified early due to regular monitoring. Project management bodies will analyse the following possibilities to ensure that planned work can be completed: 1) involve further partners with available resources, 2) rearrange resources among partners, 3) ensure involvement of organizations' internal resources, 4) re-plan work.
Project schedule is partly not appropriate (e.g., due to changes) – Common project outcome and impact cannot be achieved as planned. Likelihood: Low; Severity: High.	All	Project management continuously monitor and measures performed work. Corrective actions (e.g., change of the project plan) will be swiftly performed if necessary. In crucial cases, the coordinator will work on the plan adaptation in close cooperation with EU funding bodies.
Technical Risks		
Selected raw materials or layer thickness for fiber coating does not allow the technological application Likelihood: Low; Severity: Low.	WP1	Inherent in the activity of WP1 the best compromise among interphase layer and thermal conductivity will be analysed to optimize the fiber coating. Moreover, the consortium can use different materials for required fiber coating properties.
Application of solid-state technology to improve heating efficiency not feasible Likelihood: Low; Severity: Medium.	WP2 WP3	Back-up strategy: CEM-WAVE consortium will improve the current MW technology to realize a hybrid process in order to obtain CMCs with tailored properties for the use case and, at the same, time reducing process cost and improving efficiency with respect to conventional technologies.
Final CMC porosity <10% not reached with MW-CVI technology Likelihood: Medium; Severity: Low.	WP3	Back-up strategy: Further coating or further densification cycle will be carried out. The addition of these steps do not affect the CMCs final cost
Proposed technology (MW-CVI) is not feasible for fabrication of damage free oxide and non-oxide CMCs with high thickness. Likelihood: Medium; Severity: Medium.	WP3	Experience with MW heating in other sectors shows that a proper choice of MW inlets will enable a complete "illumination" of all the components to be fabricated, whatever their degree of complexity. Also, the use of multi-physics modelling and advanced MW heating should reduce the risk of temperature inhomogeneities leading to microcracks and other forms of damage. In case of difficulties, modelling will be further carried out to find possible technical solutions, like the excitation of additional EM modes to mitigate the problems possibly experienced.
In the CEM-WAVE schedule, the testing machines used for the characterization of the CMCs are not available (breakage of parts, maintenance, etc.) Likelihood: Low; Severity: Low.	WP4	Due to the wide availability of equipment within the consortium the scheduled experiments will be performed at another beneficiary avoiding delays.
The parameters optimized through modeling do not reflect the experimental data obtained Likelihood: High; Severity: Low	WP5	Modeling based on more suitable equations and on AI advanced tools, starting from the experimental data collected, will allow the simulations carried out to be updated and verified.
Joining technique to assemble CMCs is not feasible Likelihood: Low; Severity: High.	WP6	CEM-WAVE consortium is able to face this risk by using a previously realised metal/CMC joining process satisfying the end user requirements.
Sensors embedded in the process and in the production line of the end user may not be effective for evaluating the parameters and life of the piece. Likelihood: Medium; Severity: Low.	WP3 WP7	During the process, the consortium uses the material itself as a sensor by live monitoring the change of its properties. The monitoring during the lifetime of the component can, however, be performed effectively with modelling activities.
LCA and LCC reports low feasibility of the new processes and products. Likelihood: High; Severity: Low	WP8	Low feasibility of CEM-WAVE products & processes will be analysed constantly to identify, among other criteria, the best CMCs properties considering also the CEM-WAVE user case.
Products with similar assets reaches the market during the project duration. Likelihood: Low; Severity: Medium	WP9	This risk is mitigated by our combination of novel approaches, aimed at being protected which design will be facilitated by novel modelling. The partners will perform systematic technology watch to early identify new products and therefore devise if needed

		areas of intervention to go beyond, e.g. by higher technological content or lower cost.
Discrepancies in the consortium for the technical approaches. Likelihood: Medium; Severity: Low.	WP10	Regular communication among WP leaders to solve all technical issues. The CO, with support from the Project Management Team and the assistance of General Assembly, will control this issue.

3.3 Consortium as a whole

In order to achieve the objectives of CEM-WAVE, a multidisciplinary and complementary consortium has been built. Each one of the partners and members of the consortium has the proven and strong innovation and industrial experience and the capacity to achieve the objectives and milestones set out in the CEM-WAVE project.

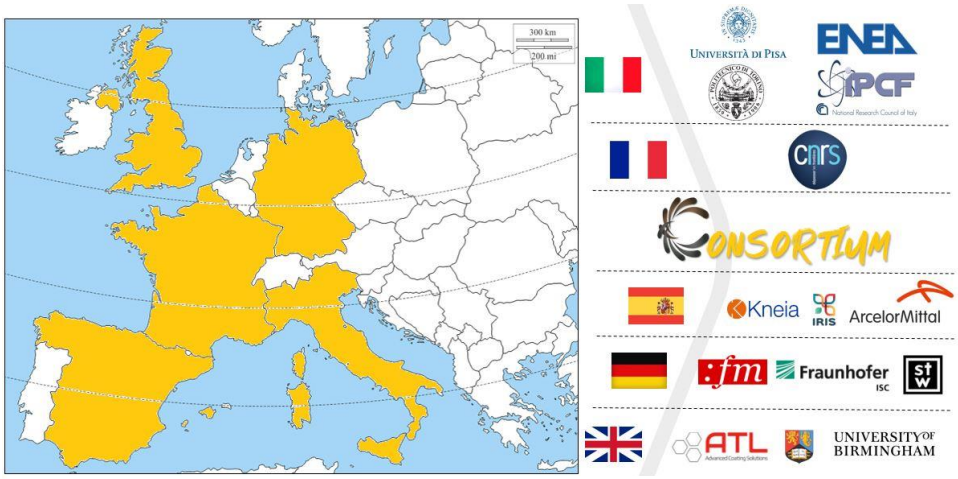


Figure 3.3 *– Map of the CEM-WAVE consortium*

The CEM-WAVE consortium consists of **13 partners from 5 European countries** (Italy, Germany, United Kingdom, France and Spain; see *Figure 3.3*), gathering the necessary infrastructure, knowledge, expertise, services and industry value chain to reach the project's objectives and the expected impacts of the topic ***LC-SPIRE-08-2020 - Novel high performance materials and components***. The CEM-WAVE consortium benefits from a strong industrial and commercial involvement since 1 Large Companies and 5 SMEs collaborate with a total of 6 RTD institutions. All these partners are familiar with European research projects and are used to interact with large groups. Moreover, most of these partners have already worked together with many other partners of this consortium in the frame of past and on-going research projects (Section 1.3). The CEM-WAVE consortium is composed by a strong and strategic consortium with a wide know-how and expertise in the different areas covered in the project.

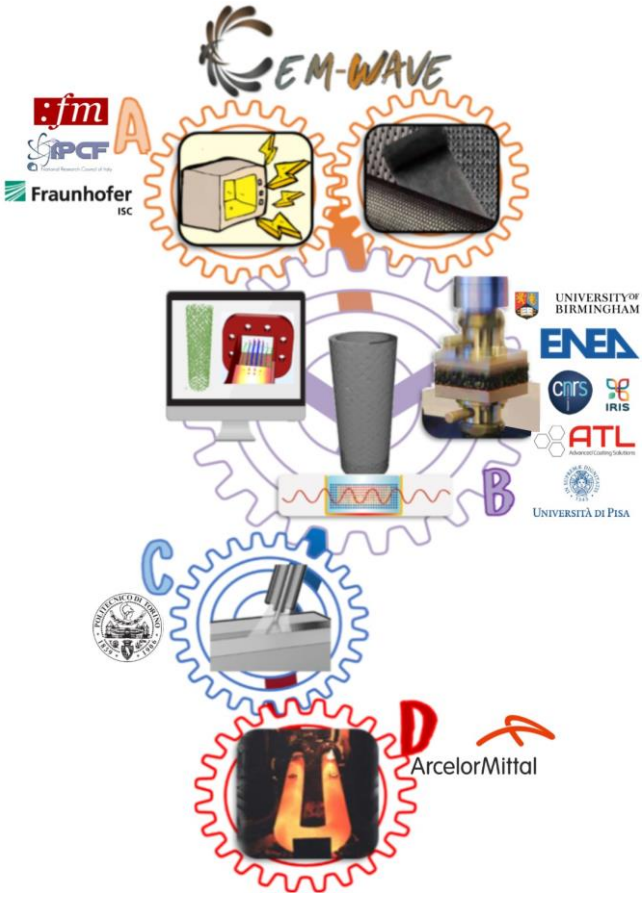


Figure 3.4 *– CEM-WAVE streams and competences: a) set-up; b) Processing, characterisation and modelling; c)joining and coating; d) use cas.e*

The consortium has been coherently built on the solid cluster of scientific and technical competencies covering all relevant fields: a) set-up of materials and innovative MW technologies (FHF, IPCF-CNR, FM i.e. WP1 and 2); b) processing, characterisation and modelling (UNIPI, UoB, CNRS, ENEA, ATL, i.e. WP3, 4 and 5); c) joining and coatings (POLITO, FHF, i.e. WP6); d) validation in future (lab) and actual (industrial) conditions (AMIII, CNRS, IRIS, i.e. WP7) (See *Figure. 3.4*). In its ambition to make a quantum leap in creating marketable products, the CEM-WAVE consortium has identified the essential need to consider regulative constraints (end-users), financial aspects (KNEIA), as well as environmental, economic and energy efficiency studies (R-TECH) together with all partners involved in the value chains, at an early development stage of these products In order to communicate and disseminate the project outcomes to targeted and larger audiences the company KNEIA, which is specialized in communication and dissemination activities, will work in close collaboration with the coordinator UNIPI, and all the consortium. On the whole, the CEM-WAVE consortium gathers all actors needed in a complementary and synergistic way to achieve its ambitious objectives. CEM-WAVE is a project with strong cooperation between RTO and industries. The WP structure and timing has been shaped with inputs from the industry actors who strive at gaining competitive advantage in

betting on novel CMCs enabled innovations. The major presence of industrial organizations (**1 Large Company and 5 SMEs**) in the project guarantees the success of the CEM-WAVE project development and the achievement of objectives and results from TRL 3 to TRL 5. As represented in Section 4, all of them have an unbeatable resume and background in the covered areas of the project. There is no overlapping of activities in the execution of the project since each partner has been assigned with a well-defined role in the research and innovation activities according to its background and tasks. Anyway, the consortium has collectively a broad expertise that will ensure, on the one hand, the proper achievement of the requested TRL, and on the other hand, a backup strategy in case one partner will leave the project (since another partner might perfectly be in charge of the tasks). The balance between the different organizations, is the most suitable for the development of the purpose of the project and ensure the achievement of the objectives and the compliance of the KPI, according to the requirements of the call ***LC-SPIRE-08-2020***. As shown below in section 3.4, the budget is adequate according to the project size and requirements, which will guarantee the achievement of the proposed objectives.

3.4 Resources to be committed

Table 3.3a *- Summary of staff effort*

	WP1	WP2	WP3	WP4	WP5	WP6	WP7	WP8	WP9	WP10	Total PM per Participant
UNIPI	5,5	5,5	**41**	39,5	2,5	1	7	2	9	**23**	**136**
IPCF-CNR	4	**27**	14	6	0	0	6	1	1	1	**60**
FM	0	10	5	0	0	0	4,5	0,5	1	1	**22**
FHF	**13,5**	0	1	0	0	6,02	0	0,5	1	1	**23,02**
UoB	1	1	16,1	2,6	1	1	1	0,5	1,4	0,8	**26,4**
ATL	0	23	6	0	0	0	0	1	1	1	**32**
ENEA	1	0	0	**12**	0	2	3	0,5	1,5	0,5	**20,5**
CNRS	0	0	11,2	7,5	**18,6**	1,5	5	0,5	1	1	**46,3**
POLITO	1	0	2	12	0	**24**	2	0,5	2	1	**44,5**
IRIS	0	3	21	0	0	0	14	1	2	2	**43**
AMIII	2	0	0	1	0	0	**27**	1	1	1	**33**
R-TECH	0	0	0	0	0	0	0	**19**	1,2	1	**21,2**
KNEIA	0	0	0	0	0	0	0	1,5	**39**	1	**41,5**
Total PM	**28**	**69,5**	**117,3**	**80,6**	**22,1**	**35,52**	**69,5**	**29,5**	**62,1**	**35,3**	**549,42**

Table 3.3b *- 'Other direct cost' items*

Participant 1/UNIPI	**Cost (€)**	**Justification**
Travel	32.000	10.000,00 € Attendance to conferences; 2.000,00 € Site visits; 20.000,00 € Consortium meetings and Advisory boards travels to consortium meetings
Equipment	18.000	18.000,00 € Muffle Furnace with Integrated Exhaust Gas Cleaning. Cost: 30000; Depreciation time: 5 years. Actual use on the project 3 years (from M7 to M42)
Other goods and services	130.000	100.000,00 € Non oxide fibers for SiC-based preform's preparation; 25.000,00 € Reagents and Consumables (H2, N2, MTS, etc.); 5.000,00 € Certificate on the financial statement
Total	**180.000**	
Participant 3/FM	**Cost (€)**	**Justification**
Travel	10.500	Management meetings (7x1500 euro each).
Other goods and services	120.000	70.000,00 € for the Solid-State Microwave Generators 0-6 kW (2.45 GHz) and 4 kW (915 MHz), 3-Stub-Tunner, Microwave Components (Isolators, Filters, Waveguides, Coax Cables, Connectors, Absorber Load), Software; 30.000,00 € for IR Temperature Measuring System with corresponding Software; 20.000,00 € Microwave Measurement Systems (2 High-Power Meters, 2 Directional Couplers, Leakage Detectors, potable Spektrum Analyzer.
Total	**130.500**	
Participant 4/FHF	**Cost (€)**	**Justification**
Travel	22.500	17.000,00 € Consortium, Work Meetings and site visits (total of 17); 4.000,00 € Attendance to Conferences (total of 2); 1.500,00 € Visit on trading show fairs (JEC, Hannover Messe).
Equipment	7.500	Slurry impregnation unit.
Other goods and services	140.000	70.000,00 € Ceramic fibres; 10.000,00 € Powders and precursors for slurries; 20.000,00 € Tools and sintering devices; 20.000,00 € Precursors and chemicals for wet-chemical coating; 20.000,00 € Precursors and chemicals for EBC coating.
Total	**170.000**	
Participant 5/UoB	**Cost (€)**	**Justification**
Travel	11.850	4.740,00 € Attendance at consortium meetings by postdoc & PI; 5.925,00 € Attendance at conferences, workshops, etc. and for dissemination; 1.185,00 € Travel & subsistence associated with using facilities not at UoB.
Equipment	59.250	Parts for use in microwave CVI tor producing oxide composites

Other goods and services	71.100	35.550,00 € Chemicals, powders, gases, etc.;11.850,00 € Spare parts, etc, for equipment items; 23.700,00 € Characterisation costs.
Total	**142.200**	
Participant 6/ATL	**Cost (€)**	**Justification**
Travel	13.000	8.000,00 € Consortium meetings (8); 5.000,00 € Service Visits to UNIPI (2).
Other goods and services	30.000	4.000,00 € Consumables wire, tube, pipe, fittings etc.; 10.000,00 € Repair and service costs for equipment at UNIPI; 16.000,00 € Replacement parts for equipment at UNIPI.
Total	**43.000**	
Participant 7/ENEA	**Cost (€)**	**Justification**
Travel	14.595	3.495,00 € Attendance to Conferences and congresses; 11.100,00 € Attendance to workshops, fairs and project meetings.
Equipment	13.200	11.200,00 € MTS upgrade (depreciation: 2 years of use); 2.000,00 € Laboratory consumables necessary to perform Thermo-physics measurements.
Other goods and services	50.000	2.000,00 € Consumables for CVD; 5.000,00 € Consumables for SEM characterisations; 24.000,00 € Consumables for CVI (technical gases) and joinings development; 14.000,00 € Furnaces, pilot lines and MTS calibration and upgrades; 5.000,00 € vacuum sensors calibration.
Total	**77.795**	
Participant 8/CNRS	**Cost (€)**	**Justification**
Travel	22.000	Consortium and Work Meetings (10.000 €) and conference (12.000 €);
Equipment	28.000	3.000,00 € Flow meters; 25.000,00 € Cluster purchase.
Other goods and services	100.000	10.000,00 € Open source gold route publication costs (5 articles); 20.000,00 € holder manufacturing, manufacture of a calorimeter, oxygen and methane; 63.000,00 € ceramic inlays; 5.000,00 € software license; 2.000,00 € fluids for corrosion test.
Total	**150.000**	
Participant 9/POLITO	**Cost (€)**	**Justification**
Travel	23.000	9.500,00 € Consortium Meetings: kick-off meeting, face to face meetings (two per year): total of about 8 meetings. Participants: 2 people. Average travel cost 600 euro; 11.00,00 € Attendance to Conferences (about 8). Participants: 2 people. 2.500,00 € Attendance to project workshop (total of 2). Participants: 2 people. Average travel cost: 600 euro.
Other goods and services	41.500	15.500,00 € Laboratory consumable; 5.000,00 € Spare parts for furnaces; 9.000,00 € Starting materials (e.g. brazing alloys); 3.000,00 € Technical gases; 4.000,00 € Consumables for characterization (e.g. FESEM, micro-CT); 5.000,00 € Open access: two publications.
Total	**64.500**	
Participant 10/IRIS	**Cost (€)**	**Justification**
Travel	21.000	6.000,00 € visit to partners to install monitoring devices; 3.000,00 € travel to trade fairs and conferences; 12.000,00 € travel to project meetings.
Other goods and services	32.000	20.000,00 € monitoring devices and mechanical interface with process; 10.000,00 € complementary monitoring devices where WP3's are not adjustable and new mechanical interface with pipes; 2.000,00 € software licence, printing of materials and registration to events.
Total	**53.000**	
Participant 11/AMIII	**Cost (€)**	**Justification**
Travel	10.000	Attending project meetings (12 meetings*2 person/meeting).
Other goods and services	120.000	Reconditioning of the plant for including H2 (60.000€); Civil works and construction of the foundation for testing the sample (60.000€).
Total	**130.000**	
Participant 12/R-TECH	**Cost (€)**	**Justification**
Travel	20.000	Kick-off meeting, 6 project meeting, 2 participants per meeting, attendance to conferences (2x).
Other goods and services	10.000	LCA Material data.
Total	**30.000**	
Participant 13/KNEIA	**Cost (€)**	**Justification**
Travel	15.000	12.000,00 € Consortium Meetings (total of 8 travel/meetings - 1500 per trip); 3.000,00 € Attendance to Conferences (total of 2 trips).
Other goods and services	31.800	1.000,00 € Project logo and Identity Set; 4.200,00 € Project website; 3.100,00 € Dissemination materials (poster +leaflets+ pen drives); 8.500,00 € Video; 14.000,00 € Conference organization; 1.000,00 € Conference attendance fees.
Total	**46.800**	

No large research infrastructure use is foreseen in the project.

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